263-13399Y

TMP POWER UNIT (for TMP-xx03 Series)

EI-R04M

Instruction Manual

(Original Instructions)

Read the instruction manual thoroughly before you use the product.
Keep this instruction manual for future reference.

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Introduction

Thank you for choosing the EI-R04M Power Supply Unit for Turbo Molecular Pump (hereafter referred to as "power supply unit" or "EI-R04M"). Please read this instruction manual carefully before using the power supply unit, and save the instruction manual for future reference.

This instruction manual explains detailed operations of the power supply unit and cables. For instructions regarding the pump unit, please refer to the instruction manual for the pump unit to be used.

Standard type is explained in this manual. For special order type, please refer to the outlines and constructions of each specification.

Copyrights and Disclaimers

An original language of this manual is English. The manuals of the same model described in other languages are the translations of the English manual.

In an effort to improve the product, this document may be revised in the future without notice.

Every effort has been made to prepare an accurate and complete manual, but if an error or omission should be discovered, revisions might not be possible immediately.

Shimadzu does not take responsibility for any effects that may result from the use of this manual.

Precautions for Safe Operation

The instruction manual's nomenclature for warnings and precautions complies with the following safety warning symbols.

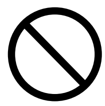
WARNING Indicates a potentially hazardous situation which, if not avoided, could result in serious injury or possibly death.

CAUTION Indicates a potentially hazardous situation which, if not avoided, may result in minor to moderate injury or equipment damage.

NOTICE Emphasizes additional information that is provided to ensure the proper use of this product.

WARNING

Turbo molecular pump repair and/or power supply repair can be very hazardous.
Only trained technicians who are authorized by Shimadzu may do service of products.



WARNING

Neither overhaul nor modify the pump proper and power supply unit without admission. Doing so would impair safety of the pump proper, or cause injury by electrical shock.

WARNING

Decisions on system compatibility should be made by the system designer or the person deciding the specifications after conducting tests as necessary. The responsibility for guaranteeing the expected performance and safety of the system lies with the person who decides system compatibility.

CAUTION

The power input voltage of the power supply unit EI-R04M is 200 to 240 VAC. Connect the power supply unit to the voltage indicated on the side panel label only. Connection of the power supply unit to the incorrect input voltage can cause damage to the equipment. Supply the power via a circuit breaker (rating 15 A). Please provide PE (Protective Earth) connection to the terminal of a "PE" marked wire in final application. When power cable whose length is longer than 5 m is used to input AC power, protective earth line has to be connected to the PROTECTIVE EARTH TERMINAL (Fig. 2-2 (20)) which is on rear panel of the power supply unit with the cable which has shorter than 5 m and thicker than AWG14.

CAUTION

If an EI-R04M power supply unit is used in combination with an existing pump that was operated in combination with a power supply unit not having the variable speed function, the variable speed function cannot be used (the "xx" number indicates the model of the corresponding pump).

If the power supply unit is to be combined with an existing pump, modification and operational inspections are necessary. Please contact Shimadzu for detailed information.

CAUTION

The following "CAUTIONS" are to prevent operation anomalies.

○ Operating Precautions

(1) Do not interrupt the electrical power operating the turbo molecular pump while the turbo molecular pump is in operation.

(2) Do not connect or disconnect the turbo molecular pump control cable during the time the power supply is "ON".

(3) Do not operate any equipment (i.e. drill motor, welding machine, etc.) that produces electro-magnetic pollution, noise, etc., in the immediate proximity of an operating turbo molecular pumping system (pump, power supply, cables, etc.).

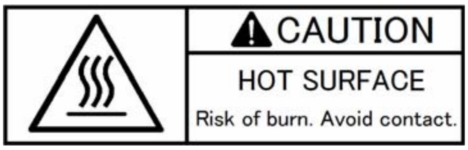
(4) When using the variable speed function to change the pump rotation rate, use a rotation rate that does not cause resonance with other devices installed at the site.

(5) Be careful to prevent a rapid pressure rise or air rush during operation.

NOTICE

Guaranteeing the expected performance covers only a combination of TMPs and controllers sold by Shimadzu.

○ Explanation of Label



(1) HOT SURFACE: Risk of burn.

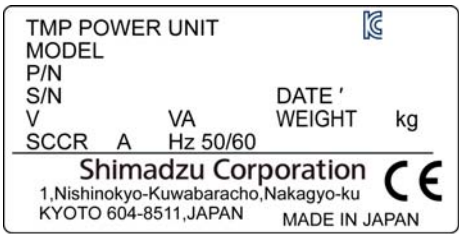
- Keep off from touching surface of the pump as it is heated.
- Keep off from touching upper side surface of the power supply while the pump is in deceleration as it is heated.

(2) Do not remove cover, or else it may cause some changes inside and it is failed.

(3) Be sure to use specified cable for this power supply. If not, it may cause connector be broken and power supply itself failed.

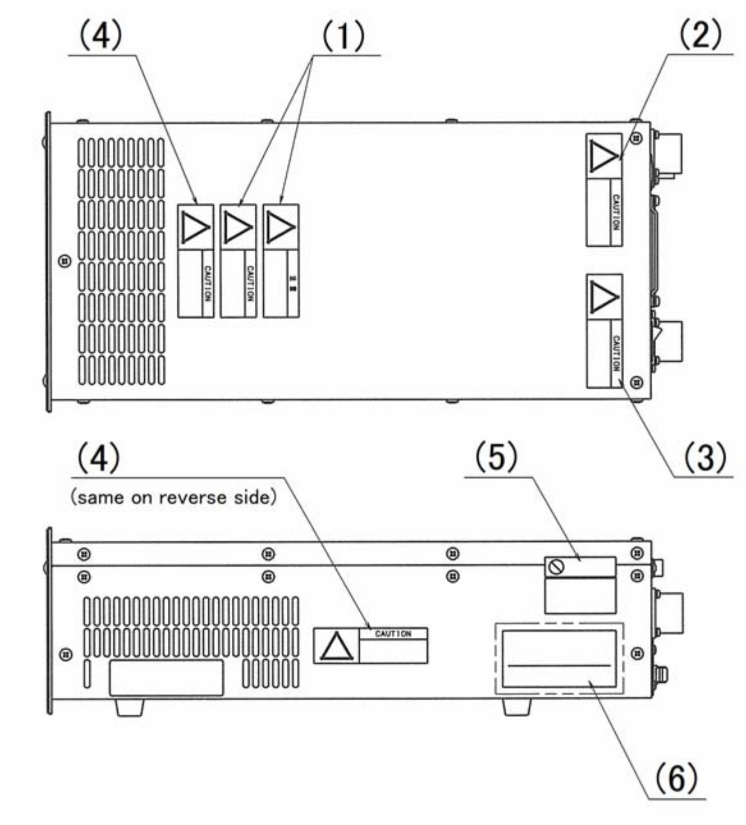
(4) Do not shut off ventilation, or else the inside of power supply get heated and it is failed.

(5) SECURITY Seal
This label certificates that the product was made or maintained by Shimadzu or by Shimadzu authorized facility.
In case "this label is removed" or "there is mark showing once this label has been removed", Shimadzu warranty shall not be applied to the product.



(6) Name Plate

○ Location of Label

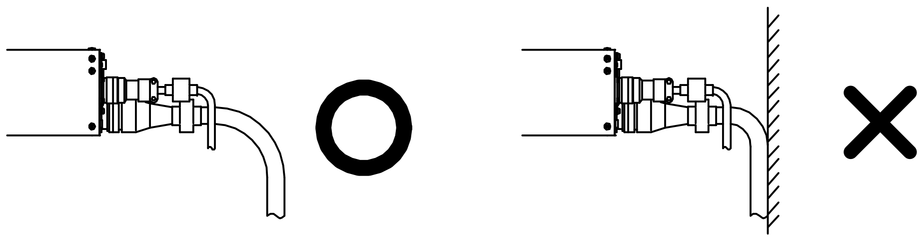


○ Installation Precautions

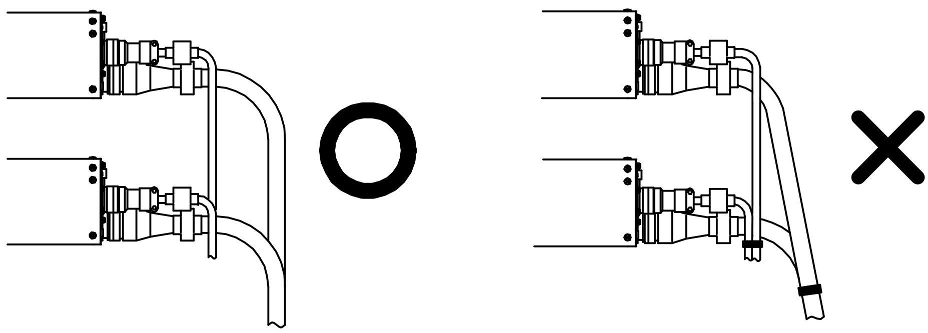
Do not apply abnormal loads to the turbo molecular pump control cable plug and/or connector. Abnormal loads may cause cable disconnection.

(1) Do not pull the turbo molecular pump control cable by the connector or plug.

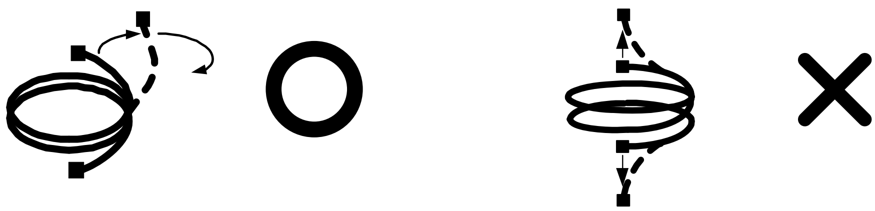
(2) When installing the power supply unit into equipment, do not allow any electrical cables to be in tension or to have very tight bending radii.



(3) Do not bundle the turbo molecular pump control cable with any cables.



(4) Unbend the turbo molecular pump cable not to make any kink during connection.



○ Part Replacement

The lifetime of parts are specified as below.

The request for changing parts exceeding the estimated lifetime should be made to Shimadzu or an approved service company in order for safety and adequate performance of the pump and power supply unit.

Table 1 Estimated Part Life

Part list	Estimated part life
Transformer	10 years
Electrolytic condenser	5 years
Cooling fan	5 years
Button-type battery	10 years

○ Warranty Period

12 months on new TMP's from the date of shipment from Shimadzu, or from any of its worldwide sales offices.

○ Conditional Warranty

During the warranty period and under normal operation, if the TMP fails to meet its product specification due to defects in material and/or workmanship, Shimadzu will, at its discretion, either repair it or exchange it with a new one for free.

○ Scope of the Warranty

The warranty covers only TMPs, controllers and accessories sold by Shimadzu.

○ Warranty of Repaired or Replacement Parts

In-warranty repaired or replacement parts are warranted only for the remaining unexpired portion of the original warranty period applicable to the parts that have been repaired or replaced.

○ Exemption from the Warranty

During the warranty period, Shimadzu will charge for repair or exchange in the following cases:

1) Failure caused by natural disasters or fire.
2) Failure or functional deterioration due to the following:
   a) Pumping of special gases and materials.
   b) Dropping of foreign objects through the TMP's protective net.
   c) TMP is operated differently than what is prescribed in the instruction manual.
   d) When Shimadzu determines through failure analysis that the cause of failure was due to abnormal operation or external circumstances. Our engineers judge that the cause of the trouble is an irregular operation.
3) Warranty is voided if the "Security Seal" on the product has been removed, or there is residue or evidence the seal has been tempered with.

○ Disposal of Products and Parts

Please contact Shimadzu for proper disposal of its products or parts. There is a possibility to pollute the environment with the material of the parts, when you dispose this product in an inappropriate way.

○ LIMITATION OF LIABILITY

EXCEPT AS STATED HEREIN, SHIMADZU MAKES NO WARRANTY, EXPRESSED OR IMPLIED (EITHER IN FACT OR BY OPERATION OF LAW), STATUTORY OR OTHERWISE: AND, EXCEPT AS STATED HEREIN, SHIMADZU SHALL HAVE NO LIABILITY FOR SPECIAL OR CONSEQUENTIAL DAMAGES OF ANY KIND OR FROM ANY CAUSE ARISING OUT OF THE SALE, INSTALLATION, OR USE OF ANY OF ITS PRODUCTS.

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Appendix A COMMUNICATIONS

1

OUTLINE AND DESCRIPTIONS

1.1 Outline

The turbo molecular pump is a vacuum pump. The turbo molecular pump is used with a backing vacuum pump to create a high vacuum in a vacuum chamber.

Typical Applications; Semiconductor equipments,
Industrial equipments,
R&D applications,
The other ultra high vacuum applications.

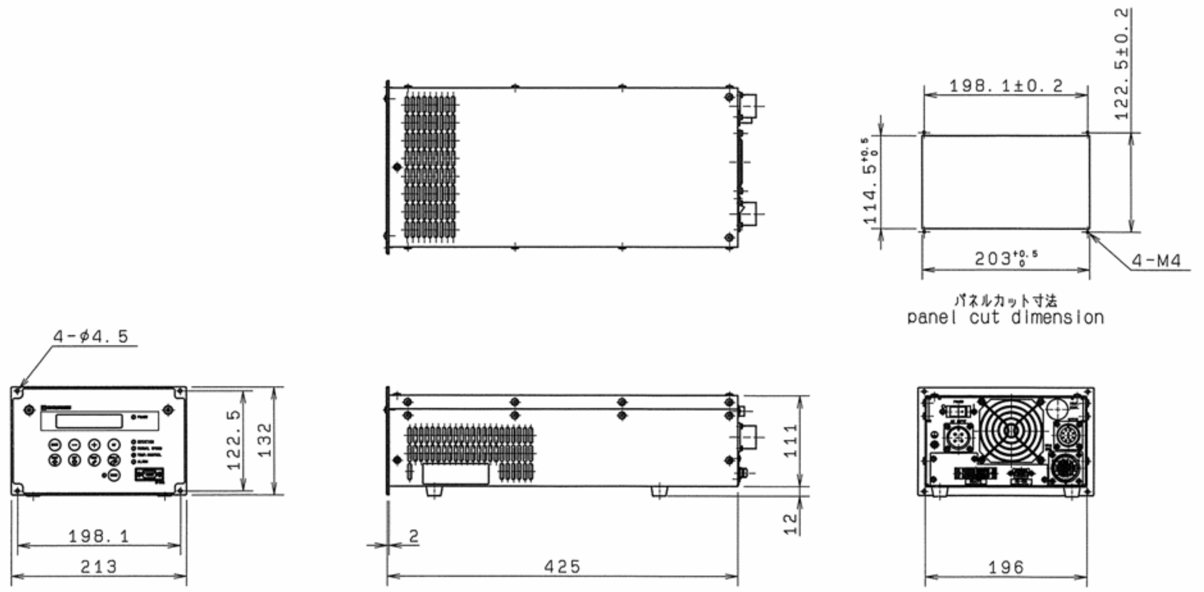
The turbo molecular pump (one standard set) consists of the following items.

・Pump Unit	1
・Power Supply Unit	1
・Control Cable	1
・Motor Cable	1
・Standard Accessories	1 Set

The cable length must be specified for the control cable and motor cable (Refer to Section 1.2.2 and Section 1.2.3).

1.2 Descriptions

1.2.1 Power Supply Unit



Description	Part number
EI-R04M	262-78982-02

Fig. 1-1 Outside Dimensions of Power Supply Unit

1.2.2 Control Cable

The cable can be selected from the following.

Description	Note	Part number
Control cable	3 meters length, straight plugs for both sides.	263-16200-03
	5 meters length, straight plugs for both sides.	263-16200-05
	7 meters length, straight plugs for both sides.	263-16200-07
	10 meters length, straight plugs for both sides.	263-16200-10
	15 meters length, straight plugs for both sides.	263-16200-15
	20 meters length, straight plugs for both sides.	263-16200-20
	30 meters length, straight plugs for both sides.	263-16200-30

1.2.3 Motor Cable

The cable can be selected from the following.

Description	Note	Part number
Motor cable	3 meters length, straight plugs for both sides.	262-76409-03
	5 meters length, straight plugs for both sides.	262-76409-05
	7 meters length, straight plugs for both sides.	262-76409-07
	10 meters length, straight plugs for both sides.	262-76409-10
	15 meters length, straight plugs for both sides.	262-76409-15
	20 meters length, straight plugs for both sides.	262-76409-20
	30 meters length, straight plugs for both sides.	262-76409-30

1.2.4 Standard Accessories

	Description	Q'ty	Notes	Part number
1	Power cable	1	5 meters length	262-76773-05
2	Remote-control connector	1	MR-34MG (Pin type connector) MR-34L4 (Connector hood)	070-50791-63 070-50792-75
3	Instruction manual	1	This manual	263-13399

2

IDENTIFICATION AND FUNCTION

2.1 Power Supply Unit

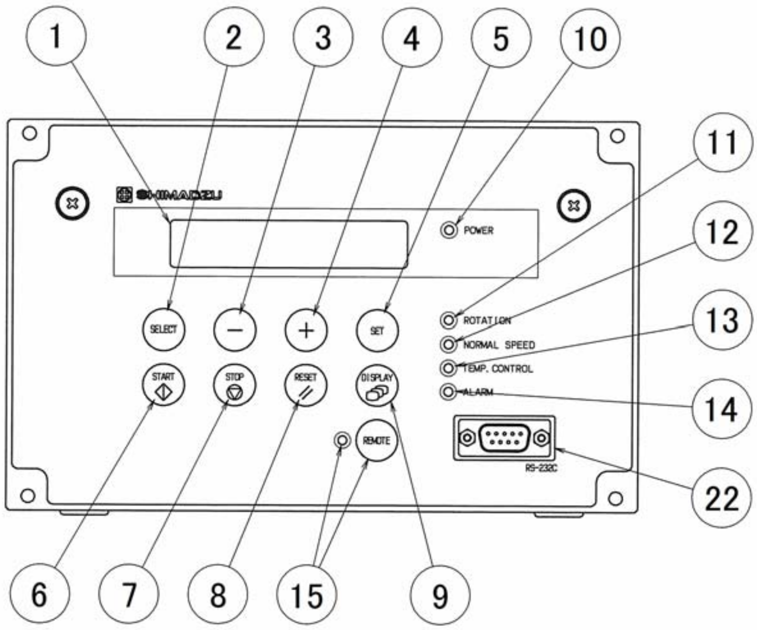


Fig. 2-1 Front Control Panel

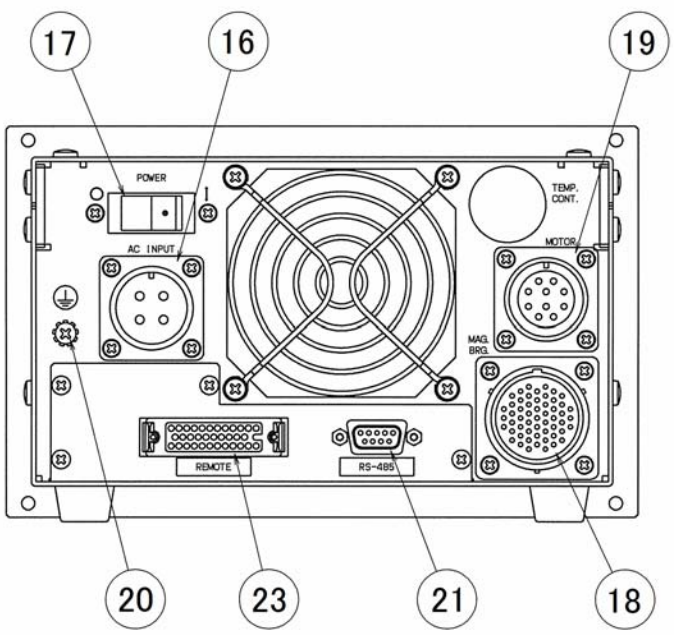


Fig. 2-2 Rear Panel

(1) LCD .... Display operation monitor, alarm contents, settings (Refer to Section 6.6 "Software Operation").

(2) SELECT KEY .... LCD display operation key, select menu.

(3) - KEY.. LCD display operation key, previous menu or subtraction.

(4) + KEY .... LCD display operation key, next menu or addition.

(5) SET KEY .... LCD display operation key, set menu.

(6) START SWITCH .... Push to start rotation of the pump's rotor.

(7) STOP SWITCH .... Push to apply the brake to stop rotation of the pump's rotor.

(8) RESET SWITCH .... Push to stop the buzzer after an alarm or warning occurs. After remedying the cause of the alarm, push the RESET switch again to turn off the ALARM lamp. However, the buzzer sounds again if the RESET switch is pushed again before the cause of the alarm is remedied.

(9) DISPLAY SWITCH.................................. LCD display operation key, change display mode.

(10) POWER LAMP...................................... Power ON indicator lamp (green).

(11) ROTATION LAMP.................................. Operation indicator lamp indicating that the pump's rotor is running (green).

(12) NORMAL SPEED LAMP........................ Operation indicator lamp indicating that the pump's rotor is rotating normally (green).

(13) TEMP. CONTROL LAMP....................... This lamp is always turned off in EI-R04M.

(14) ALARM LAMP.. Alarm/warning lamp (yellow). Refer to Section 7.5 "Alarm Detection Capabilities". Lights when an alarm occurs or flashes to give a warning.

(15) REMOTE SWITCH/LAMP...................... When the REMOTE switch is pushed, the REMOTE lamp turns on or off. When the lamp ON, the signal from the REMOTE CONNECTOR and the command from the serial interface are available. When the lamp OFF, START and STOP switch on the front panel are available.

(16) AC INPUT CONNECTOR .... Power cable receptacle.

(17) POWER SWITCH .... Power switch.

(18) MAG.BRG. CONNECTOR...................... Control cable receptacle.

(19) MOTOR CONNECTOR.......................... Motor cable receptacle.

(20) EARTH GROUND .... Electrical grounding terminal.

(21) RS-485 CONNECTOR........................... Serial interface connector (Note 1).

(22) RS-232C CONNECTOR .... Serial interface connector (Note 1).

(23) REMOTE CONNECTOR......................... Remote-control connector.

(Note 1) Refer to APPENDIX-A "COMMUNICATIONS" for details.

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3

CONSTRUCTION AND PRINCIPLE

3.1 Power Supply Unit

The power supply unit is composed of the magnetic bearing control system and the high frequency motor system and does not use back up batteries for electrical power failure.

The magnetic bearing control system controls the levitation of the rotor inside the turbo molecular pump. The system detects the rotor position by an electrical signal received from the gap sensors and maintains the levitation by regulating the current to the magnetic bearings.

The high frequency motor system rotates the rotor at a rated rotational speed. This frequency power system converts AC/single phase commercial power to controlled DC/three phase pulsed power. The DC/three phase pulsed power drives the DC motor that is an integral part of the rotor. If the electrical power is interrupted while the rotor is in a high-speed rotation, then the motor becomes a generator to power the magnetic bearing system during a power failure deceleration mode; therefore, the need for a battery backup system is eliminated.

The power supply unit is equipped with an RS-232C and an RS-485 serial interface and with Contact input/output to operate the turbo molecular pump from an external source. The operational status can be monitored and the history retrieved through the RS-232C and RS-485. Refer to APPENDIX-A "COMMUNICATIONS" for instructions to remotely operate the turbo molecular pump using the RS-232C and RS-485 serial interface.

The power supply unit, control cable, motor cable and the pump unit are all mutually compatible.

4

SPECIFICATIONS

4.1 Power Supply Unit

Power supply unit model		EI-R04M					
Suitable pump		TMP-803,1003 M/MC/LM/LMC	TMP-1103 MP/MPC/ LMP/LMPC	TMP-1303,1503 LM/LMC	TMP-2003 M/MC/ LM/LMC	TMP-2203 M/MC/ LM/LMC	TMP-W2403LMC
Exchangeable compatibility		Control cable, Motor cable, Pump main unit and Power supply unit are interchangeable.					
Magnetic bearing		5 axis control. No Battery is required (When power failure occur, magnetic bearing is support by regenerative power from motor).					
Speed control		Feedback control					
Speed variation		Speed is variable between 25 % and 100 % of the rated speed (set as 0.1 %).					
Display	LCD	20 characters×2 lines (With LED back light)					
	LED	POWER / ROTATION / NORMAL SPEED / ALARM / REMOTE / TEMP.CONTROL (not in use)					
Communi-cation	Contact	REMOTE (MR 34pin) Input: START / STOP / RESET / LOW SPEED Output: ROTATION / ACC. / BRAKE / NORMAL / REMOTE / ALARM / WARNING					
	Serial	Front panel: RS-232C (D-sub 9 pin male, Screw lock size: M2.6) Rear panel: RS-485 (D-sub 9 pin female, Screw lock size: M2.6)					
Alarm detection	Alarm	Pump temperature, Pump start-up failure, Overload, Overspin for motor, Failure of magnetic bearing, Power supply malfunction (Overtemperature inside power supply, Fail drive circuit), Power failure					
	Warning	Failure of magnetic bearing, Power supply malfunction (Overtemperature inside power supply)					
Protection	Alarm	ALARM LED lights, buzzer sounds, alarm description displayed on LCD. Power failure: Decelerates while maintaining magnetic levitation by regenerative breaking power. Levitation is stopped after deceleration to a fixed low speed, and the rotor is supported by the touch-down bearing. Once power is restored, restart is possible after resetting. When other alarms occur: Stops operation or decelerates. Magnetic levitation is continued.					
	Warning	ALARM LED flashes, buzzer sounds, warning description displayed on LCD. Pump operation continues.					
Momentary power failure (Note 1)		If the electrical power is recovered in 1 second or less, then the power supply operation prior to the electrical power failure is continued. No change output signal. Otherwise, the turbo molecular pump rotor is decelerated. If the power is disrupted over 1 second, the brake will engage. The pump can be restarted after the reset operation.					
Alarm history		Stores the date, time and detected event information for the last 99 alarm events.					
Input electric power	Voltage (Note 2)	Single phase 200 to 240 VAC (50 / 60 Hz)					
	Maximum power	1.0 kVA			1.2 kVA		1.5 kVA
	Insulation withstand voltage	1500 V, 1 minute					
	Short circuit current ratings (SCCR)	200 A					
Mass		8 kg					
Environ-mental conditions	Temperatures	Operation: 0 to 45 degrees C. / Storage: -25 to 70 degrees C. (No dew condensation)					
	Relative humidity	40 to 80 %RH					
Installation conditions (Refer to UL/EN 61010-1 standard)		Use: Indoor, Altitude max: 2000 m Overvoltage category III, Pollution degree 2 IP classification 20					

(Note 1) The time can be changed to 2 seconds from 1 second.

(Note 2) Temporary voltage fluctuation range: ± 10 %
Temporary frequency fluctuation range: ± 2 Hz

<table>
<tr><td colspan="2">Power supply unit model</td><td colspan="7">EI-R04M</td></tr>
<tr><td colspan="2">Suitable pump</td><td>TMP-H3153LMC</td><td>TMP-3203M/MC/LM/LMC</td><td>TMP-3403LMC</td><td>TMP-3413LMC</td><td>TMP-W3503LMC</td><td>TMP-H3603LMC</td><td>TMP-4203LMC</td></tr>
<tr><td colspan="2">Exchangeable compatibility</td><td colspan="7">Control cable, Motor cable, Pump main unit and Power supply unit are interchangeable.</td></tr>
<tr><td colspan="2">Magnetic bearing</td><td colspan="7">5 axis control. No Battery is required (When power failure occur, magnetic bearing is support by regenerative power from motor).</td></tr>
<tr><td colspan="2">Speed control</td><td colspan="7">Feedback control</td></tr>
<tr><td colspan="2">Speed variation</td><td colspan="7">Speed is variable between 25 % and 100 % of the rated speed (set as 0.1 %).</td></tr>
<tr><td rowspan="2">Display</td><td>LCD</td><td colspan="7">20 characters×2 lines (With LED back light)</td></tr>
<tr><td>LED</td><td colspan="7">POWER / ROTATION / NORMAL SPEED / ALARM / REMOTE / TEMP.CONTROL (not in use)</td></tr>
<tr><td rowspan="2">Communi-cation</td><td>Contact</td><td colspan="7">REMOTE (MR 34 pin)
Input: START / STOP / RESET / LOW SPEED
Output: ROTATION / ACC. / BRAKE / NORMAL / REMOTE / ALARM / WARNING</td></tr>
<tr><td>Serial</td><td colspan="7">Front panel: RS-232C (D-sub 9 pin male, Screw lock size: M2.6)
Rear panel: RS-485 (D-sub 9 pin female, Screw lock size: M2.6)</td></tr>
<tr><td rowspan="2">Alarm detection</td><td>Alarm</td><td colspan="7">Pump temperature, Pump start-up failure, Overload, Overspin for motor, Failure of magnetic bearing, Power supply malfunction (Overtemperature inside power supply, Fail drive circuit), Power failure</td></tr>
<tr><td>Warning</td><td colspan="7">Failure of magnetic bearing,
Power supply malfunction (Overtemperature inside power supply)</td></tr>
<tr><td rowspan="2">Protection</td><td>Alarm</td><td colspan="7">ALARM LED lights, buzzer sounds, alarm description displayed on LCD.
Power failure:
Decelerates while maintaining magnetic levitation by regenerative breaking power. Levitation is stopped after deceleration to a fixed low speed, and the rotor is supported by the touch-down bearing.
Once power is restored, restart is possible after resetting.
When other alarms occur:
Stops operation or decelerates. Magnetic levitation is continued.</td></tr>
<tr><td>Warning</td><td colspan="7">ALARM LED flashes, buzzer sounds, warning description displayed on LCD.Pump operation continues.</td></tr>
<tr><td colspan="2">Momentary power failure (Note 1)</td><td colspan="7">If the electrical power is recovered in 1 second or less, then the power supply operation prior to the electrical power failure is continued. No change output signal. Otherwise, the turbo molecular pump rotor is decelerated. If the power is disrupted over 1 second, the brake will engage. The pump can be restarted after the reset operation.</td></tr>
<tr><td colspan="2">Alarm history</td><td colspan="7">Stores the date, time and detected event information for the last 99 alarm events.</td></tr>
<tr><td rowspan="4">Input electric power</td><td>Voltage (Note 2)</td><td colspan="7">Single phase 200 to 240 VAC (50 / 60 Hz)</td></tr>
<tr><td>Maximum power</td><td>1.5 kVA</td><td>1.2 kVA</td><td>1.3 kVA</td><td>1.4 kVA</td><td colspan="3">1.5 kVA</td></tr>
<tr><td>Insulation withstand voltage</td><td colspan="7">1500 V, 1 minute</td></tr>
<tr><td>Short circuit current ratings (SCCR)</td><td colspan="7">200 A</td></tr>
<tr><td colspan="2">Mass</td><td colspan="7">8 kg</td></tr>
<tr><td rowspan="2">Environ-mental conditions</td><td>Temperatures</td><td colspan="7">Operation: 0 to 45 degrees C. / Storage: -25 to 70 degrees C. (No dew condensation)</td></tr>
<tr><td>Relative humidity</td><td colspan="7">40 to 80 %RH</td></tr>
<tr><td colspan="2">Installation conditions (Refer to UL/EN 61010-1 standard)</td><td colspan="7">Use: Indoor, Altitude max: 2000 m
Overvoltage category III, Pollution degree 2
IP classification 20</td></tr>
</table>

(Note 1) The time can be changed to 2 seconds from 1 second.

(Note 2) Temporary voltage fluctuation range: ± 10 %
Temporary frequency fluctuation range: ± 2 Hz

4

4.2 Standards Fulfilled

Safety	EN 61010-1 UL 61010-1 SEMI S2 EN 1012-2
EMC	EN 61326-1 Class A EN 61000-3-2 EN 61000-6-2 SEMI F47 KN 11 KN 61000-6-2

5

INSTALLATION

5.1 Installation of the Power Supply Unit

5.1.1 Location of the Power Supply Unit

Install and anchor the power supply unit inside a rack, which shall be located at a place where it is not exposed to direct sun ray and well ventilated. Avoid to locate it at the following places.

- Place where it is very humid, dusty and, in addition, oil smoke, vapor, water, etc, are exist.
- Place where the power supply unit is exposed to direct sun ray and abnormally high temperature.
- Place with high amplitude of vibration and impact.
- Near chemically active gas and explosive/combustible gas.
- Place with strong magnetic field and electric field, noisy place, and place with strong radioactive ray.
- Unventilatable place.

5.1.2 Installation of the Power Supply Unit

When mounting the power supply unit onto the customer's rack, use the front panel installation holes. The installation panel and screws are to be supplied by the customer.

Prepare the shelf to support the weight of the power supply unit in the rack.

How to install the unit onto a rack

(1) Turn the power supply unit over and remove the four rubber pads.

(2) Pass the power supply unit through the cutout hole in the installation panel and secure it with four screws (Refer to Fig. 5-2).

(3) In order to ensure that the interior of the power supply unit is cooled sufficiently, leave a gap inside the rack of at least 30 mm above, 5 mm on either side, and 50 mm to the rear of the power supply unit (Refer to Fig. 5-1).

(4) The space required for installing the cables is shown in Fig. 5-5.

(5) Reattach the rubber pads if necessary.

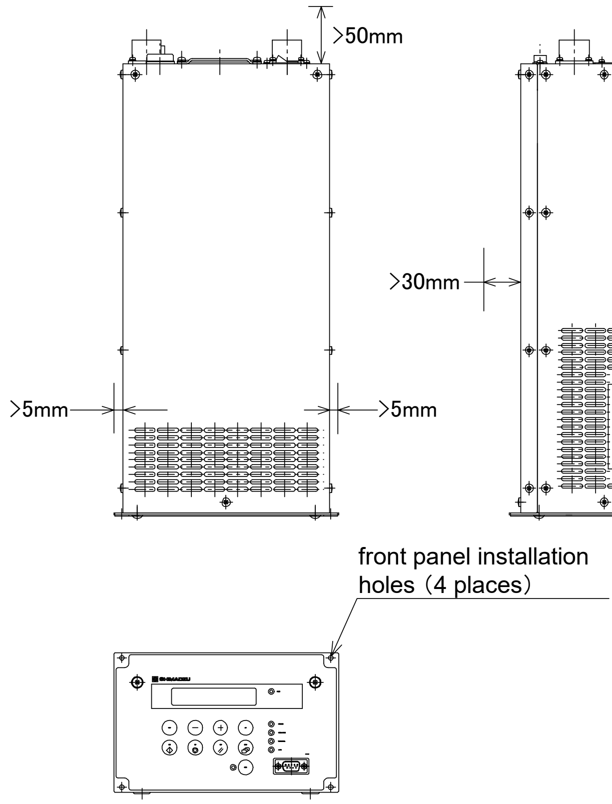


Fig. 5-1 Leave Enough Space Around the Power Supply Unit

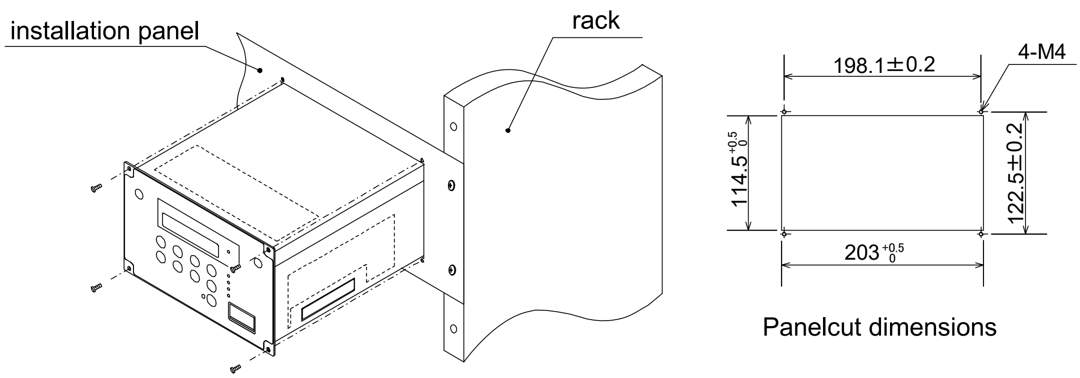


Fig. 5-2 Mount the Power Supply Unit

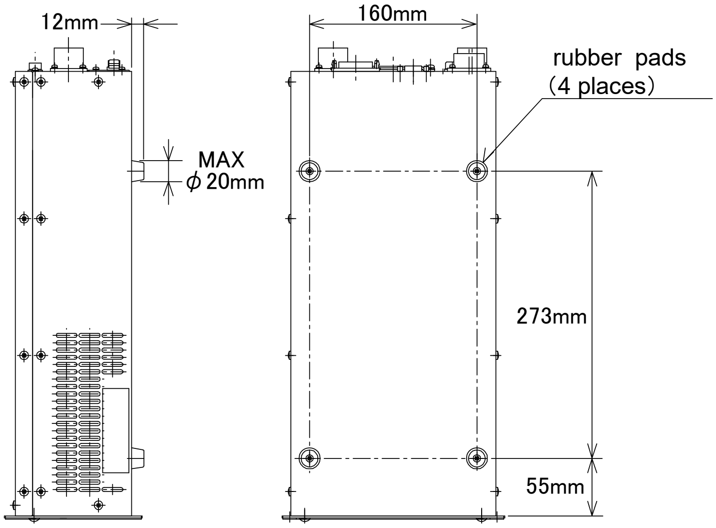
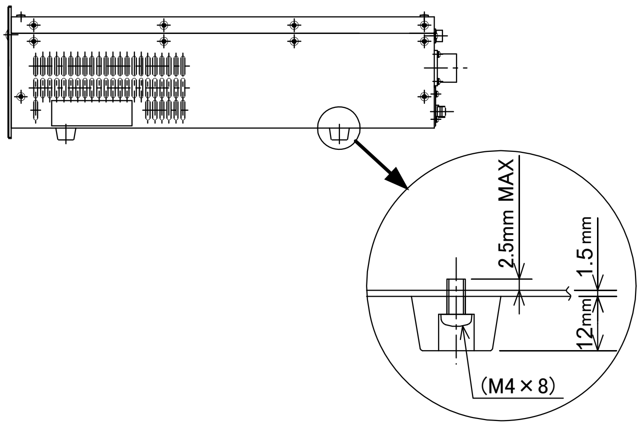


Fig. 5-3 Location of Rubber Pads



(Note 1) Use the prescribed screws to attach the rubber pads. Using the wrong screws can lead to damage or failure of the power supply unit.

FIg. 5-4 Attaching the Rubber Pads

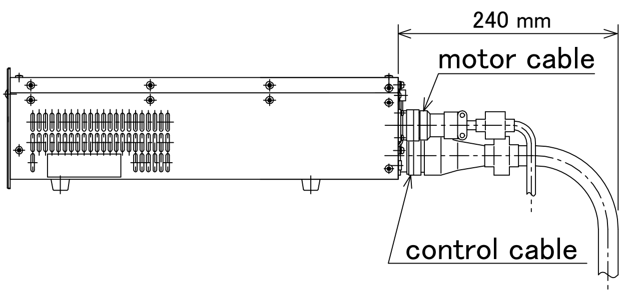


Fig. 5-5 Space to Connect the Cable

5.2 Connection of Power Cable

CAUTION

The power input voltage of the power supply unit EI-R04M is 200 to 240 VAC. Connect the power supply unit to the voltage specified on the side panel label only. Connection of the power supply unit to the incorrect input voltage can cause damage to the equipment. Supply the power via a circuit breaker (rating 15 A). Please provide PE (Protective Earth) connection to the terminal of a "PE" marked wire in final application. When power cable whose length is longer than 5 m is used to input AC power, protective earth line has to be connected to the PROTECTIVE EARTH TERMINAL (Fig. 2-2 (20)) which is on rear panel of the power supply unit with the cable which has shorter than 5 m and thicker than AWG14.

CAUTION

Electrical energy isolation (Lockout / Tagout) should be achieved by opening the main disconnect device or circuit breaker of host equipment, thereby removing power from this product. The main disconnect device or circuit breaker of host equipment should be suitably located and easily reached, and it should be marked as the disconnecting device for the equipment.

Check that the part number stuck on the cable is same as the part number shown in the power cable of Section 1.2.4 "Standard Accessories".

Connection ofer Cable:

(1) Connect the power cable terminal to the terminal board of user's power distribution board for equipment. The wire with [PE] mark is for earth use and other remaining two wires are for single phase AC power (Refer to Fig. 5-7, Table 5-1).
First, connect the wire [PE] mark. Next connect the other two wires.

(2) Turn off the POWER switch (Fig. 2-2 (17)) on the rear panel of the power supply unit. Or otherwise be sure to check that it is in off.

(3) Connect the power cable connectors to the power supply unit AC INPUT connector (Fig. 2-2 (16)).

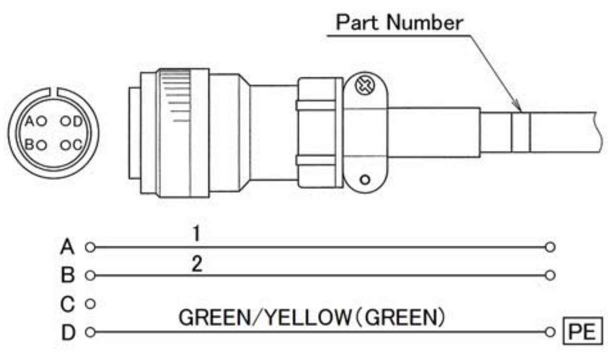


Fig. 5-6　Power Cable (Connector)

REFERENCE
For the specified power voltage, refer to the side panel of the power supply unit.

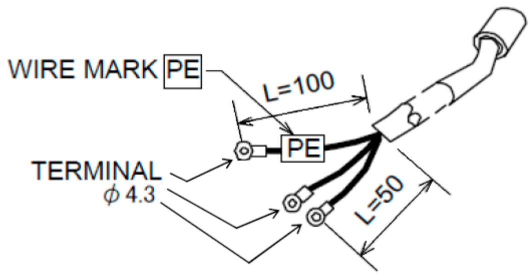


Fig. 5-7　Power Cable

Table 5-1　Power Cable CONNECTION

Wire color of power cable		Green/Yellow	Black	White
Location	EU	PE	N	L
	US	GND	L2	L1

5.3 Connection of the Pump to the Power Supply Unit

NOTICE

Insert straight the control cable connector after checking its key direction. Inserting it in oblique direction would cause damage of the connector pins. After the insertion, turn the cable connector clockwise until the rotation lock clicks.

NOTICE

Don't disconnect each cable while the pump is running. Particularly before disconnecting the control cable, check complete shutdown of the pump by ROTATION lamp goes out and, thereafter, turn off the POWER switch.

Control Cable:

Use the control cable that conformed CE marking. It has ferrite core at power supply side. Control cables which are identified with "262-75369A" are available for the use. But, If no ferrite core is fixed arround a control cable, it is not conformed CE marking.

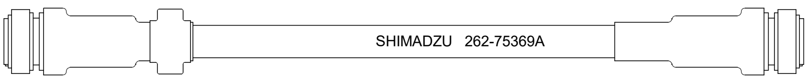


Fig. 5-8 Control Cable

Connecting Sequence (Refer to Fig. 2-2, Fig. 5-8 and Fig. 5-9):

(1) Turn off the POWER switch (Fig. 2-2 (17)) on the rear panel of the power supply unit. Or otherwise check that it is off.

(2) Connect the power supply unit to the MAG.BRG. connector (Fig. 2-2 (18)) and MOTOR connector (Fig. 2-2 (19)) of the pump proper with the control cable.

(3) For remote operation of the turbo molecular pump or intake of status signal, etc., connect the RS-485 connector (Fig. 2-2 (21)) or RS-232C connector (Fig. 2-1 (22)), Remote-control connector (Fig. 2-2 (23)). When using Remote-control connector, make wiring connection as instructed in Section 6.7 "Remote-control Connector".

All interfaces are SELV (Safety extra-low Voltage).

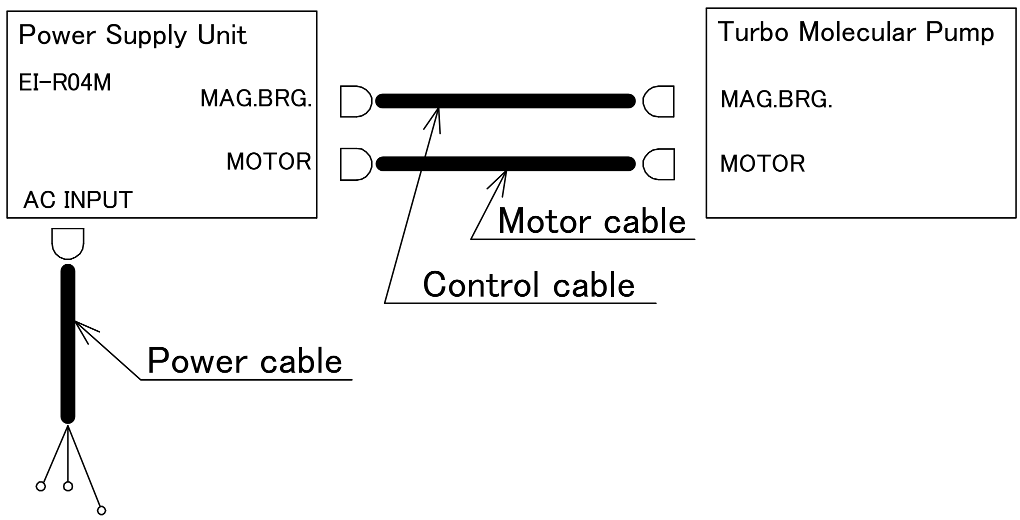


Fig. 5-9　Connection of Cables

OPERATION

6.1 Outline

CAUTION

Neither disconnect and reconnect each cable while the pump is running. Particularly for unplugging the control cable from the receptacle, check complete shutdown of the pump by ROTATION lamp goes out and, thereafter, turn off the POWER switch.

CAUTION

Do not turn the power off while the pump is running. The touch-down bearing may need to be replaced if the power is turned off repeatedly during operation. When the power is turned off while the pump is rotating, power from regenerative braking maintains the magnetic levitation until the rotational speed drops. Levitation then stops and the touch-down bearing supports the rotor. Consequently, repeated touch-down operations can reduce the life of the bearing.

6.1.1 Introduction

(Refer to Fig. 2-1 and Fig. 2-2)

The LCD (Fig. 2-1 (1)) displays the SHIMADZU EI-R04M when the power supply unit POWER switch (Fig. 2-2 (17)) is turned on.

It then displays "SELF CHECKING" and the power supply unit conducts self-diagnosis. If the result is good, the LCD changes into monitor mode (Refer to Section 6.6 "Software Operation" (1)), and the power supply is operatable. But if an alarm is delected, the LCD changes into alarm mode (Refer to Section 6.6 "Software Operation" (2)), and displays detected alarm.

6.1.2 Operation Flowchart

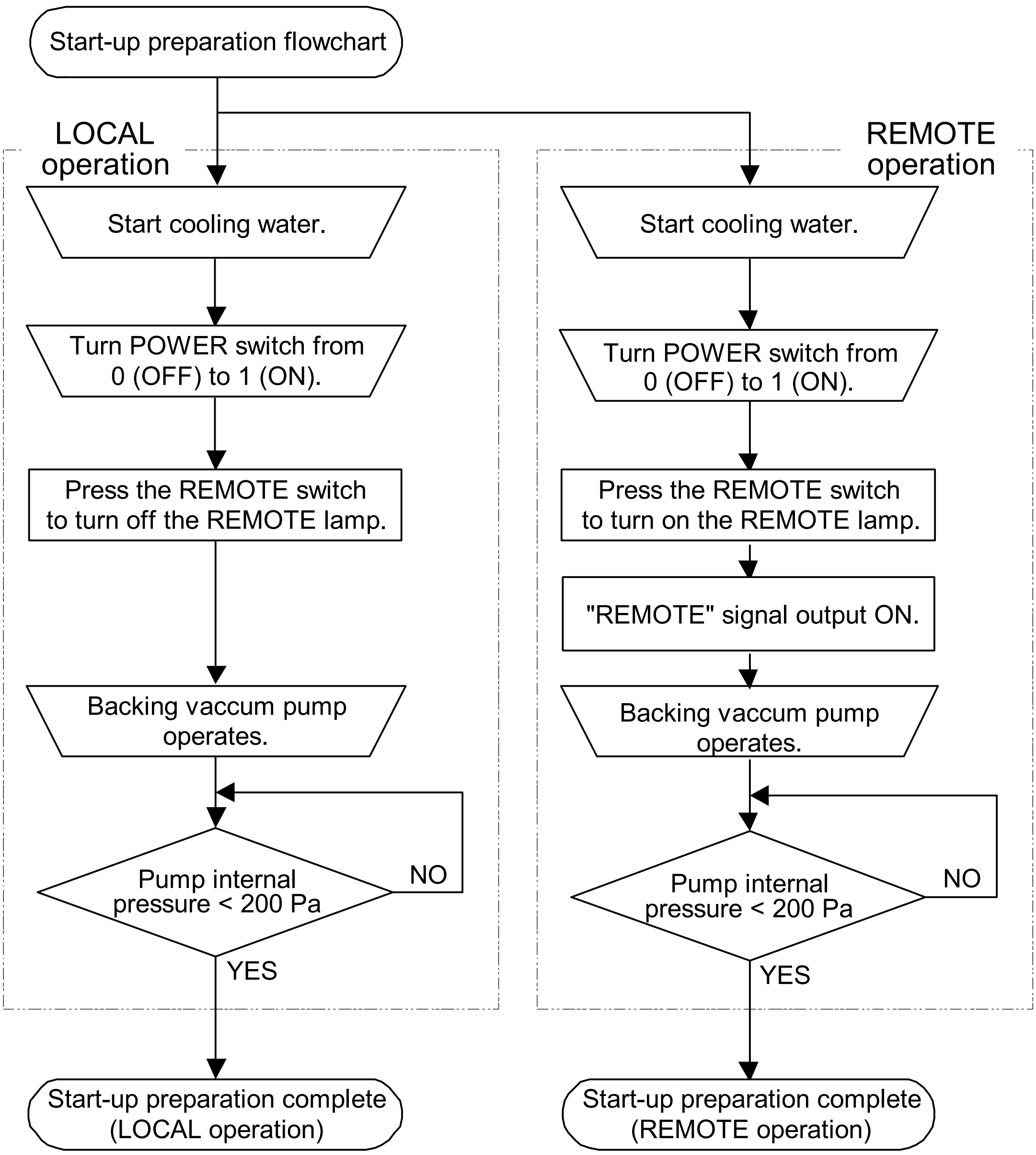


Fig. 6-1 Start-up Preparation Flowchart

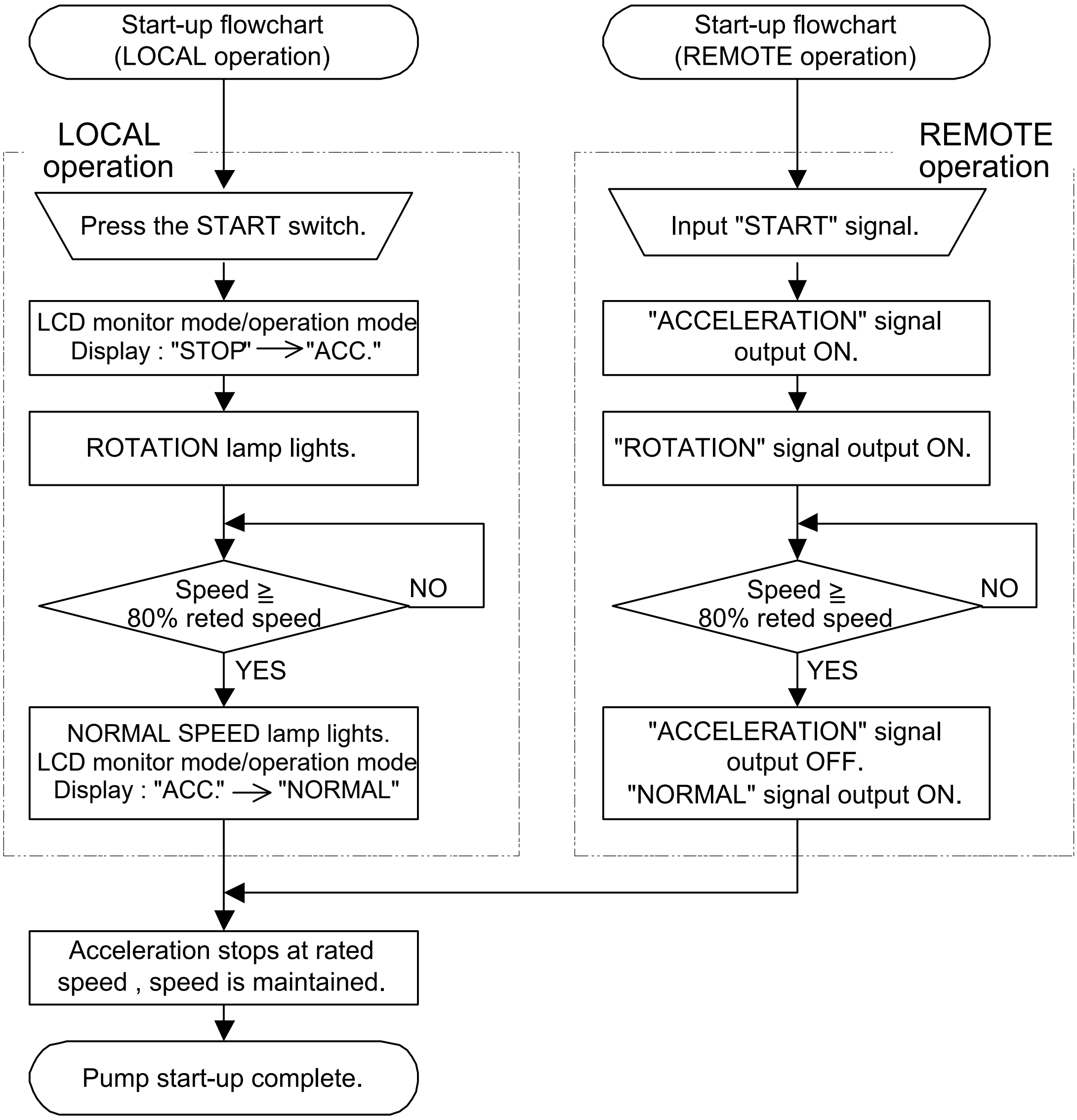


Fig. 6-2 Start-up Flowchart

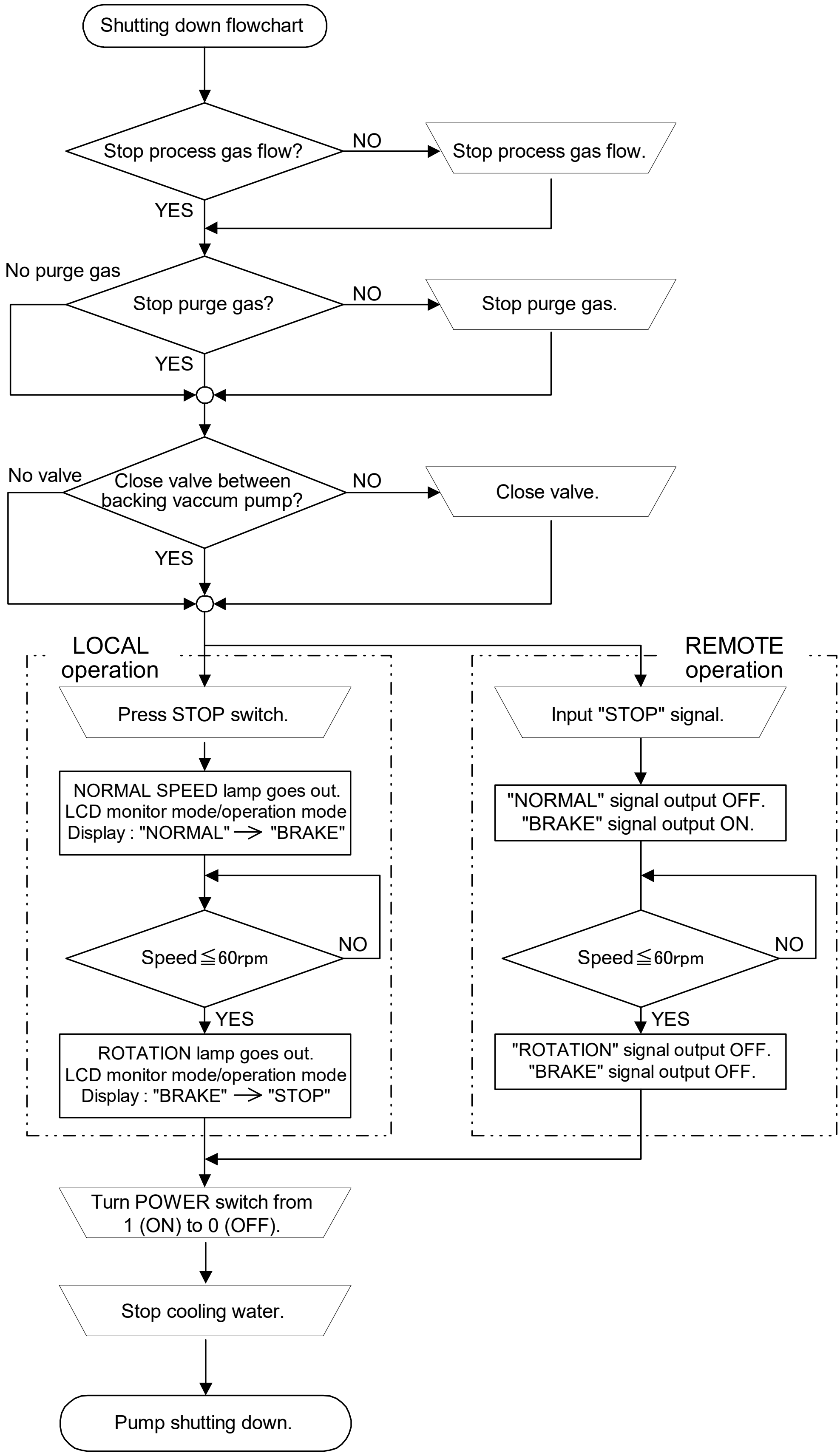
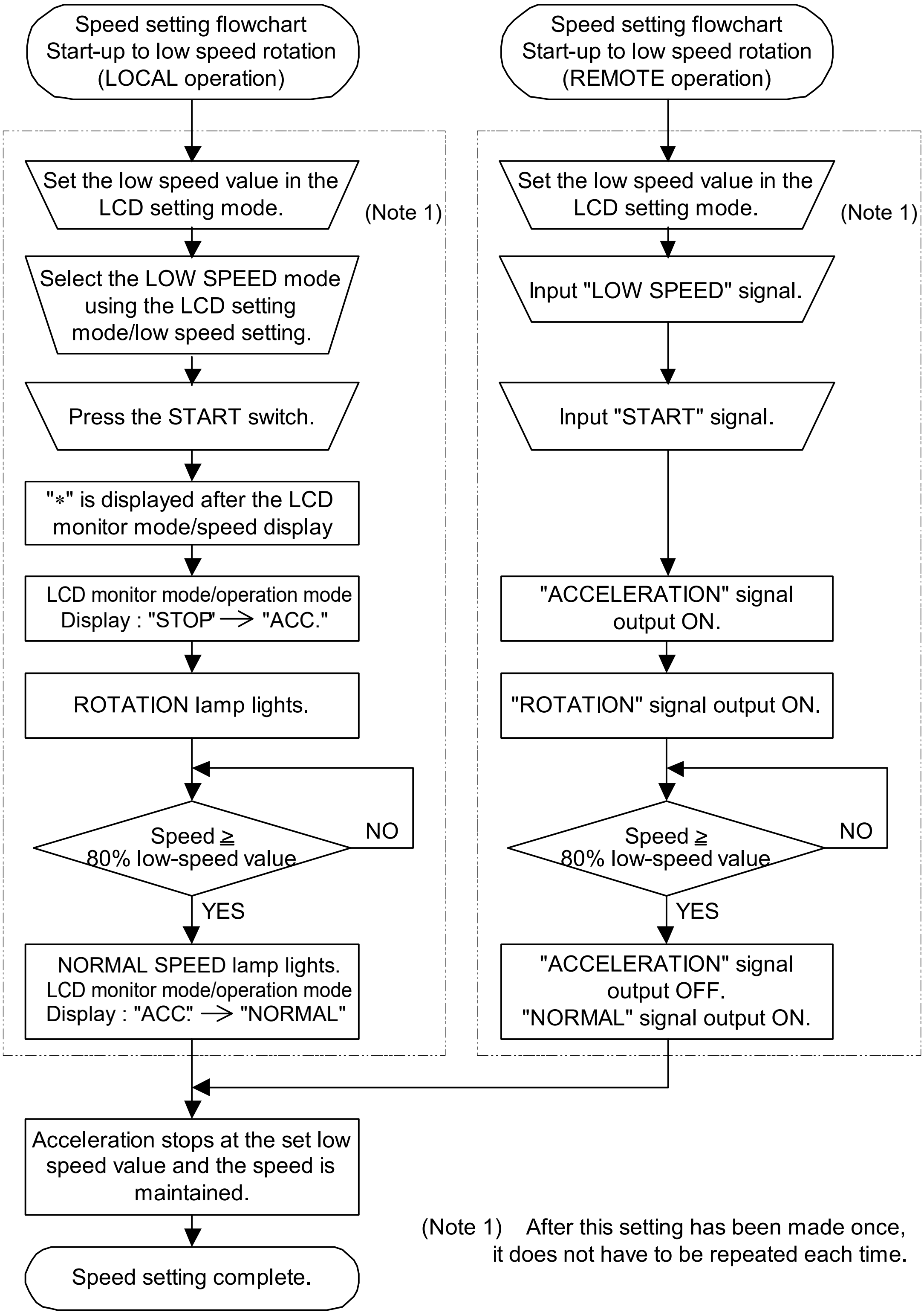


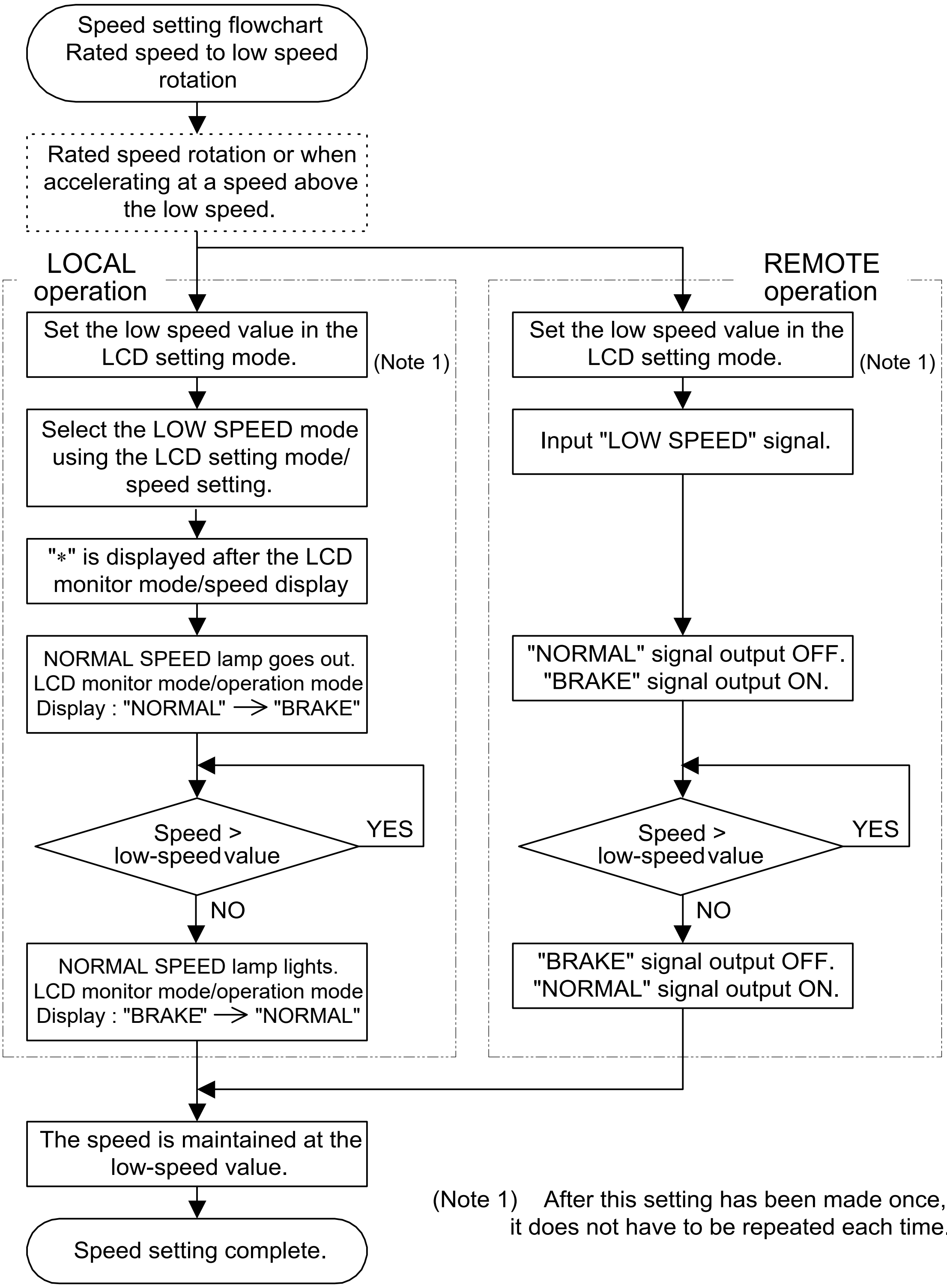
Fig. 6-3 Shutting Down Flowchart



(Note 1) After this setting has been made once, it does not have to be repeated each time.

Start-up to Low Speed Rotation

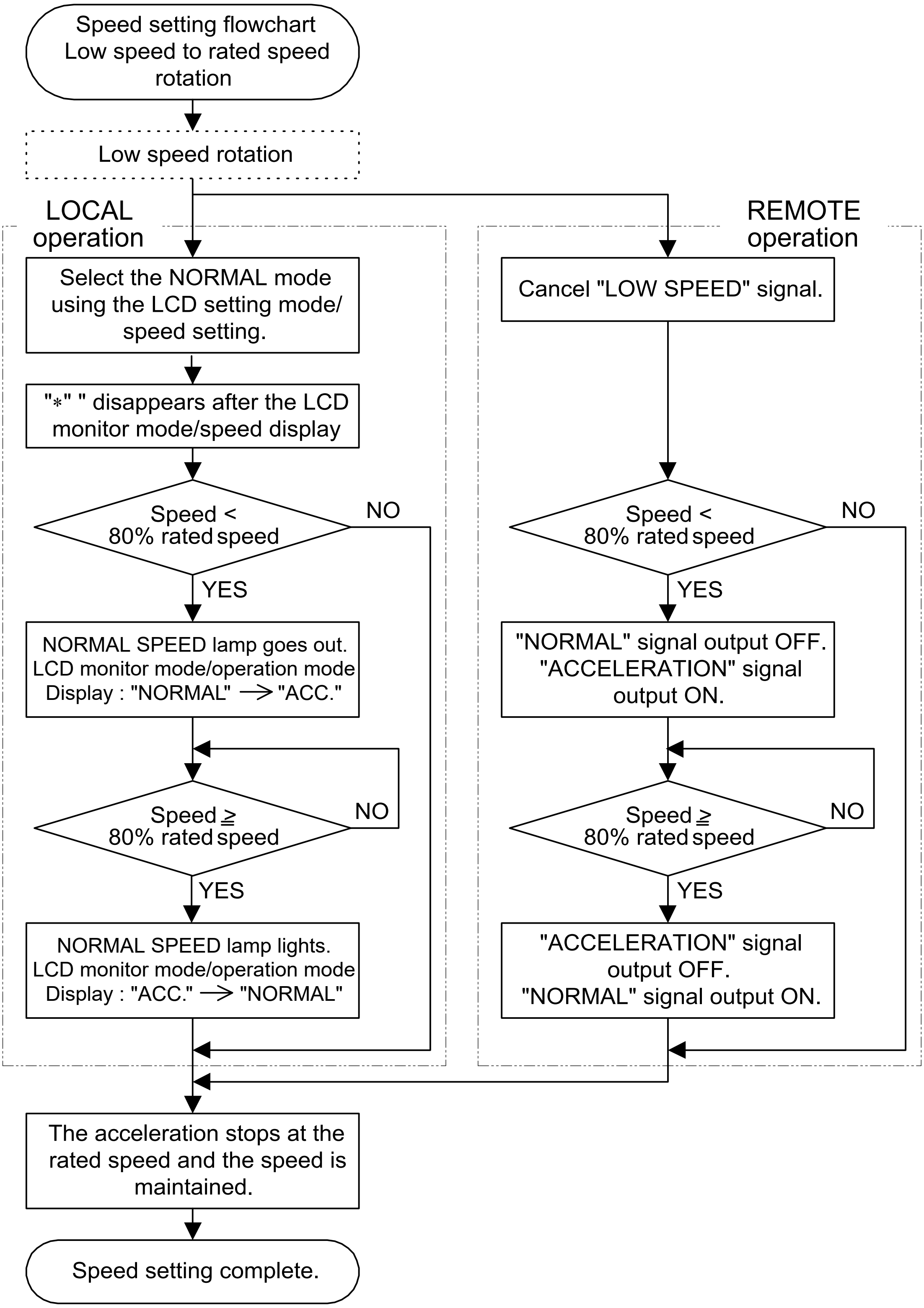
Fig. 6-4 Speed Setting Flowchart



(Note 1) After this setting has been made once, it does not have to be repeated each time.

Rated Speed to Low Speed Rotation

Fig. 6-5 Speed Setting Flowchart



Low Speed to Rated Speed Rotation

Fig. 6-6 Speed Setting Flowchart

6.2 Start-up Preparation

(Refer to Fig. 6-1)
Push the REMOTE switch (Fig. 2-1 (15)) to either LOCAL or REMOTE mode. REMOTE lamp OFF in LOCAL mode, and REMOTE lamp ON in REMOTE mode.

LOCAL (REMOTE lamp OFF)	The pump can be started/stopped by pushing START/STOP switch on the front panel of the power supply unit.
REMOTE (REMOTE lamp ON)	the pump runs only according to input signal from the remote-control connector (Fig. 2-2 (23)) or serial interface connector (Fig. 2-2 (21), Fig. 2-1 (22)).

Refer to APPENDIX-A "COMMUNICATIONS" for serial interface connector.

6.2.1 Start-up Preparation Sequence in LOCAL Mode

(1) Feed the cooling water into the cooling line.
(2) Turn on the POWER switch (Fig. 2-2 (17)) of the power supply unit and check if the POWER lamp (Fig. 2-1 (10)) lights. And the rotor of the turbo molecular pump is levitated by the magnetic bearing.
(3) Evacuate the turbo molecular pump by using a backing vacuum pump.
(4) Start-up preparation is complete if the pressure in the turbo molecular pump reduces below 200 Pa.

6.2.2 Start-up Preparation Sequence in REMOTE Mode

(1) Feed the cooling water into the cooling line.
(2) Turn on the POWER switch (Fig. 2-2 (17)) of the power supply unit and check that "REMOTE" signal (Refer to Table 6-3) from the remote-control connector (Fig. 2-2 (23)) is ON. Under this condition, the rotor of the turbo molecular pump is levitated by the magnetic bearing.
(3) Evacuate the turbo molecular pump by using a backing vacuum pump.
(4) Start-up preparation is complete if the pressure in the turbo molecular pump reduces below 200 Pa.

NOTICE

When turning the POWER switch for the power supply unit on or off, a "clunk" sound may be heard from inside the pump. This sound is from the rotor inside the pump being levitated or de-levitated. This is normal.

6.3 Start-up

(Refer to Fig. 6-2)

6.3.1 Start-up Sequence in LOCAL Mode

(1) Start-up begins when the Section 6.2.1 "Start-up Preparation Sequence in LOCAL Mode" is complete.
(2) Push the START switch (Fig. 2-1 (6)).
(3) "ACC." is displayed on the pump monitor mode / operation mode LCD and pump acceleration starts. The ROTATION lamp (Fig. 2-1 (11)) lights.
(4) When the rotational speed reaches 80% rated value, the NORMAL SPEED lamp (Fig. 2-1 (12)) lights and the pump monitor mode / operation mode LCD changes from "ACC." to "NORMAL".
    Pump start-up is complete.

6.3.2 Start-up Sequence in REMOTE Mode

(1) Start-up begins when the Section 6.2.2 "Start-up Preparation Sequence in REMOTE Mode" is complete.
(2) "START" signal (Refer to Table 6-3) is input from the remote-control connector (Fig. 2-2 (23)).
(3) The pump acceleration starts when the remote-control connector "ACCELERATION" signal (Refer to Table 6-3) turns on. The "ROTATION" signal (Refer to Table 6-3) turns on.
(4) When the rotational speed reaches 80 % rated value, the remote-control connector "ACCELERATION" signal (Refer to Table 6-3) turns off and the "NORMAL" signal (Refer to Table 6-3) turns on.
    Pump start-up is complete.

6.4 Shutting Down

CAUTION

After having operated the turbo molecular pump for evacuation of corrosive gas, keep the pump internal as vacuumed even after shutdown. Inflow of water content in the air to the pump internal would cause rapid corrosion trouble of the pump internals. The pump corrosion may result in damaging the vacuum vessel interior and other units, causing pressure fluctuation by stopping the pump and dispersal of parts.

CAUTION

When increasing internal pressure of the turbo molecular pump up to around the atmospheric pressure by use of inert gas, etc., adjust the pressure reducing valve so that the internal pressure of the same pump does not exceed 20 kPa [GAUGE].

CAUTION

Electrical energy isolation (Lockout / Tagout) should be achieved by opening the main disconnect device or circuit breaker of host equipment, thereby removing power from this product. The main disconnect device or circuit breaker of host equipment should be suitably located and easily reached, and it should be marked as the disconnecting device for the equipment.

For shut-down of the turbo molecular pump, follow the sequence below (Refer to Fig. 6-3).

6.4.1 Preparations Prior to Shutting Down Operation

(1) Check that process gas inflow is in complete stop. When main valve is provided between the turbo molecular pump and vacuum chamber, close the valve, too.

(2) When purge gas is being fed into the turbo molecular pump, stop the gas feed, too.

(3) When forevacuum valve is provided between the turbo molecular pump and backing vacuum pump, close the valve, too.

6.4.2 Shutting Down Sequence in LOCAL Mode

(1) Push the STOP switch (Fig. 2-1 (7)) and check that the indication on the pump monitor mode / operation mode LCD switches from "NORMAL" to "BRAKE".
Then the NORMAL SPEED lamp (Fig. 2-1 (12)) goes off.
(2) Wait until the ROTATION lamp (Fig. 2-1 (11)) goes out. The pump monitor mode / operation mode LCD switches from "BRAKE" to "STOP".
(3) Turn off the power supply unit POWER switch (Fig. 2-1 (17)).
(4) Stop the cooling water flow.
(5) If you want to isolate this product from electrical supply, unplug AC INPUT connector.

6.4.3 Shutting Down Sequence in REMOTE Mode

(1) Input the "STOP" signal (Refer to Table 6-3) from the remote-control connector (Fig. 2-2 (23)) and check that the "BRAKE" signal (Refer to Table 6-3) is ON.
(2) Wait until the "ROTATION" signal (Refer to Table 6-3) turns off. At this time, the "BRAKE" signal (Refer to Table 6-3) also turns off.
(3) Turn off the power supply unit POWER switch (Fig. 2-2 (17)).
(4) Stop the cooling water flow.
(5) If you want to isolate this product from electrical supply, unplug AC INPUT connector.

When the turbo molecular pump is turned off after pumping a corrosive gas, maintain a vacuum inside the turbo molecular pump or purge the interior of the pump with an inert gas.

Further, in such a case when a hydraulic rotary vacuum pump is used as backing vacuum pump and there is possible reverse flow and diffusion of oil from the backing vacuum pump, return the pump internal pressure to atmospheric pressure using dry nitrogen gas, after complete shut-down of the pump [ROTATION lamp (Fig. 2-1 (11)) goes out], to prevent the turbo molecular pump from being contaminated with oil vapor.

For shutting down the turbo molecular pump in running at high speed with infeed of dry nitrogen gas to the pump, keep the nitrogen gas flow rate at 1500 mL/min maximum.

REFERENCE

The ROTATION lamp (Fig. 2-1 (11)) goes out or "ROTATION" signal (Refer to Table 6-3) turns off when the pump rotational speed is 60 rpm or less. Turning off the POWER switch (Fig. 2-2 (17)) permits the pump rotor to be supported by the touch-down bearings.

6.5 Variable Speed Operation

CAUTION

When using the variable speed function to change the pump rotation rate, use a rotation rate that does not cause resonance with other devices installed at the site.

CAUTION

If an EI-R04M power supply unit is used in combination with an existing pump that was operated in combination with a power supply unit not having the variable speed function, the variable speed function cannot be used.

If the power supply unit is to be combined with an existing pump, modification and operational inspections are necessary. Please contact Shimadzu for detailed information.

6.5.1 Outline

(1) The rotational speed settings function sets the rotational speed by selecting between the NORMAL speed mode or LOW SPEED mode.

(2) Select the NORMAL mode or LOW SPEED mode by LOCAL operation using the setting mode / rotational speed settings / speed setting set value on the LCD display or by REMOTE operation using the remote-control connector "LOW SPEED" signal input (Refer to Table 6-3).

(3) The NORMAL mode or LOW SPEED mode selection can be made before or after start-up (Refer to the Speed Setting Flowcharts in Fig. 6-4, Fig. 6-5, and Fig. 6-6).

(4) Set the low speed value between 25 % and 100 % of the rated speed in 0.1 % increment with the setting mode / rotational speed settings / low speed setting set value on the LCD display.

(5) The low speed value can be set while the pump is rotating in the LOW SPEED mode. The pump then accelerates or decelerates to the new set value and maintains the set speed.

(6) The time required for the speed to change is the same as the time for normal acceleration or deceleration.
For example, if the low speed value is set to 80 % and the LOW SPEED mode is selected during normal rotation, the time for the speed to drop to 80 % is approximately one-fifth of the time required to stop from rated speed only as a guide.

6.5.2 Operation from Start-up to Low Speed Rotation

This is the procedure until low-speed rotation is achieved when the speed setting is made with the pump stopped (Refer to Fig. 6-4).

LOCAL Operation

(1) Start-up begins when the Section 6.2.1 "Start-up Preparation Sequence in LOCAL Mode" is complete.

(2) Set the low speed value with the setting mode / rotational speed settings / low speed setting set value on the LCD display (Refer to Section 6.6 "Software Operation" (4)). After this setting has been made once, it does not have to be repeated each time.

(3) Select the LOW SPEED mode using the setting mode / rotational speed settings / low speed setting set value (Refer to Section 6.6 "Software Operation" (4)).

(4) Push the START switch (Fig. 2-1 (6)).

(5) While the speed is changing, "*" is displayed after the monitor mode / speed display (Refer to Section 6.6 "Software Operation" (1)).

(6) ACC is displayed on the pump monitor mode / operation mode LCD and the pump starts to accelerate. After a few seconds the ROTATION lamp (Fig. 2-1 (11)) lights.

(7) When the rotational speed reaches 80 % of low-speed value, the NORMAL SPEED lamp lights (Fig. 2-1 (12)) and the pump monitor mode / operation mode LCD switches from "ACC." to "NORMAL".

(8) When the pump speed reaches the set low speed value, acceleration stops and the pump speed is maintained.

(Note) The same operation occurs if the LOW SPEED mode is selected after the START switch is pushed but before the pump speed reaches the set low speed value.

REMOTE Operation

(1) Start-up begins when the Section 6.2.2 "Start-up Preparation Sequence in REMOTE Mode" is complete.

(2) Set the low speed value with the setting mode / rotational speed settings / low speed setting set value on the LCD display (Refer to Section 6.6 "Software Operation" (4)). After this setting has been made once, it does not have to be repeated each time.

(3) Input the "LOW SPEED" signal (Refer to Table 6-3) from the remote-control connector (Fig. 2-2 (23)).

(4) Input the "START" signal from the remote-control connector (Refer to Table 6-3).

(5) The pump starts to accelerate when the "ACCELERATION" signal (Refer to Table 6-3) from the remote-control connector (Fig. 2-2 (23)) turns on. After a few seconds the "ROTATION" signal turns on.

(6) When the rotational speed reaches 80 % of low speed value, the remote-control connector "ACCELERATION" signal turns off and the "NORMAL" signal turns on.

(7) When the pump speed reaches the set low speed value, acceleration stops and the pump speed is maintained.

(Note) The same operation occurs if the "LOW SPEED" signal is input after the "START" signal is input but before the pump speed reaches the set low speed value.

6.5.3 Operation from Rated Speed Rotation to Low Speed Rotation

This is the procedure to select low speed operation during rated speed rotation or when accelerating at a speed above the low speed (Refer to Fig. 6-5).

LOCAL Operation

(1) Set the low speed value with the setting mode / rotational speed settings / low speed setting set value on the LCD display (Refer to Section 6.6 "Software Operation" (4)). After this setting has been made once, it does not have to be repeated each time.

(2) Next, select the LOW SPEED mode using the setting mode / rotational speed settings / speed setting set value on the LCD display (Refer to Section 6.6 "Software Operation" (4)).

(3) While the speed is changing, "*" is displayed after the monitor mode / speed display (Refer to Section 6.6 "Software Operation" (1)).

(4) The pump monitor mode / operation mode LCD switches from "NORMAL" to "BRAKE" and the pump starts to decelerate. Then the NORMAL SPEED lamp (Fig. 2-1 (12)) goes off.

(5) When the rotational speed reaches the set low speed value, the NORMAL SPEED lamp lights and the pump monitor mode / operation mode LCD switches from "BRAKE" to "NORMAL".

(6) The pump stops decelerating and the pump speed is maintained.

(Note) A normal start-up and normal operation occurs if the mode is reverted to NORMAL speed before the pump speed reaches the set low speed value.

REMOTE Operation

(1) Set the low speed value with the setting mode / rotational speed settings / low speed setting set value on the LCD display (Refer to Section 6.6 "Software Operation" (4)). After this setting has been made once, it does not have to be repeated each time.

(2) Input the "LOW SPEED" signal (Refer to Table 6-3) from the remote-control connector (Fig. 2-2 (23)).

(3) The pump starts to decelerate when the remote-control connector "NORMAL" signal (Refer to Table 6-3) turns off and the "BRAKE" signal (Refer to Table 6-3) turns on.

(4) When the pump speed reaches the set low speed value, the remote-control connector "BRAKE" signal turns off and the "NORMAL" signal turns on.

(5) The pump stops decelerating and the pump speed is maintained.

(Note) A normal start-up and normal operation occurs if the "LOW SPEED" signal is cancelled before the pump speed reaches the set low speed value.

6.5.4 Operation from Low Speed Rotation to Rated Speed Rotation

This is the procedure to select normal speed operation during low speed rotation.
(Refer to Fig. 6-6)

LOCAL Operation

(1) Select the NORMAL mode using the setting mode / rotational speed settings / speed setting set value on the LCD display (Refer to Section 6.6 "Software Operation" (4)).

(2) The "*" disappears after the monitor mode / speed display (Refer to Section 6.6 "Software Operation" (1)).

(3) If the set low speed value did not exceed 80 % rated speed, the pump monitor mode / operation mode LCD switches from "NORMAL" to "ACC." and the pump starts to accelerate. The NORMAL SPEED lamp (Fig. 2-1 (12)) goes off.

(4) When the rotational speed reaches 80 % rated speed, the NORMAL SPEED lamp lights and the pump monitor mode / operation mode LCD switches from "ACC." to "NORMAL".

(5) If the set low speed value exceed 80 % rated speed, the LCD display remains unchanged and the pump accelerates.

(6) When the rated speed is reached, the pump stops accelerating and the pump speed is maintained.

REMOTE Operation

(1) Cancel the "LOW SPEED" signal (Refer to Table 6-3) inputted in the remote-control connector (Fig. 2-2 (23)).

(2) If the set low speed value did not exceed 80 % rated speed, the remote-control connector "ACCELERATION" signal turns on and (Refer to Table 6-3) and the pump starts to accelerate. The "NORMAL" signal turns off.

(3) When the rotational speed reaches 80 % rated speed, the remote-control connector "ACCELERATION" signal turns off and the "NORMAL" signal turns on (Refer to Table 6-3).

(4) If the set low speed value exceed 80 % rated speed, remote-control signals remain unchanged and the pump accelerates.

(5) When the rated speed is reached, the pump stops accelerating and the pump speed is maintained.

6.6 Software Operation

The software operation functions are listed in Table 6-1.

Table 6-1 Software Operation Functions

Reference flowchart		Function	Description
(1) Monitor mode		Operation mode User memo Operation status Motor speed Motor current Magnetic bearing displacement monitor Magnetic bearing unbalance monitor (Note 1)	Monitors pump operation status
(2) Alarm mode		Alarm and warning display	Displays details of alarms and warnings.
		Alarm and warning history and details	Displays the alarm and warning history.
Menu mode	(3) Timer	Run time Maintenance call time Number of power failure touch-downs Number of high-speed touch-downs Number of magnetic bearing warnings	Resets times and counters.
	(4) Setting	User memo setting	Inputs user memo.
		RS-232C setting	Sets the communication environment.
		RS-485 setting	Sets the communication environment.
		Speed setting	Sets the pump speed display format.
		Power failure detect setting	Sets the power failure detect.
		Remote-control signal setting	Sets the operations of the remote-control signals.
		Warning output setting	External output setting for warnings
		Default setting	Reverts to the default settings.
	(5) Self test	LCD contrast adjustment	Displays the LCD contrast adjustment.
		Self diagnosis	Self diagnosis of the magnetic bearing sensor.
		Corresponding model	Displays thecorresponding model.

(Note 1) When detected alarm below, the monitor function do not operate.
Alarm code: 46, 47(rotational speed error), 51 to 55 (excessive vibration of the magnetic bearing), 66 (magnetic bearing control error) (Refer to Table 7-6).

A flowchart of the entire LCD display is shown below.
SELECT , + , − , SET in the flowchart indicate keys on the front panel of power supply unit, and other ☐ indicated contents of the LCD display.

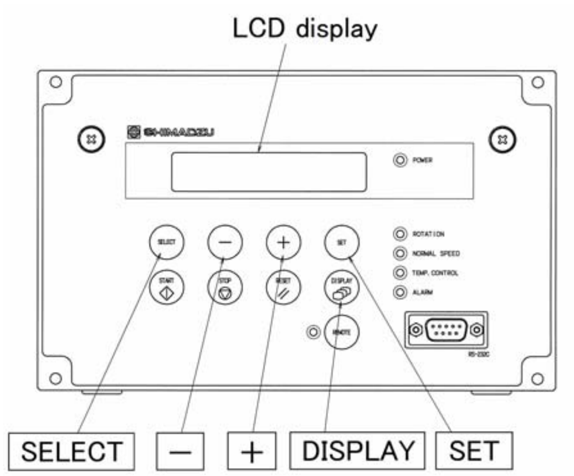
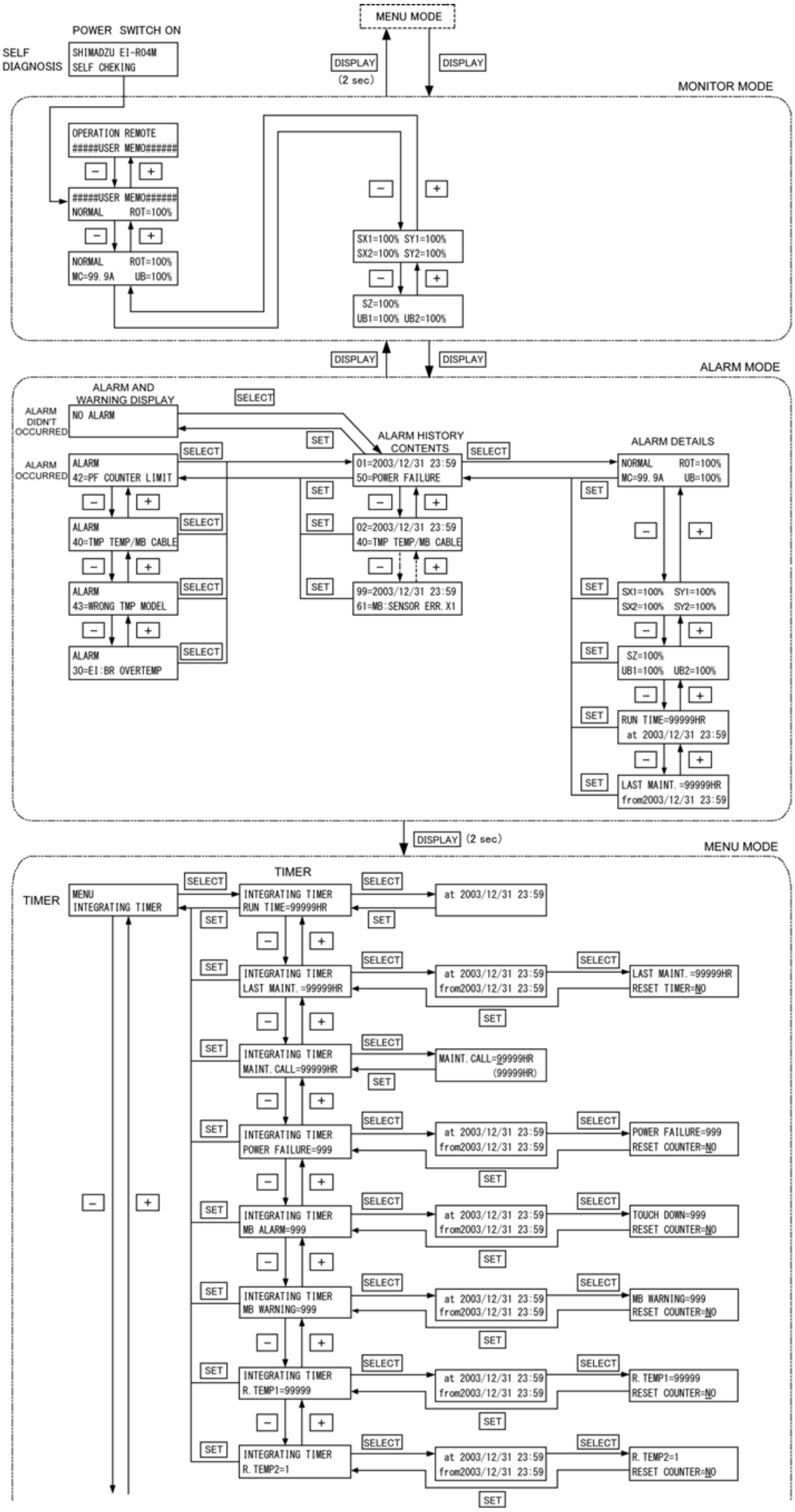


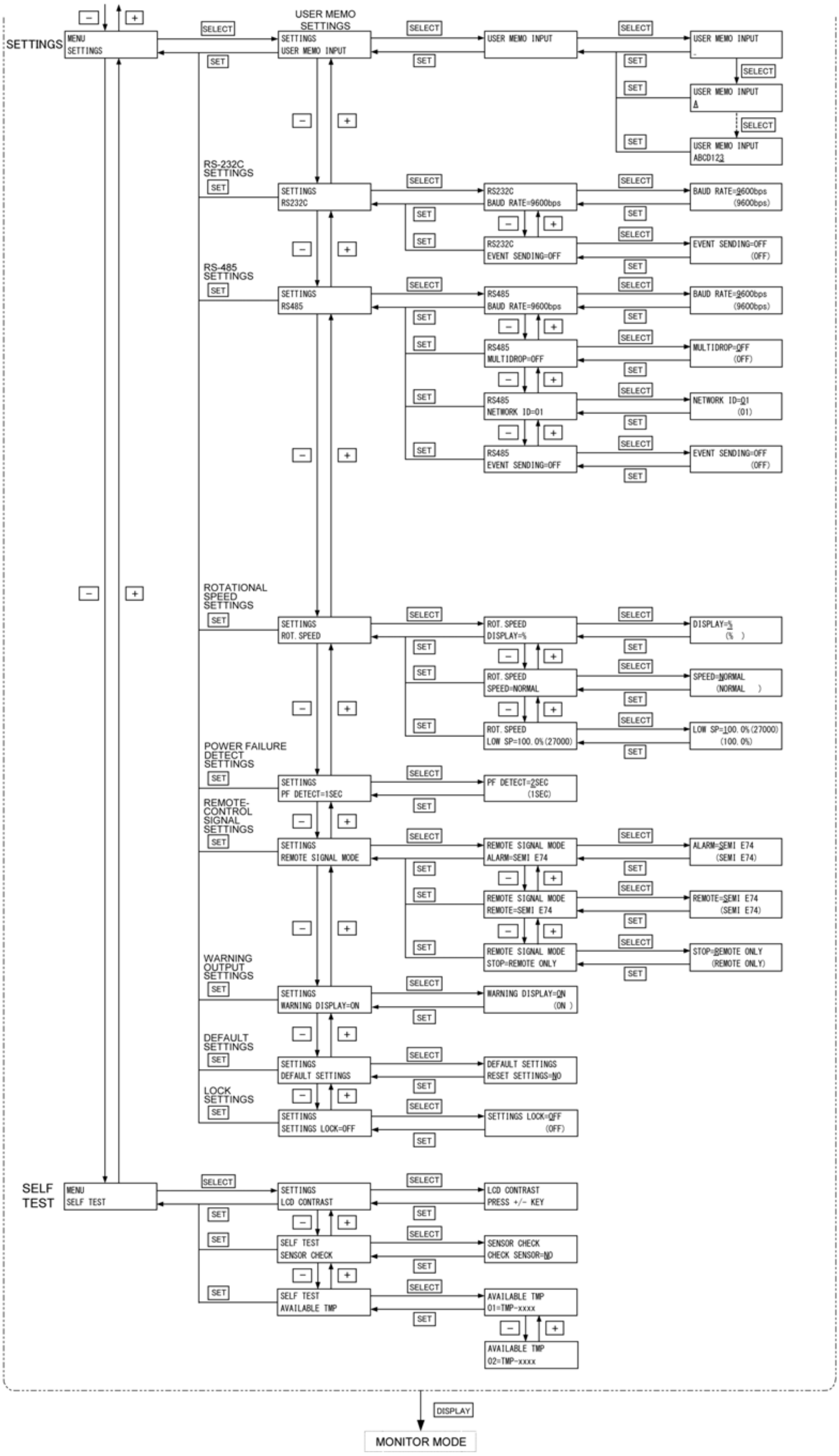
Fig. 6-7 Power Supply Unit Front Panel



Detail
→ P. 43

Detail
→ P. 45

Detail
→ P. 46



SETTINGS
MENU SETTINGS
USER MEMO SETTINGS
SETTINGS USER MEMO INPUT
USER MEMO INPUT
USER MEMO INPUT _
USER MEMO INPUT A
USER MEMO INPUT ABCD123
RS-232C SETTINGS
SETTINGS RS232C
RS232C BAUD RATE=9600bps
BAUD RATE=9600bps (9600bps)
RS232C EVENT SENDING=OFF
EVENT SENDING=OFF (OFF)
RS-485 SETTINGS
SETTINGS RS485
RS485 BAUD RATE=9600bps
BAUD RATE=9600bps (9600bps)
RS485 MULTIDROP=OFF
MULTIDROP=OFF (OFF)
RS485 NETWORK ID=01
NETWORK ID=01 (01)
RS485 EVENT SENDING=OFF
EVENT SENDING=OFF (OFF)
ROTATIONAL SPEED SETTINGS
SETTINGS ROT. SPEED
ROT. SPEED DISPLAY=%
DISPLAY=% (%)
ROT. SPEED SPEED=NORMAL
SPEED=NORMAL (NORMAL)
ROT. SPEED LOW SP=100. 0% (27000)
LOW SP=100. 0% (27000) (100. 0%)
POWER FAILURE DETECT SETTINGS
SETTINGS PF DETECT=1SEC
PF DETECT=2SEC (1SEC)
REMOTE-CONTROL SIGNAL SETTINGS
SETTINGS REMOTE SIGNAL MODE
REMOTE SIGNAL MODE ALARM=SEMI E74
ALARM=SEMI E74 (SEMI E74)
REMOTE SIGNAL MODE REMOTE=SEMI E74
REMOTE=SEMI E74 (SEMI E74)
REMOTE SIGNAL MODE STOP=REMOTE ONLY
STOP=REMOTE ONLY (REMOTE ONLY)
WARNING OUTPUT SETTINGS
SETTINGS WARNING DISPLAY=ON
WARNING DISPLAY=ON (ON)
DEFAULT SETTINGS
SETTINGS DEFAULT SETTINGS
DEFAULT SETTINGS RESET SETTINGS=NO
LOCK SETTINGS
SETTINGS SETTINGS LOCK=OFF
SETTINGS LOCK=OFF (OFF)
SELF TEST
MENU SELF TEST
SETTINGS LCD CONTRAST
LCD CONTRAST PRESS +/- KEY
SELF TEST SENSOR CHECK
SENSOR CHECK CHECK SENSOR=NO
SELF TEST AVAILABLE TMP
AVAILABLE TMP 01=TMP-xxxx
AVAILABLE TMP 02=TMP-xxxx
SELECT
SET
DISPLAY
MONITOR MODE

In the next tables, they are shown detailed flowchart of LCD display.

(1) MONITOR MODE

In monitor mode operation status of pump can be identified. If key DISPLAY is pushed in menu mode, the LCD changes into monitor mode. The LCD changes automatically into monitor mode after start or stop operation.

Mode	Operation and LCD display	Description of display
MONITOR 1	OPERATION REMOTE #####USER MEMO######	Operation Mode (Note 1) USER MEMO (Note 3)
MONITOR 2	#####USER MEMO###### NORMAL ROT=100%	USER MEMO (Note 3) Pump Operation Mode (Note 2) Motor Rotational Speed (Note 4)
MONITOR 3	NORMAL ROT=100% MC=99.9A UB=100%	Pump Operation Mode (Note 2) Motor Rotational Speed (Note 4) Motor Current Unbalance Monitor Value
MONITOR 4	SX1=100% SY1=100% SX2=100% SY2=100%	Magnetic Bearing Sensor Input Value X1,Y2,X2,Y2 Axis
MONITOR 5	SZ=100% UB1=100% UB2=100%	Magnetic Bearing Sensor Input Value Z Axis Unbalance Monitor Value 1,2 Axis

First, Monitor 2 display following to initial display after power switch turning on.

(Note 1) Operation Mode

LCD display	Operation
LOCAL	Control by a manual switch operation on the front panel
REMOTE	Control by a remote-control signal operation
RS-232C	Control by a RS-232C communication
RS-485	Control by a RS-485 communication or optional communication.

Refer to Section 6.2 "Start-up Preparation" for details on changing the operation mode.

(Note 2) Pump Operation Mode

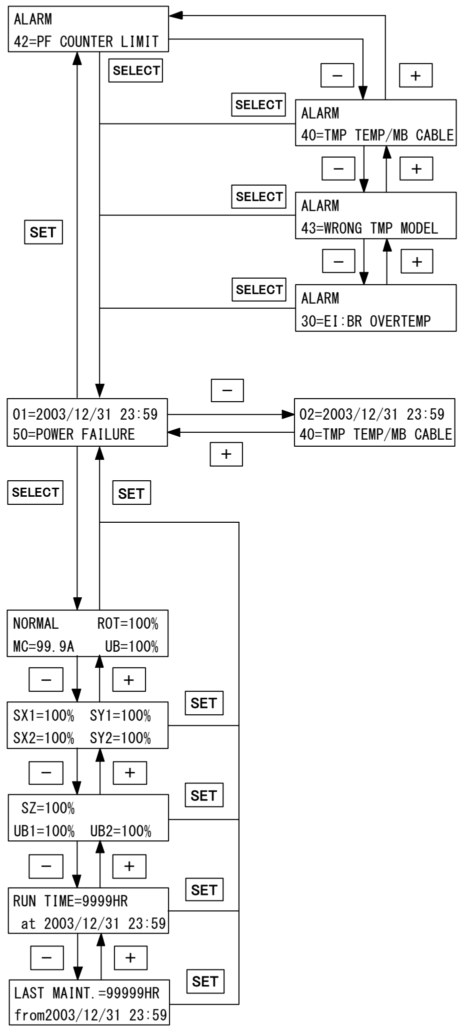
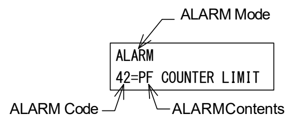
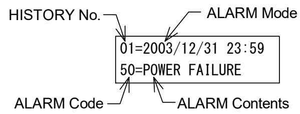
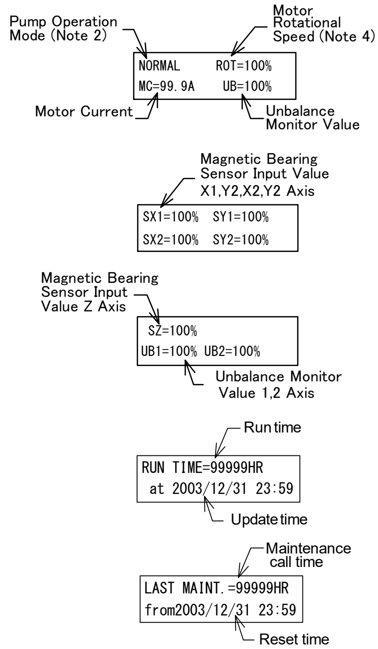
LCD display	Pump operation
NORMAL	Normal rotation
ACC.	Motor acceleration
BRAKE	Motor brake deceleration
STOP	Motor stop
E-STOP	Error occurs (stop)
E-BRAKE	Error occurs (motor deceleration)
E-IDLE	Error occurs (free operation = motor stop)

(Note 3) Any character can be entered in the USER MEMO from the menu mode "SETTINGS / USER MEMO INPUT". Use it for displaying the chambers connected to the pump etc.

(Note 4) The Motor Rotational Speed display can be selected from %, rpm and rps in the menu mode "SETTINGS / ROT.SPEED / DISPLAY".
An asterisk (∗) is displayed after the speed display during variable speed operation.

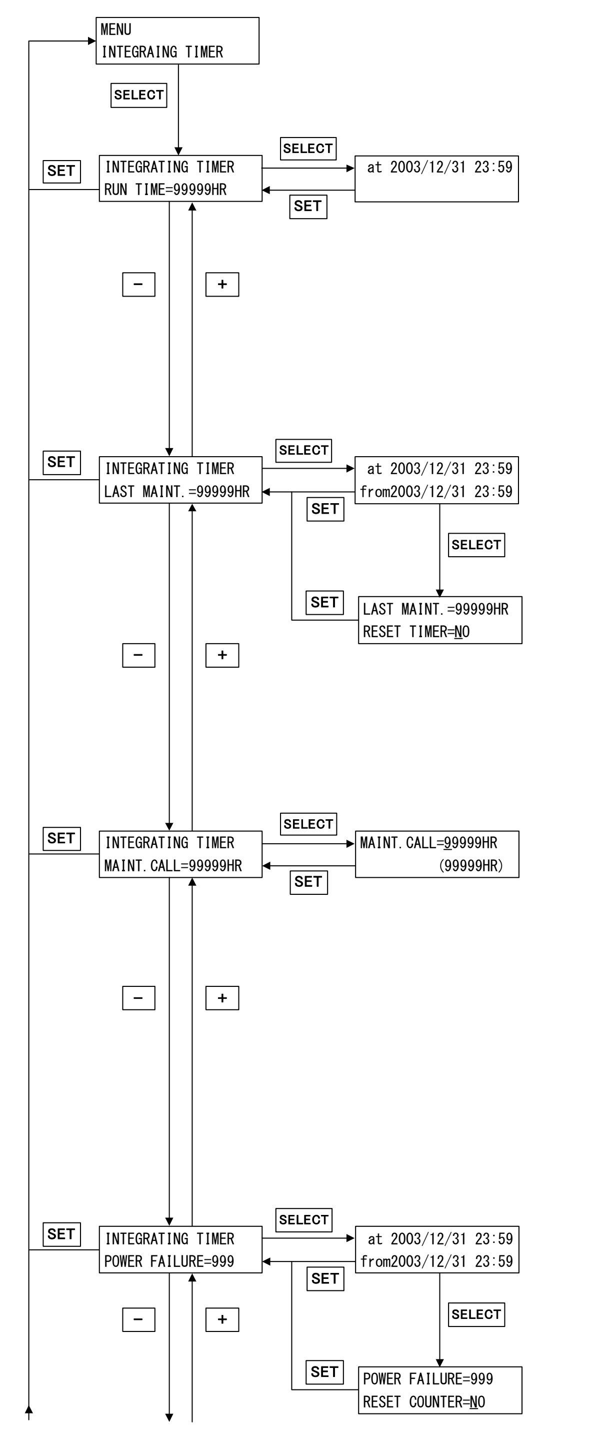
(2) ALARM MODE

Alarm mode is a mode to display detected alarm contents and alarm history. If key DISPLAY is pushed in monitor mode, the LCD changes into alarm mode. The LCD changes automatically changes into alarm mode when an alarm is detected. Refer to Table 7-6 "Table of Alarms" and Table 7-7 "Table of Warnings" about alarm code.

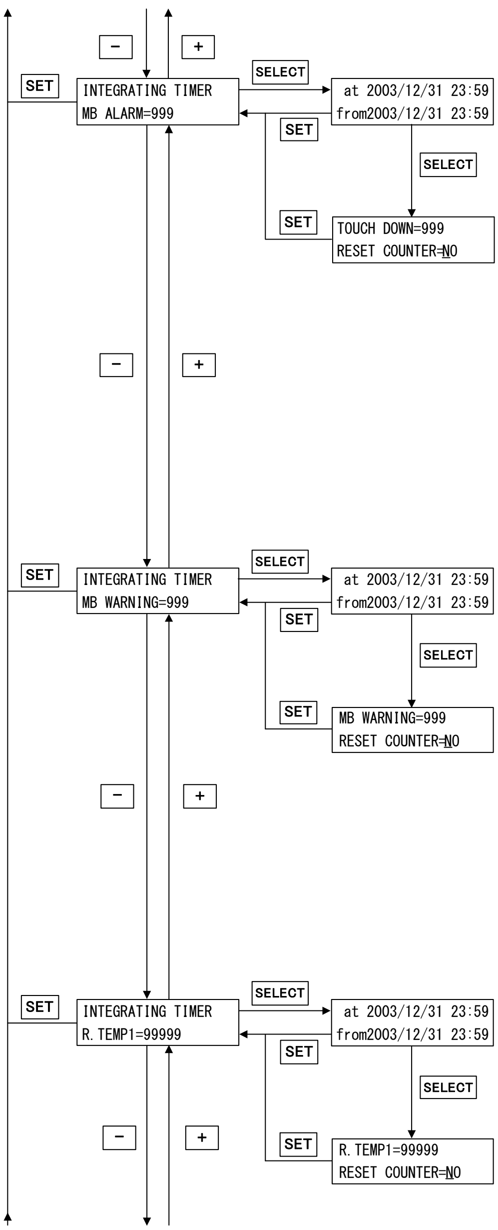
Mode	Operation and LCD display	Description of display
ALARM LIST		The currently detected error is displayed. If there is more than one error, all the errors can be displayed using the + and − keys  ALARM No.2 ALARM No.3 ALARM No.4
ALARM HISTORY		Maximum ALARM No.99  Dates and times are recorded as Greenwich Mean Time (GMT).
ALARM HISTORY DETAILS		The pump's operating status when an error is detected is displayed. 

6

(3) MENU MODE / TIMER

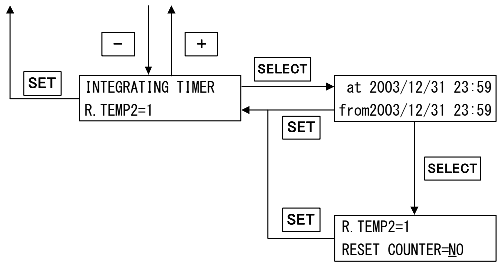


Mode	Operation and LCD display	Description of display
TIMER		MENU MODE MENU INTEGRATING TIMER TIMER MODE TITLE
RUN TIME		The pump's cumulative run time is displayed. INTEGRATING TIMER RUN TIME=99999HR Run time Update time at 2003/12/31 23:59
MAINTENANCE TIMER		The pump's cumulative run time is displayed. Reset is possible at the same time as, for example, maintenance. INTEGRATING TIMER LAST MAINT.=99999HR Maintenance call time Update time at 2003/12/31 23:59 from2003/12/31 23:59 Reset time Select Reset Timer LAST MAINT.=99999HR RESET TIMER=NO Press the + or − key to select between NO or YES
		If the cumulative value for the maintenance call time exceeds the set value, alarm code 99 (maintenance call warning) occurs. Setting the set value to 0 disables this function. INTEGRATING TIMER MAINT.CALL=99999HR Settings of Maintenance call time The value is changed using the + and − keys. MAINT.CALL=99999HR (99999HR) A cursor moves to the right when pushes SELECT key.
POWER FAILURE TOUCH DOWN COUNTER		The number of power failure touch-downs is counted. If this number exceeds 250, alarm code 12 (number of power failure touch-downs alarm) occurs. Refer to Section 7.5.3 for the appropriate remedy. INTEGRATING TIMER POWER FAILURE=999 Number of POWER FAILURE Update time at 2003/12/31 23:59 from2003/12/31 23:59 Reset time Select Reset Counter POWER FAILURE=999 RESET COUNTER=NO Press the + or − key to select between NO or YES

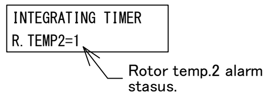


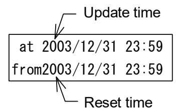
HIGH SPEED TOUCH DOWN COUNTER	The number of alarms that occur for the magnetic bearing system at speeds exceeding 80 % of the rated speed is counted. If this number exceeds 10, alarm code 11 (number of high-speed touch-downs alarm) occurs. INTEGRATING TIMER MB ALARM=999 Number of TOUCH DOWN Update time at 2003/12/31 23:59 from2003/12/31 23:59 Reset time Select Reset Counter TOUCH DOWN=999 RESET COUNTER=NO Press the + or - key to select between NO or YES
MB WARNING COUNTER	The number of warning that occur for the magnetic bearing system is counted. INTEGRATING TIMER MB WARNING=999 Number of Magnetic Bearing Warning Update time at 2003/12/31 23:59 from2003/12/31 23:59 Reset time Select Reset Counter MB WARNING=999 RESET COUNTER=NO Press the + or - key to select between NO or YES
ROTOR TEMPERA-TURE COUNTER 1	The cumulative value that rotor temperature is out of specification temperature is displayed. But, this function is invalid now because there is no appropriate pump for this function yet. INTEGRATING TIMER R. TEMP1=99999 Rotor temp.1 counter. Update time at 2003/12/31 23:59 from2003/12/31 23:59 Reset time Select Reset counter. R. TEMP1=99999 RESET COUNTER=NO Press the + or - key to select between NO or YES.

ROTOR TEMPERA- TURE COUNTER 2

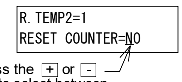


Alarm status generated when the rotor temperature is out of specification temperature is displayed.
But, this function is invalid now because there is no appropriate pump for this function yet.



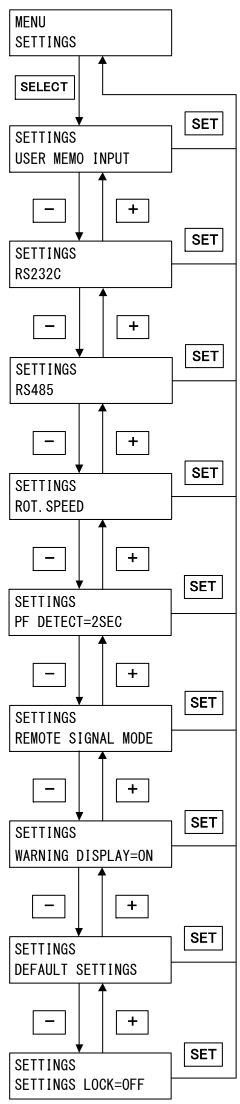
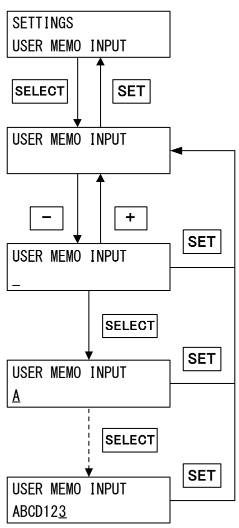
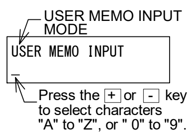


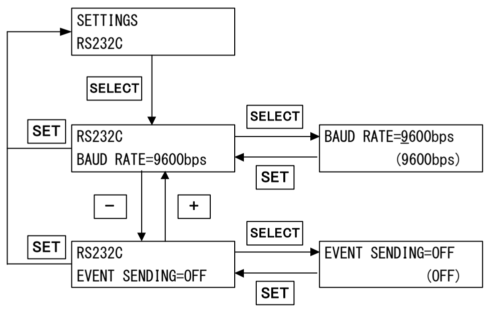
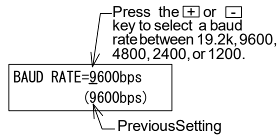
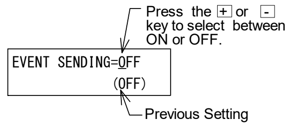
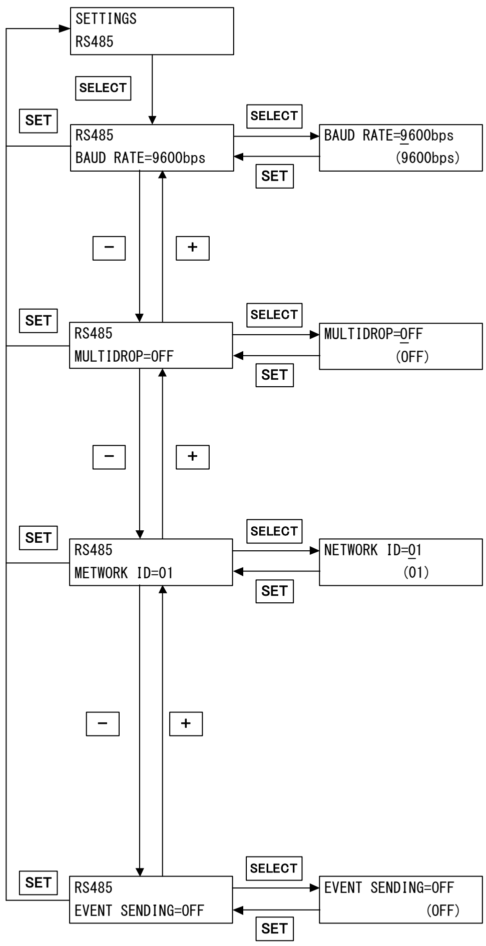
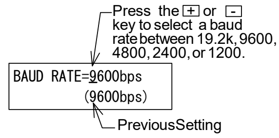
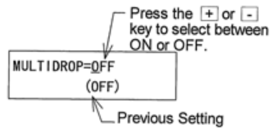
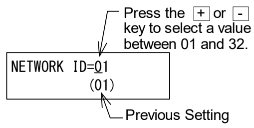
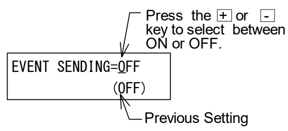
Select Reset Counter

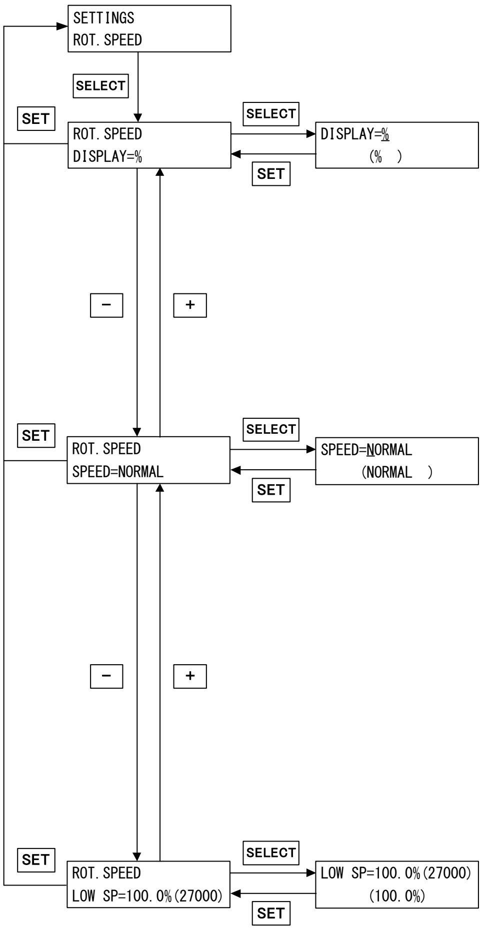
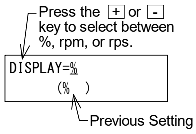
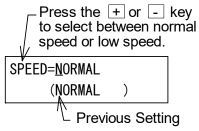
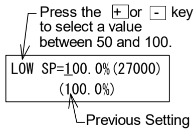
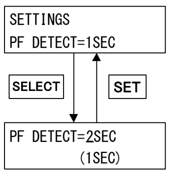
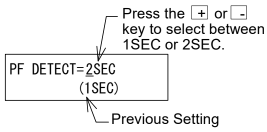


Press the + or - key to select between NO or YES.

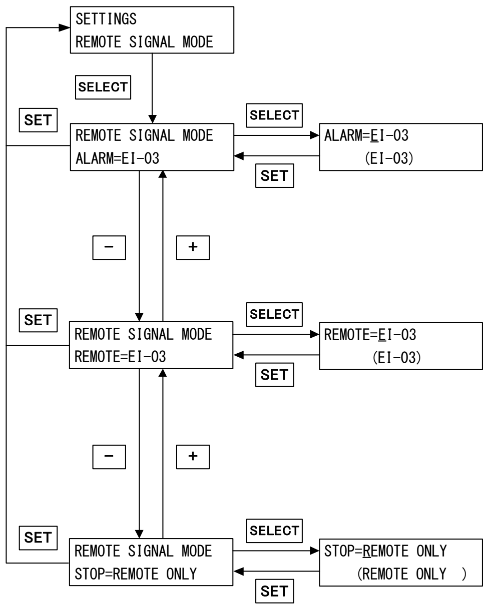
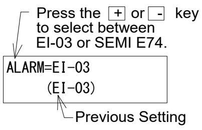
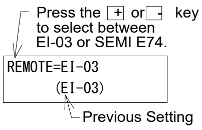
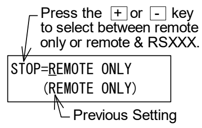
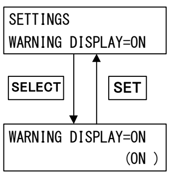
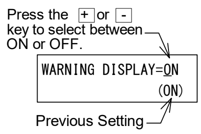
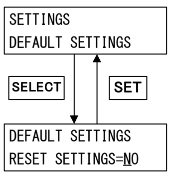
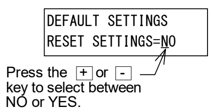
(4) MENU MODE / SETTINGS

Mode	Operation and LCD display	Description of display
SETTINGS		Select settings mode
USER MEMO SETTINGS		Select user memo settings
RS-232C SETTINGS		Select RS-232C settings
RS-485 SETTINGS		Select RS-485 settings
ROTATIONALSPEED SETTINGS		Select rotational speed settings
POWER FAILURE DETECT SETTINGS		Select power failure detect settings
REMOTE-CONTROL SIGNAL SETTINGS		Select remote-control signal settings
WARNING OUTPUT SET-TINGS		Select warning output setting
DEFAULT SETTINGS		Select default settings
LOCK SETTINGS		Select lock settings
USER MEMO SETTINGS		Setting of any 20 characters is possible (Ex.Connection name of Turbo Molecular Pump). Characters are displayed during monitor mode.  A cursor moves to the right when pushes SELECT key.

Mode	Operation and LCD display	Description of display
RS-232C SETTINGS		Baud rate setting 
		Event Transmission Settings (Note 1)  The event sending function is enabled or disabled. If set to ON, an event command is sent from the power supply unit when an event occurs.
RS-485 SETTINGS		Baud rate setting 
		<u>Multi-drop setting</u> (Note 1)  Set ON to select the multi-drop mode.
		<u>Network ID setting</u> (Note 1)  Setting required in multi-drop mode.
		Event Transmission Settings (Note 1)  The event sending function is enabled or disabled. If set to ON, an event command is sent from the power supply unit when an event occurs.

Mode	Operation and LCD display	Description of display
ROTATIONAL SPEED SETTINGS		Speed display setting  The format of the motor speed display in the pump monitor mode is changed.
		Speed setting  The pump rotates at the rated speed when NORMAL is selected. The pump rotates at the speed set with the low speed setting when LOW SPEED is selected. Refer to Section 6.5 "Variable Speed Operation"
		Low speed setting  When the speed setting is set to LOW speed or the remote-control signal designates low speed, the pump continues operation at the speed set by the low speed setting.
POWER FAILURE DETECT SETTINGS		 Set the time to detect power failure alarm.

6

Mode	Operation and LCD display	Description of display
REMOTE-CONTROL SIGNAL SETTINGS		"ALARM" signal setting of remote-control signal (Note 2)  "REMOTE" signal setting of remote-control signal (Note 2)  "STOP" signal setting of remote-control signal (Note 2) 
WARNING OUTPUT SETTINGS		 The setting changes the operation when warnings of alarm codes 86 to 98 (Refer to Table 7-7 "Table of Warnings") occur, as follows: ON: Buzzer ON, ALARM LED flashes, Error event sent to RS-232C and RS-485, The remote-control connector "WARNING" signal turn ON OFF: Buzzer OFF, ALARM LED OFF, No error event sent to RS- 232C and RS-485, The remote-control connector "WARNING" signal turn off
DEFAULT SETTINGS		 Select YES and push the SET key to return all the above items to their default settings. (Note 3)

Mode	Operation and LCD display	Description of display
LOCK SETTINGS	SETTINGS SETTINGS LOCK=OFF SELECT SET SETTINGS LOCK=OFF (OFF)	Press the + or - key to select between ON or OFF. SETTINGS LOCK=OFF (OFF) Previous Setting Set ON to disable changes on settings above mentioned.

(Note 1) Refer to APPENDIX-A "COMMUNICATIONS" for details.

(Note 2) Remote-control signal settings mode.

Signals		Description	Pin No.	
ALARM			(20)–(22)	(21)–(22)
	EI-03 (*1)	Alarm occurrence	open	close
		Power OFF and no failure	close	open
	SEMI E74 (*2)	Power OFF and alarm occurrence	close	open
		No failure	open	close
WARNING			(11)–(13)	(12)–(13)
	EI-03 (*1)	Warning occurrence	open	close
		Power OFF and no failure	close	open
	SEMI E74 (*2)	Power OFF and warning occurrence	close	open
		No failure	open	close
REMOTE	EI-03 (*1)	“REMOTE” signal is OFF in power failure regeneration braking even if remote-controlled operation is available.		
	SEMI E74 (*2)	“REMOTE” signal is always ON when remote-controlled operation is available.		
STOP	REMOTE ONLY	“STOP” signal ((16)-(14) open) is effective for only the time when remote-controlled operation is available.		
	REMOTE & RS-XXX	“STOP” signal ((16)-(14) open) is effective in operation by RS-232C or RS-485 set it in the cases that wants to use hardware inter rock.		

*1: When set to EI-03, behavior of remote-control signals is the same as SHIMADZU Turbo Molecular Pump power supply "EI-xx03ME", "EI-xx03MED" series.

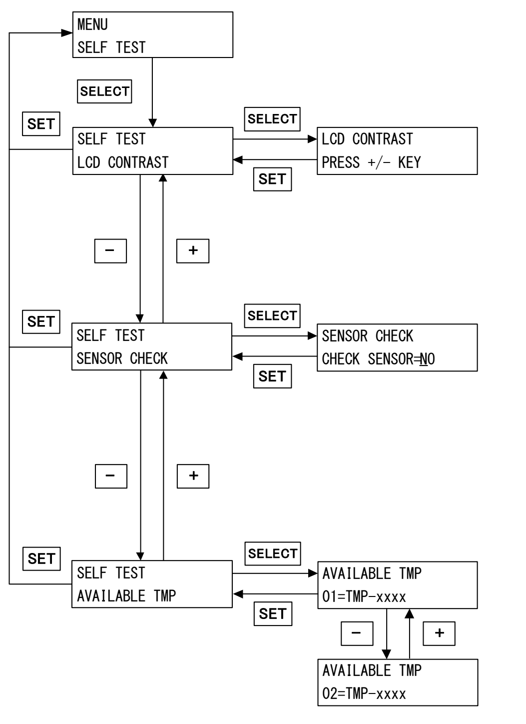
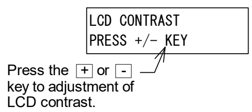
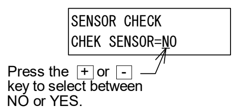
*2: When set to SEMI E74, behavior of remote-control signals conform to SEMI E74 standard "Specification for vacuum Pump Interface-Turbo Molecular Pumps".
Refer to the same standard about the shape of connectors and the pin configuration.

(Note 3) Default parameters

Function	Default settings
User memo	SHIMADZU EI-R04M
RS-232C	BAUD RATE = 9600 bps EVENT SENDING = ON
RS-485	BAUD RATE = 9600 bps MULTI DROP = OFF NETWORK ID = 01 EVENT SENDING = ON
Variable speed	DISPLAY = % SPEED = NORMAL LOW SP = 100 %
Power failure detect	PF DETECT = 1 sec
Remote-control signal	ALARM = EI-03 REMOTE = EI-03 STOP = REMOTE ONLY
Warning output setting	WARNING DISPLAY = ON
Lock	SETTINGS LOCK = OFF

6

(5) MENU MODE / SELF TEST

<table>
<tr><th>Mode</th><th>Operation and LCD display</th><th>Description of display</th></tr>
<tr><td>LCD CONTRAST ADJUSTMENT</td><td rowspan="3">
</td><td>
</td></tr>
<tr><td>SELF DIAGNOSIS</td><td>Select YES and push the SET key to execute self diagnosis of the magnetic bearing sensor. If an abnormality is discovered by self diagnosis, alarm codes 81 to 85 occur. Refer to Section 7.5.3 for the appropriate remedy. This function can be used only when the pump is stopped.

</td></tr>
<tr><td>CORRE-SPONDING MODEL</td><td>Displayed corresponding model that be available at this Power Supply Unit. Push the + or − key to display all corresponding model.</td></tr>
</table>

6.7 Remote-control Connector

6.7.1 Specifications

The controller is provided with remote-control connector for connection with remote operation, alarm signals, etc. Use this connector and a cable with shield as necessary. The shield of the cable should be connect to case of Remote-connector (Refer to Fig. 6-8, Fig. 6-9, Fig. 6-10 and Table 6-2, Table 6-3). For remote-controlled operation, push the REMOTE switch (Fig. 2-1 (15)) to REMOTE lamp ON.

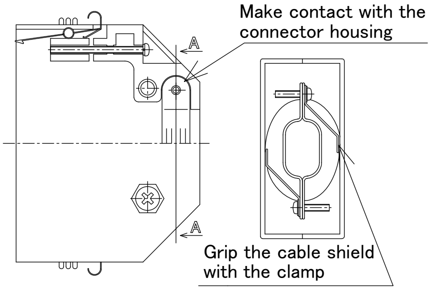


Fig. 6-8 Remote-control Connector

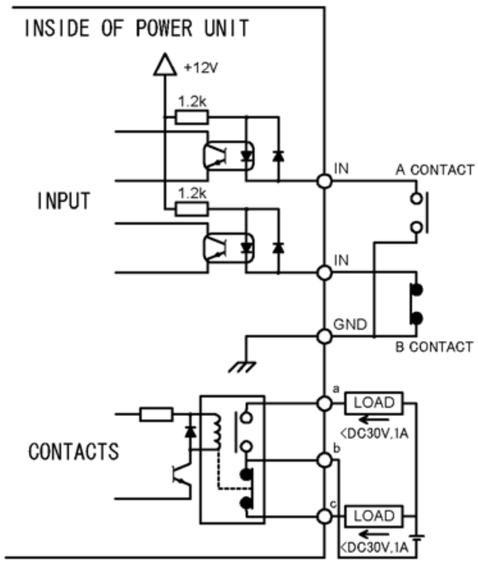


Fig. 6-9 Remote-control Circuit

Table 6-2 START/STOP According to Remote-control Signals

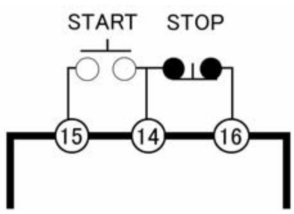
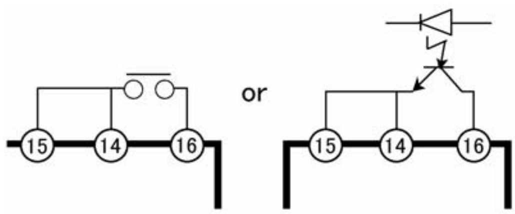
Connection method	By momentary type START/STOP switch	By alternate type switch
Wiring connection		
Control	Pump start by short-circuiting (15) and (14). Pump stop by opening (16) and (14).	Pump start, with the contact close or photo transistor ON ((16) to (14) short-circuit). Pump stop, with the contact open or photo transistor OFF ((16) to (14) open).
Electric capacity	[Contact] It is connected to +12 V circuit and subject to stable open-close of 5 VDC, 1 mA. Voltage...30 VDC or more, Current...10 mA or more [Photo transistor] Select a photo transistor with a collector-emitter voltage limit of 30 VDC and an on-state collector current of 10 mA or more.	
Input rating	Direct forward current 50 mA; DC reverse voltage 5 V	

Table 6-3 Remote-control Signals (Default Settings)

Classifi-cation	Signals	Pin No. (Note 1)	Operation (Note 2)	Electric spec.
Input	START	15	Starting operation on short-circuiting between GND and pin No.15. (Note 3)	Contact input
	STOP	16	Pump stop by opening GND and pin No.16. (Note 3) (Note 6)	
	RESET (Note 4)	17	Resetting operation on short-circuiting between GND and pin No.17.	
	LOW SPEED	33	Variable Speed Operation on short-circuiting between GND and pin No.33.	
	GND	14		
Output	ROTATION	29 4 30	During rotation; (29) - (30): open → close (make contact) (4) - (30): close → open (break contact)	Contact output Contact capacity (resistance load) 30 VDC 1 A
	NORMAL	25 2 26	During normal rotation; (25) - (26): open → close (make contact) (2) - (26): close → open (break contact)	
	ACCELERATION	23 1 24	During acceleration; (23) - (24): open → close (make contact) (1) - (24): close → open (break contact)	
	BRAKE	27 3 28	During deceleration; (27) - (28): open → close (make contact) (3) - (28): close → open (break contact)	
	REMOTE	31 5 32	Remote-controlled operation is available; (31) - (32): open → close (make contact) (5) - (32): close → open (break contact) (Note 6)	
	ALARM (Note 5)	21 20 22	Against alarm; (21) - (22): open → close (make contact) (20) - (22): close → open (break contact) (Note 6)	
	WARNING (Note 5)	12 11 13	Against warning; (12) - (13): open → close (make contact) (11) - (13): close → open (break contact) (Note 6)	

(Note 1) Don't connect any pins other than specified above.

(Note 2) Approximately 6 seconds is required for the remote control signal to turn on after the POWER switch is turned on.

(Note 3) "STOP" signal is prior to "START" signal.

(Note 4) One reset signal is received each time when the contact closes. Repeatedly short and open the contact to input multiple reset signals. Refer to Section 7.5.3 "Reset Procedure" for details about the error-reset procedure.

(Note 5) Refer to Table 7-6 "Table of Alarms" and Table 7-7 "Table of Warnings" about alarm and warning.

(Note 6) It is possible to change movement by remote-control signal settings of settings mode (Refer to Section 6.6 "Software Operation" (4)).

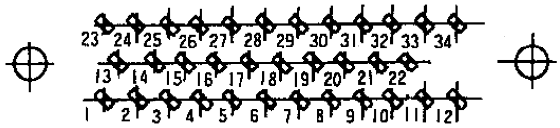


Fig. 6-10 Arrangement of Remote-control Connector Pins

(The power supply unit rear panel attachment connectors, as viewed from the front)

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7

TROUBLESHOOTING

7.1 Nothing Happens After an Operation is Made

Table 7-1 Nothing Happens After an Operation is Made

	Problem	Possible causes	Corrective action
1	Power ON/OFF switch in the ON position but the turbo molecular pump fails to operate.	Electrical power cable not properly connected.	Properly connect the electrical power cable.
		Electrical power outside power supply unit's power range.	Operate within power supply unit's power range.
2	START switch is pushed but turbo molecular pump does not accelerate.	REMOTE lamp ON.	Push the REMOTE switch to turn off the REMOTE lamp.
		Other causes.	Check the power supply unit's ALARM lamp is not ON. If an alarm is indicated, correct the malfunction and reset the power supply unit.
3	Remote "START" signal active but the turbo molecular pump does not accelerate.	REMOTE lamp OFF.	Push the REMOTE switch to turn on the REMOTE lamp.
		"STOP" signal active.	Deactivate "STOP" signal.
		Other problems.	Check the unit's ALARM lamp is not ON. If an alarm is indicated, correct the malfunction and reset the power supply unit.
4	STOP switch is pushed but the turbo molecular pump does not decelerate.	REMOTE lamp ON.	Push the REMOTE switch to turn off the REMOTE lamp.
5	Remote "STOP" signal activated but the turbo molecular pump does not decelerate.	REMOTE lamp OFF.	Push the REMOTE switch to turn on the REMOTE lamp.

7.2 Power Failures

When a power interruption occurs, the motor inside the turbo molecular pump immediately begins regenerative braking. The magnetic bearing will use this generated electricity to keep functioning and the rotor inside the turbo molecular pump will continue being levitated. The rotation will slow down due to the regenerative braking and eventually the rotor will be supported by the touchdown bearing. Table 7-2 shows the number of revolutions and period of time that will elapse before being supported by the touchdown bearing, when the power fails at the rated speed.

Table 7-2 Time and Rotational Speed During a Power Interruption Before being Supported by Touchdown Bearing

Pump model	Power supply model	Rotational speed before support by touchdown bearing	Period of time before support by touchdown bearing (Note 1)
TMP-803M/MC/LM/LMC TMP-1003M/MC/LM/LMC	EI-R04M	7200 rpm	about 4 minutes
TMP-1103MP/MPC/LMP/LMPC			about 4.5 minutes
TMP-1303LM/LMC TMP-1503LM/LMC		7800 rpm	about 5.5 minutes
TMP-2003M/MC/LM/LMC		5400 rpm	about 10 minutes
TMP-2203M/MC/LM/LMC			about 13 minutes
TMP-W2403LMC			about 12 minutes
TMP-H3153LMC			about 20 minutes
TMP-3203M/MC/LM/LMC			about 15.5 minutes
TMP-3403LMC			about 17 minutes
TMP-3413LMC			about 16 minutes
TMP-W3503LMC			
TMP-H3603LMC			about 20 minutes
TMP-4203LMC		4200 rpm	about 22 minutes

(Note 1) The time is typical for regenerative braking from the rated speed.
Actual time will vary depending on vacuum conditions inside the pump and the rotational speed when the power fails.

7.2.1 Power Failure Counter-operation

Table 7-3 shows the counter-operations against power supply failure which occurred while the pump rotor is normally rotating.

Table 7-3 Counter-operations Against Power Supply Failure

Interruption time	1 second or less (Note 1)		Over 1 second (Note 1)	
Interrupt/re-supply	During interruption	After re-supply	During interruption	After re-supply
Pump status				
Magnetic levitation	Levitation goes on.	Levitation goes on.	Levitation goes on.	Levitation goes on.
Run	Decelerates	Returns to before-power-failure running condition	Decelerates	Decelerates (Note 1)
Indicator lamp				
ROTATION	Lamp ON goes on	Lamp ON goes on	Lamp ON goes on	Lamp ON goes on
NORMAL SPEED	Before-power-failure indication goes on.	Before-power-failure indication goes on.	Turns off	Lamp OFF goes on
ALARM	Before-power-failure indication goes on.	Before-power-failure indication goes on.	Turns on	Lamp ON goes on
Remote-control output signals (Note 3)				
ROTATION Pin no. (29)-(30) (4)-(30)	"CLOSE" goes on "OPEN" goes on	"CLOSE" goes on "OPEN" goes on	"CLOSE" goes on "OPEN" goes on	"CLOSE" goes on "OPEN" goes on
NORMAL Pin no. (2)-(26) (25)-(26)	Before-power-failure condition goes on. Before-power-failure condition goes on.	Before-power-failure condition goes on. Before-power-failure condition goes on.	Contact close Contact open	"CLOSE" goes on "OPEN" goes on
ACCELERATION Pin no. (1)-(24) (23)-(24)	Before-power-failure condition goes on. Before-power-failure condition goes on.	Before-power-failure condition goes on. Before-power-failure condition goes on.	Contact close Contact open	"CLOSE" goes on "OPEN" goes on
BRAKE Pin no. (27)-(28) (3)-(28)	Before-power-failure condition goes on. Before-power-failure condition goes on.	Before-power-failure condition goes on. Before-power-failure condition goes on.	Contact close Contact open	"CLOSE" goes on "OPEN" goes on
REMOTE Pin no. (31)-(32) (5)-(32)	Before-power-failure condition goes on. Before-power-failure condition goes on.	Before-power-failure condition goes on. Before-power-failure condition goes on.	Contact open Contact close	Return to Before-power-failure condition. Return to Before-power-failure condition.
ALARM Pin no. (21)-(22) (20)-(22)	"OPEN" goes on "CLOSE" goes on	"OPEN" goes on "CLOSE" goes on	Contact close Contact open	By resetting, "CLOSE" → "OPEN" "OPEN" → "CLOSE"
WARNING Pin no. (12)-(13) (11)-(13)	"OPEN" goes on "CLOSE" goes on	"OPEN" goes on "CLOSE" goes on	"OPEN" goes on "CLOSE" goes on	By resetting, "OPEN" goes on "CLOSE" goes on
Buzzer	Does not sound	Does not sound	(Note 2) Before resetting: Sounds After resetting: Released (reset)	(Note 2) Before resetting: Sounds After resetting: Released (reset)

(Note 1) Replace 1 second by 2 seconds, if "2sec" is selected in POWER FAILURE DETECT SETTINGS.

(Note 2) For restarting after the power re-supply (restoration), push the RESET switch twice and, thereafter, perform the start-up operation. First pushing of the RESET releases the buzzer and second pushing of the RESET releases "ALARM" signal.

(Note 3) The pin numbers are shown in Fig. 6-10 "Arrangement of Remote-control Connector Pins".

(Note 4) The ALARM, WARNING and REMOTE signal activity will change depending on the SETTINGS / REMOTE SIGNAL MODE menu settings on the LCD display. For more information, refer to Section 6.6 "Software Operation".

7.3 Vacuum Pressure Rise

A rapid rise of vacuum pressure in the turbo molecular pump causes the internal motor of the turbo molecular pump to start braking and the ALARM lamp (Fig. 2-1 (14)) lights.
Be careful to prevent a rapid pressure rise or air rush during operation.

7.4 Abnormal Noise and/or Vibration

Should the turbo molecular pump ever generate abnormal noise and/or vibration, the turbo molecular pump operation is to be stopped immediately.
But there is possible that a race of touch-down bearing may make sounds for seconds when the pump internal pressure gets back to atmospheric pressure using air (or non-activity gas). This phenomena is not abnormal and make no damage to the pump, because the air whirlpool sometimes occurs and then makes the touch-down bearing rotate slightly.

NOTICE

When turning the POWER switch for the power supply unit on or off, a "clunk" sound may be heard from inside the pump. This sound is from the rotor inside the pump being levitated or de-levitated. This is normal.

7.5 Alarm Detection Capabilities

The fault detection functions shown in Table 7-6 "Table of Alarms" and Table 7-7 "Table of Warnings" are incorporated for protection in the event of a problem with the turbo molecular pump or power supply unit.

When an error is detected, check the ALARM lamp (Fig. 2-1 (14)) and the alarm mode display on the front panel (Refer to Section 6.6 "Software Operation" (2)) and refer to Section 7.5.3 "Reset Procedure" for the appropriate remedy.

7.5.1 Movement in Alarm Detection Capabilities (ALARM)

1. ALARM lamp (Fig. 2-1 (14)) lights.
2. Alarm description is displayed on LCD.
3. The remote-control connector "ALARM" signal turn ON.
4. The buzzer sounds.
5. The pump start the protective operations shown in Table 7-6 "Table of Alarms".
6. The detection error is recorded in the error log.

7.5.2 Movement in Alarm Detection Capabilities (WARNING)

The warning output setting in the menu mode "SETTINGS / WARNING DISPLAY" item on the LCD changes the operation when a warning occurs.

<When the warning output setting is ON>
1. ALARM lamp (Fig. 2-1 (14)) flashes.
2. Warning description is displayed on LCD.
3. The remote-control connector "WARNING" signal turn ON.
4. The buzzer sounds.
5. Pump operation continues.
6. The detection error is recorded in the error log.

<When the warning output setting is OFF>
[When warnings of alarm codes except 86 to 98 occur.] (Refer to Table 7-7 "Table of Warnings".)
The operation is the same as the warning output setting is ON.
[When warnings of alarm codes 86 to 98 occur.] (Refer to Table 7-7 "Table of Warnings".)
(1. ALARM lamp (Fig. 2-1 (14)) does not flash.)
2. Warning description is displayed on LCD.
(3. The remote-control connector "WARNING" signal do not turn ON.)
(4. The buzzer does not sound.)
5. Pump operation continues.
6. The detection error is recorded in the error log.

7.5.3 Reset Procedure

1. The buzzer stops after the first reset operation.
2. Refer to the Troubleshooting information and eliminate the cause of the problem.
3. Conduct the reset operation again.
4. (ALARM) If the problem has been eliminated, the ALARM lamp (Fig. 2-1 (14)) goes out, the "ALARM" remote-control signals (Refer to Table 6-3) turn off after an alarm was given, the pump rotor decelerates rotational speed.
   (WARNING) When the problem is eliminated after a warning occurred, pump operation continues.
5. If the problem was not completely eliminated, the buzzer sounds again and the error is detected.

(Note) When the buzzer does not sound and if the warning output setting is OFF in the menu mode SETTINGS / WARNING DISPLAY item on the LCD, the first reset operation after a warning occurs is equivalent to the third reset operation.

Table 7-4 If the ALARM Lamp Lights

	LCD display	Possible cause	Remedy	Section
1	11=TD COUNTER LIMIT 12=PF COUNTER LIMIT	The number of high speed or power failure touch-downs has exceeded the prescribed number.	The touch-down bearing may have deteriorated. Consult Shimadzu or an approved service company regarding replacement of the touch-down bearing.	
2	13=WRONG TMP MODEL	The models of pump and power supply do not match.	Check the combination of the pump and power supply models. Corresponding model can be confirm in menu mode SELF TEST / AVAILABLE TMP. Check the connection of the magnetic bearing cable. Also check that the connector pins are not bent. Turn off the power supply before checking the magnetic bearing cable. Never disconnect the cable while the power is on.	6.6 5.3
3	14=AC LOW VOLTAGE 15=POWER FAILURE	Power failure or reduction in the power supply voltage.	Wait for the power to be restored. An unfamiliar sound will be heard a few minutes after a power failure. This sound occurs when the rotor contacts the protective bearing as magnetic levitation cannot be maintained. This is not an abnormal sound.	
		POWER switch was turned off by mistake.	Wait about 5 seconds before turning the POWER switch back on. Re-acceleration is possible after resetting and start-up.	
4	16=TMP:OVERLOAD	Drop in rotation speed during rotation at rated speed (increased internal pump pressure).	Check that the outlet and inlet pressures are below the specified maximum pressures. Check for leakage. Check that too much purge gas is not flowing. Check that process gas flow rate is not too high.	

	LCD display	Possible cause	Remedy	Section
5	21=TMP TEMP/MB CABLE	High pump unit temperature.	Check that the ambient temperature around the pump is within the specified range. Check that the temperature and flow rate of the cooling water are within the specified ranges. Check that no load in excess of the specified range is continuously applied to the pump.	
		Magnetic bearing cable is not connected correctly.	Check the connection of the magnetic bearing cable. Also check that the connector pins are not bent. Turn off the power supply before checking the magnetic bearing cable. Never disconnect the cable while the power is on.	5.3
6	22=TMP:SENSOR ERROR 46=MOTOR OVERSPEED 47=EI:R-SPEED ERROR	Rotation detection signal from the motor cannot be detected correctly.	Check that equipment causing noise is not used around the power supply unit, pump unit, motor cable, or magnetic bearing cable.	5.1 5.2 5.3
7	23=EI:MOTOR OVERCURR 34=EI:INV. OVERCURR	Overcurrent supplied to the motor.	Check the motor cable and magnetic bearing cable. (Check for connection and wiring likely to cause disconnection or short circuit)	
8	31=EI:BR OVERTEMP 32=EI:DC-DC OVERTEMP	Temperature increase in the power supply unit.	Check that the ambient temperature around the power supply unit is within the specified range. Check the ventilation of the power supply unit.	4.1 5.1
9	33=EI:FAN ERROR	Fan stopped.	Check that the fan on the power supply unit rear panel is operating. Remove any object obstructing its operation.	5.1
10	35=EI:INV. OVERVOLT 36=EI:DC-DC LOW VOLT 37=EI:DC-DC OVERCURR 38=EI:DC-DC OVERVOLT 45=EI:BRAKE OVERTIME	Defective circuit in the power supply unit.	Turn the power on again after the pump stops. The power supply unit must be repaired if the problem occurs again.	
11	43=EI:PARAM ERROR	Defective circuit in the power supply unit.	Turn the power on again after the pump stops. The power supply unit must be repaired if the problem occurs again.	
12	44=EI:CPU ERROR 66=MB:DSP ERROR 67=MB:DSP OVERFLOW	Abnormal operation of circuit in the power supply unit.	Check that the ambient temperature around the power supply unit is within the specified range. Check that equipment causing noise is not used around the power supply unit, pump unit, motor cable, or magnetic bearing cable.	4.1 5.1 5.2 5.3
13	48=EI:ACCEL OVERTIME	Rotation speed does not increase at start-up.	Check that the outlet and inlet pressures are not too high. Check for leakage. Check that too much purge gas is not flowing.	6.2

	LCD display	Possible cause	Remedy	Section
14	49=TMP:CAN NOT START	Pump does not rotate.	Adhesion of reaction products or damage to the protective bearing is the possible cause. Remove the pump from the unit and check that the rotor blades rotate smoothly at the inlet. An overhaul is required if blades do not rotate smoothly. Check the connection of the magnetic bearing cable. Also check that the connector pins are not bent. Turn off the power supply before checking the magnetic bearing cable. Never disconnect the cable while the power is on.	
15	51=MB:VIBRATION2 X1 52=MB:VIBRATION2 Y1 53=MB:VIBRATION2 X2 54=MB:VIBRATION2 Y2 55=MB:VIBRATION2 Z 56=MB:VIBRATION1 X1 57=MB:VIBRATION1 Y1 58=MB:VIBRATION1 X2 59=MB:VIBRATION1 Y2 60=MB:VIBRATION1 Z	Strong external shock or vibrations.	If the shock or vibrations are transient, re-acceleration is possible. If the shock or vibrations occur frequently, stop the pump and remove the source of the shock or vibrations; or re-examine the pump mounting method.	
16	61=MB:SENSOR ERR. X1 62=MB:SENSOR ERR. Y1 63=MB:SENSOR ERR. X2 64=MB:SENSOR ERR. Y2 65=MB:SENSOR ERR. Z	Control cable is not connected correctly.	Check the connection of the magnetic bearing cable. Also check that the connector pins are not bent. Turn off the power supply before checking the magnetic bearing cable. Never disconnect the cable while the power is on.	5.3
		The rotor does not move due to adhering matter or a damaged touch-down bearing.	Remove the pump and check that the rotor rotates smoothly. The pump requires an overhaul if the rotor does not rotate or the resistance to rotation is large.	
17	68=MB:BALANCE AXIS1 69=MB:BALANCE AXIS2	Rotor inside the pump is out of balance.	Adhesion of reaction products is a possibility. An overhaul is required.	

Table 7-5 If the ALARM Lamp Flashes

	LCD display	Possible cause	Remedy	Sec-tion
1	80=EI:CONT.TEMP.WARN	Temperature increase in the power supply unit.	Check that the ambient temperature around the power supply unit is within the specified range. Check the ventilation of the power supply unit.	4.1 5.1
2	81=MB:SELFCHECK X1 82=MB:SELFCHECK Y1 83=MB:SELFCHECK X2 84=MB:SELFCHECK Y2 85=MB:SELFCHECK Z	Rattling of the protective bearing becomes pronounced.	Deterioration of the protective bearing is likely. Overhaul as soon as possible to avoid damage to the protective bearing in the event of a power failure etc.	
3	86=MB:VIB. WARN. X1 87=MB:VIB. WARN. Y1 88=MB:VIB. WARN. X2 89=MB:VIB. WARN. Y2 90=MB:VIB. WARN. Z	Transient strong external shock or vibrations.	Continuous operation is possible. However, if the shock or vibrations occur frequently, stop the pump and remove the source of the shock or vibrations; or re-examine the pump mounting method.	
4	91=MB:BALANCE WARN.1 92=MB:BALANCE WARN.2	Rotor inside the pump is out of balance.	Adhesion of reaction products is a possibility. An overhaul is recommended.	
5	94=MB:AIR RUSH B	Atmospheric penetration has occurred.	Create a vacuum system not allowing atmospheric penetration by re-examining the operating sequence of the back pump and valves etc.	
6	95=DSP WARNING	Abnormal operation of circuit in the power supply unit.	Check that the ambient temperature around the power supply unit is within the specified range. Check that equipment causing noise is not used around the power supply unit, pump unit, motor cable, or magnetic bearing cable.	4.1 5.1 5.2 5.3
7	99=MAINTENANCE TIME	Maintenance call timer has reached the set time.	Implement maintenance works prescribed by the customer, such an overhaul. The alarm can be cancelled by resetting the maintenance call timer after implementing necessary works.	6.6

(Note) When warnings of alarm codes 86 to 95 (Refer to Table 7-7 "Table of Warnings") occur, the alarm lamp does not flash and the display appears on the LCD only if the warning output setting is OFF in the menu mode SETTINGS / WARNING DISPLAY item on the LCD.

Table 7-6 Table of Alarms

Alarm code	LCD display	Cause	Protective action
11	11=TD COUNTER LIMIT	Counts of the high speed touch-down counter exceeded the specified number.	Start-up impossible (detected during power supply self-diagnostics)
12	12=PF COUNTER LIMIT	Counts of the power failure touch-down counter exceeded the specified number.	
13	13=WRONG TMP MODEL	The combination of the pump and power supply is wrong.	
14	14=AC LOW VOLTAGE	Fall in AC input power voltage.	Regenerative braking
15	15=POWER FAILURE	Power failure.	
16	16=TMP:OVERLOAD	After accelerating to 80 % of the designated speed or low-speed setting, the speed dropped below 80 % due to overloading etc.	Deceleration
21	21=TMP TEMP/MB CABLE	Increased pump drive motor temperature. MB cable is not connected.	Free run (motor stop)
22	22=TMP:SENSOR ERROR	Pump rotation signal could not be detected correctly.	
23	23=EI:MOTOR OVERCURR	Overcurrent ran through the motor.	
31	31=EI:BR OVERTEMP	Increased temperature inside power supply unit.	Free run (motor stop)
32	32=EI:DC-DC OVERTEMP	Increased temperature inside power supply unit.	
33	33=EI:FAN ERROR	Power supply cooling fan has stopped.	
34	34=EI:INV. OVERCURR	Overcurrent ran through the motor.	
35	35=EI:INV. OVERVOLT	Defective circuit in the power supply.	
36	36=EI:DC-DC LOW VOLT	Defective circuit in the power supply.	Regenerative braking
37	37=EI:DC-DC OVERCURR	Defective circuit in the power supply.	Deceleration
38	38=EI:DC-DC OVERVOLT	Defective circuit in the power supply.	
43	43=EI:PARAM ERROR	Stored parameters are not correct.	Start-up impossible (detected during power supply self-diagnostics)
44	44=EI:CPU ERROR	Error in the CPU for inverter control.	Free run (motor stop)
45	45=EI:BRAKE OVERTIME	Pump does not stop within the specified time after the stop operation.	
46	46=MOTOR OVERSPEED	Pump rotation speed is too high.	
47	47=EI:R-SPEED ERROR	Pump rotation speed cannot be detected.	

Alarm code	LCD display	Cause	Protective action
48	48=EI:ACCEL OVERTIME	Pump does not accelerate to 80 % of the designated speed or low-speed setting within the specified time after start-up.	Deceleration
49	49=TMP:CAN NOT START	Pump fails to rotate within 2 minutes after start-up.	
51	51=MB:VIBRATION2 X1	Continuous excessive vibration of the magnetic bearing.	
52	52=MB:VIBRATION2 Y1		
53	53=MB:VIBRATION2 X2		
54	54=MB:VIBRATION2 Y2		
55	55=MB:VIBRATION2 Z		
56	56=MB:VIBRATION1 X1	Excessive magnetic bearing vibration.	
57	57=MB:VIBRATION1 Y1		
58	58=MB:VIBRATION1 X2		
59	59=MB:VIBRATION1 Y2		
60	60=MB:VIBRATION1 Z		
61	61=MB:SENSOR ERR. X1	Abnormal output signal from the magnetic bearing sensor.	
62	62=MB:SENSOR ERR. Y1		
63	63=MB:SENSOR ERR. X2		
64	64=MB:SENSOR ERR. Y2		
65	65=MB:SENSOR ERR. Z		
66	66=MB:DSP ERROR	Error in the DSP for magnetic bearing control.	
67	67=MB:DSP OVERFLOW	Overflow in the magnetic bearing control calculations.	
68	68=MB:BALANCE AXIS1	Rotor is out of balance.	
69	69=MB:BALANCE AXIS2		

Table 7-7 Table of Warnings

Alarm code	LCD display	Causes	Protective action
80	80=EI:CONT.TEMP.WARN	Increased temperature inside power supply unit.	Operation continued.
81	81=MB:SELFCHECK X1	Results of magnetic bearing sensor self-diagnostics are abnormal.	Operation is possible (detected during power supply self-diagnostics).
82	82=MB:SELFCHECK Y1		
83	83=MB:SELFCHECK X2		
84	84=MB:SELFCHECK Y2		
85	85=MB:SELFCHECK Z		
86	86=MB:VIB. WARN. X1	Vibrations of the magnetic bearing become temporarily excessive.	Operation continued.
87	87=MB:VIB. WARN. Y1		
88	88=MB:VIB. WARN. X2		
89	89=MB:VIB. WARN. Y2		
90	90=MB:VIB. WARN. Z		
91	91=MB:BAL. WARN. AXIS1	Rotor is slightly out of balance.	
92	92=MB:BAL. WARN. AXIS2		
94	94=MB:AIR RUSH B	Atmospheric penetration.	
95	95=DSP WARNING	Error in the DSP for magnetic bearing control.	
99	99=MAINTENANCE TIME	Maintenance call timer reaches its set time.	

Appendix A

COMMUNICATIONS

A1 GENERAL SPECIFICATION

The EI-R04M/MT series power supply units contain serial interfaces conforming to RS-232C and RS-485 specifications. The following functions are available by connecting a computer with communication capacity to these interfaces and creating the appropriate software.

The RS-232C and RS-485 interfaces can be used simultaneously, permitting simultaneous access from two computers. Also, the RS-485 interface permits multi-drop connections, allowing multiple power supplies to be connected to a single computer.

1. Checking current operation mode: The serial interfaces allow the user to check the status of REMOTE lamp (REMOTE/LOCAL). In REMOTE mode, the user can change the operation mode to RS-232C or RS-485.
2. Operation: Operations equivalent to the START, STOP, and RESET switches are available in the RS-232C or RS-485 operation mode. Also, the speed setting can be made using the set value write function.
3. Checking turbo molecular pump run status: The serial interfaces allow the user to check the current turbo molecular pump's running status (Normal rotation, Accelerating, Decelerating, failure occurrence, etc.).
4. Reading parameters: The serial interfaces allow the user to read a variety of turbo molecular pump parameters such as pump rotational speed and motor current which are stored in the power supply unit.
5. Receiving events: The power supply unit can transmit status commands for events such as failure occurrences, rotation start and stop, and attainment of normal rotation speed.
6. Reading history data: The serial interfaces allow the user to read the alarm history data displayed in the LCD display history mode.
7. Reading and writing timer data: The serial interfaces allow the user to read the timer and counter values displayed in the LCD display timer mode and to reset the counters.
8. Reading and writing settings data: The serial interfaces allow the user to read and change settings in the LCD display setting mode.

A2 INTERFACE SPECIFICATION

A2.1 RS-232C

A2.1.1 Transmission Specification

Interface	RS-232C
Synchronous system	Asynchronous
Transmission rate	1200, 2400, 4800, 9600 and 19.2 k bits per second (Refer to Section 6.6 "Software Operation" (4) for settings)
Character configuration	Start bit: 1 Data bits: 8 Parity: None Stop bit: 1
Flow control	None

A2.1.2 Communications Connector

Connector	Front panel RS-232C connector (Refer to Section 2.1 "Power Supply Unit")
Connector type	D-sub 9 pin Male, Screw lock size: M2.6
Pin assignment	2: RD (Receive data) 3: SD (Transmit data) 5: SG (Signal ground) *Other pins are not connected.

A2.1.3 CABLE

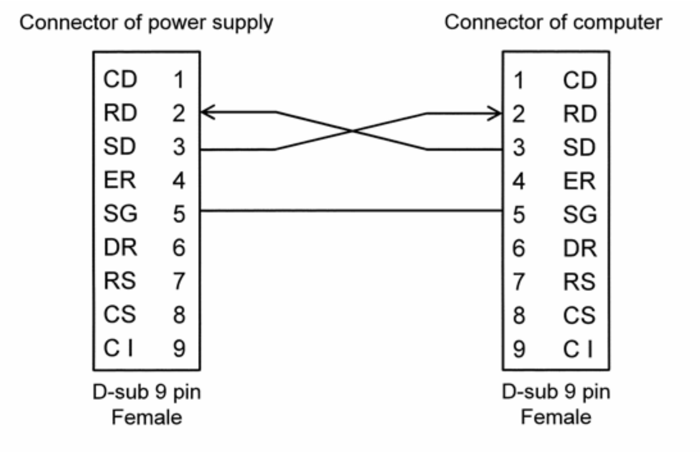
(1) Cable connection
Use the connection cable as shown in Fig. A-1 to connect the power supply and computer (Number on figure is pin number of connector).

(2) Cables used
For connection with this connector, the communication cable with shield is necessary. The shield of the cable should be connected to case.

(3) Cable length

Connection cables can be extended up to 15 meters, but may be subjects to errors depending on actual operational environment.

a. Cable wiring connections for 9-pin to 9-pin connector cables.



b. Cable wiring connections for 9-pin to 25-pin connector cables.

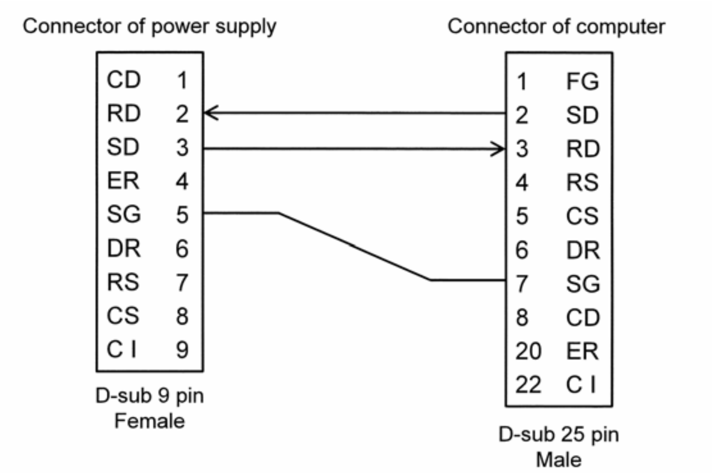


Fig. A-1 Example of RS-232 Cable Wiring Connections

A2.2 RS-485

A2.2.1 Transmission Specification

Interface	RS-485
Synchronous system	Asynchronous
Transmission rate	1200, 2400, 4800, 9600 and 19.2k bits per second (Refer to Section 6.6 "Software Operation" (4) for settings)
Character configuration	Start bit: 1 Data bits: 8 Parity: None Stop bit: 1
Flow control	None
Number of power supply	Multi-drop function OFF: 1 Multi-drop function ON: Max 32 (∗1)

∗1: There may be restrictions depending on cable length or cable type. Perform appropriate checks in the actual operating environment.

A2.2.2 Communications Connector

Connector	Rear panel RS-485 connector (Refer to Section 2.1 "Power Supply Unit")
Connector type	D-sub 9 pin Female, Screw lock size: M2.6
Pin assignment	1, 6: RxA (Receive data +) 2, 7: RxB (Receive data –) 3, 8: TxB (Transmit data –) 4, 9: TxA (Transmit data +) *Other pins are not connected.

A2.2.3 CABLE

(1) Cable Connection

a. Multi-drop function OFF

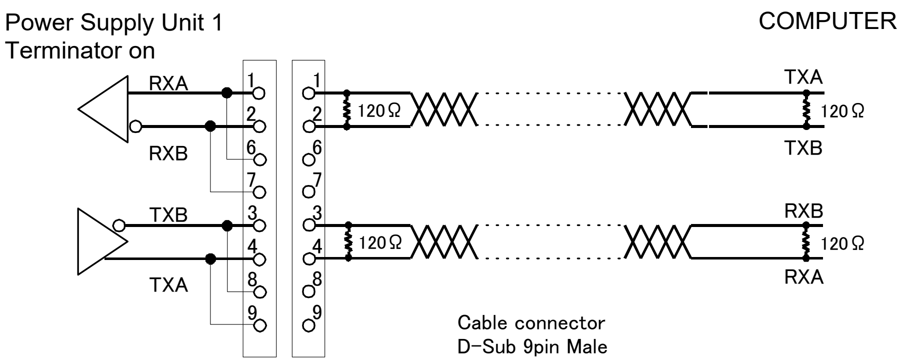


Fig. A-2 Example of RS-485 Cable Wiring Connections (Multi-drop function OFF)

b. Multi-drop function ON

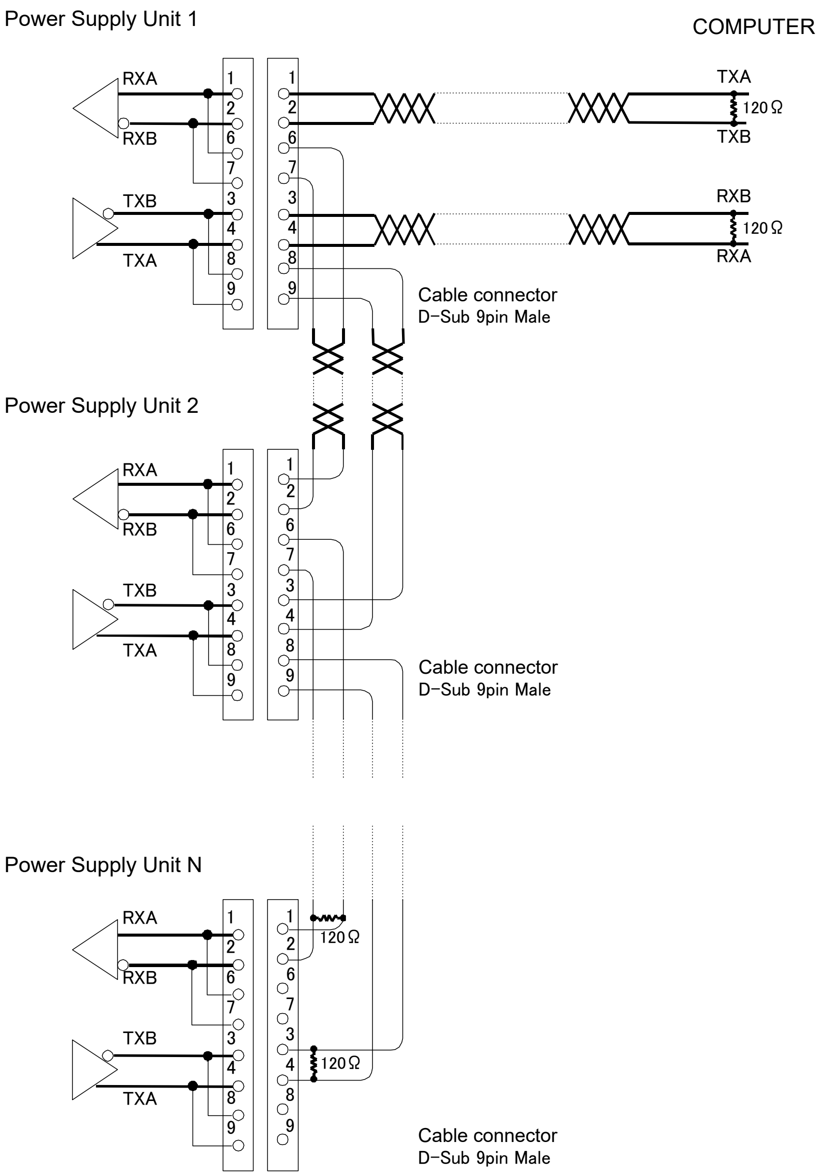


Fig. A-3 Example of RS-485 Cable Wiring Connections (Multi-drop function ON)

(2) Cables used

RS-485 is a differential transmission and use twisted-pair cables in combinations as shown in Fig. A-2 and Fig. A-3. The shield of the cable should be connected to case.

(3) Connecting the terminator

A terminator (120 Ω, 1/4 W min.) is required for connection.

Terminators are not necessary for multi-dropped turbo molecular pump with another pump or pump to which the computer connects is connected to both ends (pump 1 through N in Fig. A-3).

However, connection of the terminator may prevent communications with certain cable lengths and RS-485 device types. Connect the terminator to determine whether it is required.

(4) Cable length
Connection cables can be extended up to 1.2 kilometers, but may be subjects to errors depending on actual operational environment.

NOTICE

Serial communication specifications conform to RS-232C and RS-485.
These interfaces were tested on a typical condition, but the normal communication with all equipments are not guaranteed.

A3 POWER SUPPLY TO COMPUTER CONNECTION

A3.1 Communication Cable Connection

Turn off the power supply unit and the computer to be connected. Connect the RS-232C connector on the power supply unit front panel or the RS-485 connector on the rear panel (Refer to Section 2.1 in this document, "Power Supply Unit") to the communications port of the computer with a cable, referring to Section A2 "INTERFACE SPECIFICATION".

A3.2 Serial Communications Baud Rate Configuration

Check with a baud rate of a computer connected to, and set up a baud rate of RS-232C or RS-485. Refer to Section 6.6 "Software Operation" (4) for details about the setting method.

A3.3 RS-485 Multi-drop Settings

The RS-485 interface multi-drop function is used to connect multiple power supply units to a single computer. Turn off the multi-drop function if the RS-485 is used instead of RS-232C to extend the communication cable length.

When the multi-drop function is turned on, the network ID portion of the sent character strings (Refer to Section A4 "SERIAL COMMUNICATIONS PROTOCOL") is enabled and the event communication function that notifies the power supply unit status to the connected computer is disabled.

Set up the RS-485 as described below when using a multi-drop connection.

(1) Turn on the Multi-drop Function
Turn on the multi-drop function using setting mode/RS485 setting/multi-drop setting on the LCD display. Refer to Section 6.6 "Software Operation" (4) for details about the setting method.

(2) Setting the Network ID
The network ID is set using setting mode/RS485 setting/network ID setting on the LCD display to designate which power supply connected via the multi-drop connection the computer is sending commands to.
The network ID is set as a number between 01 and 32, and must be unique for each power supply connected to a computer. Refer to Section 6.6 "Software Operation" (4) for details about the setting method.

When the RS-485 interface multi-drop function is turned on, the event sending function (Refer to Section A6.5 "Events") will be disabled, regardless of the SETTINGS/RS485/EVENT SENDING menu settings on the LCD display.

A4 SERIAL COMMUNICATIONS PROTOCOL

Communications software, between the power supply and customer equipment should be design according to the following specifications.

A4.1 Basic Message Structure

A basic transmit and receive message begins with the characters "MJ" and ends with a carriage return code (0dH: xxH means hexadecimal code) (Refer to Table A-1).

The first message to be sent is referred as the COMMAND, while the reply to the command is referred as the ANSWER.

Table A-1 Basic Structure of Commands and Answers

Character	Hex. Code	Description	Number Of Bytes
M	4d	Command header characters	2
J	4a		
0	30	Network ID Multi-drop function OFF: 01 fixed Multi-drop function ON: 01 to 32	2
1	31		
X	xx	Command Characters	2
X	xx		
		Sub-command Characters	X
f	xx	Checksum characters (Calculation result by Section A4.7 enters it)	2
f	xx		
CR	0d	Carriage return character	1

A4.2 Character to Character Time-out: 0.1 sec.

Delays between characters, in the answer message, longer that 0.1 sec., shall be considered as a transmission line failure and special considerations should be made to re-send the message.

A4.3 Command to Answer Time-out: 1 sec.

Delays between COMMAND and ANSWER messages, longer that 1 sec., shall be considered as a transmission line failure and special considerations should be made to re-send the message.

The power supply re-sends a COMMAND, if it does not receive and ANSWER within a one second period.

A4.4 Power Supply Command Send Retry Cycles: 5

If the power supply does not receive and answer to a command, within one second, it will re-send the same command up to a maximum of five times.

A4.5 Command Transmission Specification

A command sent before an answer is sent, will be ignored by the power supply, until a reply to the first command is sent. However, this does not apply after a transmission time-out occurs between command and answer (If processing is performed normally, an answer is returned within 100 msec.).

A4.6 Receiving Sequence

The character string from the power supply unit is received after the COMMAND character string is sent. When the carriage return code (0dH) is received, the received character string is checked from the beginning and the portion from the initial command header "MJ" to the carriage return code received last is processed as an answer.

Initialize the receive buffer after the answer character string is acquired from the receive buffer. The read user memo (described below) may receive the same "MJ" as the command header character string in the answer character string. Therefore, interpret the character string from the first "MJ" appearing in the receive buffer to the carriage return as the answer character string.

A4.7 Using the Checksum Byte

Always calculate the checksum for a received character string and compare it with the checksum byte data to confirm that the character string was received correctly. Conduct error processing such as re-sending the command when a character string is received with an incorrect checksum.

Calculation Example

In the received character string "MJ01LS97$" ("$" represents the carriage return code), the check sum code is represented by the last two characters: "97".

The checksum for the received character string is calculated as follows. The result shows that the received character string is correct.

	'M'	'J'	'0'	'1'	'L'	'S'		
Checksum =	4dH +	4aH +	30H +	31H +	4cH +	53H	= 197H	= **97**H

A4.8 Outline of Multi-drop Communications

If the multi-drop function is turned on, set the network ID portion of a sent command to the network ID of the power supply unit with which communication is required.

All connected power supply units receive the sent command, but only the power supply unit with a network ID (set using setting mode/RS485 setting/network ID setting on the LCD display) that matches the network ID portion of the command returns an answer to the command it received.

Consequently, to acquire information from all connected power supply units requires repeated command/answer cycles with all power supply units while changing the network ID.

A5 TABLE OF COMMANDS

Table A-2 Table of Commands

Type	Command/ answer	Name	Command character string	Sub-command character string	
Operation mode	Command	Operation mode check	LS	None	
		On-line request	LN	None	
		Off-line request	LF	None	
	Answer	Local	LL	None	
		Remote	LR	None	
		RS-232C	LC	None	
		RS-485	LD	None	
Operation request	Command	START operation	RT	None	
		STOP operation	RP	None	
		RESET operation	RR	None	
	Answer	Acceleration start	RA	None	
		Deceleration start	RB	None	
		Buzzer off	RZ	None	
		Failure occurred	RF	aa	*1
		Failure elimination	RC	None	
		Operation invalid	RV	None	
Run status	Command	Run status check	CS	None	
	Answer	Stop	NS	aa	*1
		Acceleration	NA	aa	*1
		Normal rotation	NN	aa	*1
		Deceleration	NB	aa	*1
		Failure-Stop	FS	aa	*1
		Failure-Free run	FF	aa	*1
		Failure-Regenerative braking	FR	aa	*1
		Failure-Deceleration	FB	aa	*1
	Command	Read alarm list	CF	aa	*5
	Answer	Send alarm list	CA	aabb	*6
		No alarm list	CV	aa	*5
Parameters	Command	Read parameter	PR	aa	*2
	Answer	Send parameter	PA	aabbbb	*3
		Invalid parameter number	PV	aa	*2
Event	Command	Failure occurred	EF	aa	*1
		Rotation start	ER	None	
		Rotation stop	ES	None	
		Normal rotation	EN	None	
	Answer	Event confirmation	EC	aa	*4
Timer	Command	Read timer	TR	aa	*7
		Clear timer	TC	aa	*7
		Write timer	TW	06aaaaa	*8
	Answer	Send timer value	TA	aabbbbbccc...cccd dd...ddd	*9
		Invalid timer number	TV	aa	*7

Type	Command/ answer	Name	Command character string	Sub-command character string	
History	Command	Read alarm history	GA	aa	*10
	Answer	Send alarm history	GB	xxx...xxx	*11
		No history data	GV	aa	*10
Settings	Command	Read settings	SR	aa	*12
		Write settings	SW	aabbbb	*13
	Answer	Send settings value	SA	aabbbb	*13
		Invalid setting number	SV	aa	*12
	Command	Read user memo	SU	None	
		Write user memo	SX	xxx...xxx	*14
	Answer	Send user memo	SF	xxx...xxx	*14
Shared answer	Answer	Invalid command	AN	None	

*1 aa: Failure alarm codes (hexadecimal) corresponding to the protection system. Refer to Table 7-6 "Table of Alarms" and Table 7-7 "Table of Warnings" in this manual for details.

*2 aa: Parameter number (decimal). Refer to Table A-3 "Table of Parameters".

*3 aa: Parameter number (decimal). Refer to Table A-3 "Table of Parameters".
bbbb: Parameter value (decimal). Refer to Table A-3 "Table of Parameters".

*4 aa: Confirm event command character string.
Send the unchanged command character string of the confirmed event command.
Example: Failure occurred → "EF"

*5 aa: Alarm list number (decimal)

*6 aa: Alarm list number (decimal)
bb: Alarm code (decimal) stored in the alarm list with the requested number.
Refer to Table 7-6 "Table of Alarms" and Table 7-7 "Table of Warnings" in this manual for details.

*7 aa: Timer number (decimal). Refer to Table A-4 "Table of Timer".

*8 06 (fixed): Timer number (decimal). Refer to Table A-4 "Table of Timer".
aaaaa: Set value (decimal). Refer to Table A-4 "Table of Timer".

*9 aa: Timer number (decimal). Refer to Table A-4 "Table of Timer".
bbbbb: Timer value (decimal). Refer to Table A-4 "Table of Timer".
cccccccccc: Time when the timer updated (YYMMDDHHMM format. Stored as Greenwich Mean Time).
dddddddddd: Time when the timer reset (YYMMDDHHMM format. Stored as Greenwich Mean Time).

*10 aa: History number

*11 xxx...xxx: Refer to Table A-5 "Alarm History Data Format".

*12 aa: Settings number (decimal). Refer to Table A-6 "Table of Settings".

*13 aa: Settings number (decimal). Refer to Table A-6 "Table of Settings".
bbbb: Set value (decimal). Refer to Table A-6 "Table of Settings".

*14 xxx...xxx: Any 20-character displayable character string.

Table A-3 Table of Parameters

No.	Name	Range	Description and format
01	Model identification number	0000 to 9999	The pump model connected. Example: TMP-3203 → "3203"
03	Rotational speed	0000 to 5000	Rotational speed / 10 Example: 15000 rpm → "1500"
04	Motor current	0000 to 0150	Motor drive current x 10 Example: 2.3 A → "0023"
07	Temperature control function	0000 to 0002	Temperature control function setting status "0000": Temperature control function on "0001": Temperature control function off "0002": Power supply has no temperature control function
09	Rotational speed (%)	0000 to 0100	Percentage of rated rotational speed. Example: 80 % → "0080"
10	Rotational speed (%)	0000 to 1000	Percentage of rated rotational speed (x10). Example: 80 % → "0800"
11	Rated rotational speed	0000 to 5000	Rated rotational speed / 10 Example: 21000 rpm → "2100"
21	Axis 1 unbalance monitor amount	0000 to 100	Unbalance monitor value of magnetic bearing: Axis 1 Example: 3 % → "0003"
22	Axis 2 unbalance monitor amount	0000 to 0100	Unbalance monitor value of magnetic bearing: Axis 2 Example: 3 % → "0003”
26	MB sensor output X1	0000 to 0100	Sensor output monitor value of magnetic bearing: Axis X1 Example: 3 % → "0003"
27	MB sensor output Y1	0000 to 0100	Sensor output monitor value of magnetic bearing: Axis Y1 Example: 3 % → "0003"
28	MB sensor output X2	0000 to 0100	Sensor output monitor value of magnetic bearing: Axis X2 Example: 3 % → "0003"
29	MB sensor output Y2	0000 to 0100	Sensor output monitor value of magnetic bearing: Axis Y2 Example: 3 % → "0003"
30	MB sensor output Z	0000 to 0100	Sensor output monitor value of magnetic bearing: Axis Z Example: 3 % → "0003"

Table A-4 Table of Timer

No.	Name	Range	Description and format
01	Run time	00000 to 99999	Read the timer value in the “MENU MODE / INTEGRAL TIMER / RUN TIME” on the LCD display (Cannot be reset. Reset date is invalid). Example: 0 time → "00000", 99999 time → "99999"
02	Last maintenance time	00000 to 99999	Read the timer value in the “MENU MODE / INTEGRAL TIMER / LAST MAINT.” on the LCD display (Can be reset). Example: 0 time → "00000", 99999 time → "99999"
03	Power failure touch-down count	00000 to 00999	Read the timer value in the “MENU MODE / INTEGRAL TIMER / POWER FAILURE” on the LCD display (Can be reset). Example: 0 count → "00000", 999 count → "00999"
04	High-speed touch-down count	00000 to 00999	Read the timer value in the “MENU MODE / INTEGRAL TIMER / MB ALARM” on the LCD display (Can be reset). Example: 0 count → "00000", 999 count → "00999"
05	MB warning counter	00000 to 00999	Read the timer value in the “MENU MODE / INTEGRAL TIMER / MB WARNING” on the LCD display (Can be reset). Example: 0 count → "00000", 999 count → "00999"
06	Maintenance call time	00000 to 99999	Read or change the timer value in the “MENU MODE / INTEGRAL TIMER / MAINT.CALL” on the LCD display. Example: 0 time (function is disabled) → "00000", 99999 time → "99999"

Table A-5 Alarm History Data Format

	Item	Number of bytes	Data	Comments
1	History number	2	01 to 99	History number designated by the command.
2	Time	10	YYMMDDHHMM	Time when the failure occurred (stored as Greenwich Mean Time) YY: year, MM: month, DD: day, HH: hour, MM: minutes
3	Alarm number	2	00 to 99	Alarm number of the failure that occurred. Refer to Table 7-6 "Table of Alarms" and Table 7-7 "Table of Warnings" in this manual for details.
4	Run status	2	NS, NA, NN···	Run status when the failure occurred. Data is identical to CS command answer.
5	Rotational speed	4	0000 to 0100	Speed when the failure occurred. Format is identical to 09 in Table A-3.
6	Motor current	4	0000 to 0150	Motor current in the event of a fault. The format is the same as No.04 in Table A-3.
7	Pump temperature	2	00 to 99	This value is not significant in EI-R04M.
8	Temperature control function	2	00, 01, 02, 03	Temperature control function when the failure occurred. Format is equivalent to last 2 characters of 07 in Table A-3.
9	Temperature control set temperature	2	00 to 99	This value is not significant in EI-R04M.
10	Axis 1 unbalance monitor amount	4	0000 to 0100	Unbalance monitor value of magnetic bearing when a fault occurs: Axis 1 Format is the same as No.21 in Table A-3.
11	Axis 2 unbalance monitor amount	4	0000 to 0100	Unbalance monitor value of magnetic bearing when a fault occurs: Axis 2 Format is the same as No.22 in Table A-3.
12	MB sensor output X1	4	0000 to 0100	Sensor output monitor value of magnetic bearing when a fault occurs: Axis X1 Format is the same as No.26 in Table A-3.
13	MB sensor output Y1	4	0000 to 0100	Sensor output monitor value of magnetic bearing when a fault occurs: Axis Y1 Format is the same as No.27 in Table A-3.
14	MB sensor output X2	4	0000 to 0100	Sensor output monitor value of magnetic bearing when a fault occurs: Axis X2 Format is the same as No.28 in Table A-3.
15	MB sensor output Y2	4	0000 to 0100	Sensor output monitor value of magnetic bearing when a fault occurs: Axis Y2 Format is the same as No.29 in Table A-3.
16	MB sensor output Z	4	0000 to 0100	Sensor output monitor value of magnetic bearing when a fault occurs: Axis Z Format is the same as No.30 in Table A-3.
17	Operation time	6	000000 to 099999	Operation time when a fault occurs. Format is the same as No.01 in Table A-4.

Table A-6 Table of Settings

No.	Name	Range	Description and format
01	Temperature control on/off	0000 / 0001	Read or change the set values in the “MENU / SETTINGS / TEMP.CONTORL / TEMP.CONTROL” on the LCD display. "0000": Temperature control function on "0001": Temperature control function off (Valid only for a power supply with a temperature control function)
02	Speed display format	0000 to 0002	Read or change the set values in the “MENU / SETTINGS / ROT.SPEED / DISPLAY” on the LCD display. "0000": %, "0001": rpm, "0002": rps
03	Rotational speed	0000 / 0001	Read or change the set values in the “MENU / SETTINGS / ROT.SPEED / SPEED” on the LCD display. "0000": NORMAL, "0001": LOW SPEED
04	Low speed value	0025 to 0100	Read or change the set values in the “MENU / SETTINGS / ROT.SPEED / LOW SPEED” on the LCD display. Example: 25 % → "0025", 100 % → "0100"
05	“ALARM” signal operation setting	0000 / 0001	Read or change the set values in the “MENU / SETTINGS / REMOTE SIGNAL MODE / ALARM” on the LCD display. "0000": SEMI-E74, "0001": EI-03
06	“REMOTE” signal operation setting	0000 / 0001	Read or change the set values in the “MENU / SETTINGS / REMOTE SIGNAL MODE / REMOTE” on the LCD display. "0000": SEMI-E74, "0001": EI-03
07	“STOP” signal operation setting	0000 / 0001	Read or change the set values in the “MENU / SETTINGS / REMOTE SIGNAL MODE / STOP” on the LCD display. "0000": REMOTE ONLY, "0001": REMOTE&RSXXX
08	Low rotation speed	0250 to 1000	Read or change the set values in the “MENU / SETTINGS / ROT.SPEED / LOW SPEED” on the LCD display. Example: 25.0 % → "0250", 99.9 % → "0999"
10	Warning output setting	0000 / 0001	Read or change the set values in the “MENU / SETTINGS / WARNING DISPLAY” on the LCD display. "0000": ON, "0001": OFF
11	Power failure detect setting	0000 / 0001	Read or change the set values in the “MENU / SETTINGS / PF DETECT” on the LCD display. "0000": 2 sec, "0001": 1 sec

A6 COMMAND DESCRIPTION

A6.1 Operation Mode

Commands	LS	Operation mode check Enables operation mode verification (LOCAL / REMOTE / RS-232C / RS-485) Action: Power supply returns an ANSWER showing present operation mode.
	LN	ON-LINE request If the current operation mode is REMOTE, the operation mode is shifted to RS-232C or RS-485. This command is ineffective in other operation modes. Action: Power supply returns an ANSWER showing the present operation mode.
	LF	OFF-LINE request If the current operation mode is RS-232C or RS-485, the operation mode is shifted to REMOTE. This command is ineffective in other operation modes. Action: Power supply returns an ANSWER showing the present operation mode.
Answers	LL	Operation mode LOCAL This answer is returned when the operation mode is LOCAL. The operation mode can also be shifted to LOCAL mode by the front panel REMOTE/ LOCAL selection switch.
	LR	Operation mode REMOTE This answer is returned when the operation mode is REMOTE. The operation mode can also be shifted to REMOTE mode by the front panel REMOTE/ LOCAL selection switch or when in the RS-232C or RS-485 operation mode by the "OFF-LINE" request command.
	LC	Operation mode RS-232C This answer is returned when the operation mode is RS-232C. The operation mode is shifted to RS-232C when the "ON-LINE" request command is sending via RS-232C in the remote operation mode.
	LD	Operation mode RS-485 This answer is returned when the operation mode is RS-485. The operation mode is shifted to RS-485 when the "ON-LINE" request command is sending via RS-485 in the remote operation mode.

A6.2 Operation

Commands	RT	START Operation This command is the equivalent of pushing the front panel START switch. Action: The turbo molecular pump starts accelerating and sends the “Acceleration Start” answer.
	RP	STOP Operation This command is the equivalent of pushing the front panel STOP switch. Action: The turbo molecular pump starts decelerating and sends the “Deceleration Start” answer.
	RR	RESET Operation This command is the equivalent of pushing the front panel RESET switch. Action: This command is effective against failures. This command resets the alarm buzzer sound and returns the “Buzzer Off” answer. If the buzzer is already off, this command resets the ALARM. If the cause of the alarm is eliminated after resetting, the “Failure Elimination” answer will be returned, else the buzzer will sound again and the “Failure occurrence “ Answer is sent back.
Answers	RA	Acceleration Start This answer is returned by the power supply after the acceleration is started on a START operation.
	RB	Deceleration Start This answer is returned by the power supply after the deceleration is started on a STOP operation.
	RZ	Buzzer Off This answer is returned by the power supply after the buzzer is turned off on a RESET operation.
	RC	Failure elimination This answer is returned by the power supply after the failure cause is removed after the power supply is reset on a RESET operation.
	RF	Failure Occurrence This answer is returned by the power supply if the failure cause is not removed after the power supply is reset on a RESET operation. The alarm code of the failure that has not been eliminated is returned as a 2-character sub-command.
	RV	Operation invalid This answer is returned if the operation is invalid (START operation command sent during acceleration) or if the operation mode differs from the port that sent the command (operation mode is RS-485 but operation command was sent from the RS-232C port).

A6.3 Run Status

Commands	CS	Run Status Check This command requests the current power supply status.
Answers	NS	Stop This answer is returned when the pump stops. Equivalent to the monitor mode/STOP run status on the LCD display. For a normal status, "00" is returned as the sub-command. If a warning has occurred, the 2-character alarm code is returned as the sub-command.
	NA	Acceleration This answer is returned during pump acceleration. Equivalent to the monitor mode/ACC. run status on the LCD display. For a normal status, "00" is returned as the sub-command. If a warning has occurred, the 2-character alarm code is returned as the sub-command.
	NN	Normal rotation This answer is returned during normal pump rotation. Equivalent to the monitor mode/ NORMAL run status on the LCD display. For a normal status, "00" is returned as the sub-command. If a warning has occurred, the 2-character alarm code is returned as the sub-command.
	NB	Deceleration This answer is returned during pump deceleration. Equivalent to the monitor mode/ BRAKE run status on the LCD display. For a normal status, "00" is returned as the sub-command. If a warning has occurred, the 2-character alarm code is returned as the sub-command.
	FS	Failure-Stop This answer is returned when the pump is stopped after a failure occurs. Equivalent to the monitor mode/E-STOP run status on the LCD display. The 2-character alarm code is returned as the sub-command.
	FF	Failure-Free run This answer is returned when the pump is free-running (neither accelerating nor decelerating) after a failure occurs. Equivalent to the monitor mode/E-IDLE run status on the LCD display. The 2-character alarm code is returned as the sub-command.
	FR	Failure-Regenerative braking This answer is returned when the pump is regenerative braking after a failure occurs. Equivalent to the monitor mode/E-BRAKE run status on the LCD display. The 2-character alarm code is returned as the sub-command.
	FB	Failure-Deceleration This answer is returned when the pump is decelerating after a failure occurs. Equivalent to the monitor mode/E-BRAKE run status on the LCD display. The 2-character alarm code is returned as the sub-command.
Commands	CF	Read alarm list Reads the alarm that occurred for a designated alarm list number. The alarm list numbers are equivalent to the sequence displayed on the LCD in the LCD display alarm mode. To check all the current failures, the sub-command alarm list number is increased sequentially from 01 until the answer CV is returned.
Answers	CA	Send alarm list Returns the alarm code corresponding to the requested alarm list number. The sub-command returns a 2-character alarm list number and a 2-character alarm code.
	CV	No alarm list This answer is returned if no alarm corresponds to the requested alarm list number. The sub-command returns a 2-character alarm list number.

A6.4 Parameters

Commands	PR	Read paramater Reads the parameter value for a designated parameter number. Sends the 2-character parameter number as the sub-command. Refer to Table A-3 "Table of Parameters" for parameter number.
Answers	PA	Send parameter Returns the parameter value for the designated parameter number. The 2-character parameter number + 4-character parameter value is returned as the sub-command.
	PV	Invalid parameter number This answer is returned if the designated parameter number is invalid. Returns the 2-character parameter number as the sub-command.

A6.5 Events

For the event functions only, commands are sent from the power supply unit to the connected computer, and the answers are sent from the computer to the power supply unit.

This function can also be disabled by setting the SETTINGS/RS232C/EVENT SENDING or SETTINGS/RS485/EVENT SENDING menu setting to OFF on the LCD display.

This function will be disabled automatically if the RS485 multi-drop function is enabled.

In the default status, command EF is transmitted when either an alarm or warning occurs.

However, if the warning output setting is OFF in the menu mode SETTINGS/WARNING DISPLAY item on the LCD, command EF is not transmitted when warnings of alarm codes 86 to 98 (Table 7-7 "Table of Warnings") occur.

Commands	EF	Failure occurred Sent to the connected computer when an failure occurs. The 2-character alarm code for the failure is sent as the sub-command.
	ER	Start rotation Sent to the connected computer when pump rotation starts. Equivalent to the ROTATION lamp lighting.
	ES	Stop rotation Sent to the connected computer when pump rotation stops. Equivalent to the ROTATION lamp going out.
	EN	Normal speed Sent to the connected computer when the normal rotation speed is achieved. Equivalent to the NORMAL SPEED lamp lighting.
Answers	EC	Confirm event Return this answer to the power supply unit when an event is received from the power supply unit. The power supply unit sends the command up to five times at one-second intervals until it receives the confirm event answer. Send the 2-character command character string for the received event command as a sub-command.

A6.6 Timer

Commands	TR	Read timer Reads the timer value for a designated timer number. Sends the 2-character timer number as the sub-command. Refer to Table A-4 "Table of Timer" for timer number.
	TC	Clear timer Clears the timer value for a designated timer number. Sends the 2-character timer number as the sub-command.
	TW	Write timer Overwrites the set value for a maintenance call timer. Sends the 2-character settings number + 5-character set value data as the sub-command.
Answers	TA	Send timer value Returns the timer value for the designated timer number. The 2-character timer number + 5-character timer value is returned as the sub-command.
	TV	Invalid timer number This answer is returned if the designated timer number is invalid. Returns the 2-character timer number as the sub-command.

A6.7 History

Commands	GA	Read alarm history Reads the alarm history for a designated alarm history number. Sends the 2-character alarm history number as the sub-command.
Answers	GB	Send alarm history Returns the alarm history for the designated alarm history number. The 64-character alarm history data is returned as the sub-command in the format shown in Table A-5 "Alarm History Data Format".
	GV	No history data This answer is returned if no alarm history data corresponds to the designated alarm history number. Returns the 2-character alarm history number as the sub-command.

A6.8 Settings

Commands	SR	Read settings Reads the set value for a designated settings number. Sends the 2-character settings number as the sub-command. Refer to Table A-6 "Table of Settings" for setting number.
	SW	Write settings Overwrites the set value for a designated settings number. Sends the 2-character settings number + 4-character set value data as the sub-command. Refer to Table A-6 "Table of Settings" for setting number and set value.
Answers	SA	Send settings value Returns the set value for the designated settings number. The 2-character settings number + 4-character set value is returned as the sub-command.
	SV	Invalid setting number This answer is returned if the designated settings number is invalid. Returns the 2-character settings number as the sub-command.
Commands	SU	Read user memo Reads the character string in the user memo.
	SX	Write user memo Overwrites the character string in the user memo. Sends the 20 characters to set in the user memo as the sub-command. If less than 20 characters are set, the remaining characters are filled with spaces. Be sure to always send 20 characters.
Answers	SF	Send user memo Returns as a sub-command the set user memo character string or the 20-character user memo character string overwritten by the SX command.

A6.9 Shared Answer

Answers	AN	Invalid Command Answer returned by the power supply after it receives an invalid command.

A7 RS-232C COMMANDS / ANSWERS (SEND AND RECEIVE Examples)

Table A-7 RS-232C COMMANDS / ANSWERS (SEND AND RECEIVE Examples)

Type	Computer **(Host)** *1	Send/ **Receive** *2	Power supply	Description	Remarks
Operation Mode	MJ01LS97$	→		Operation Mode Check	
		←	MJ01LL90$	LOCAL	
			MJ01LR96$	REMOTE	
			MJ01LC87$	RS-232C	
			MJ01LD88$	RS-485	
	MJ01LN92$	→		ON-LINE Request	ON-LINE request from RS-232C communication port
		←	MJ01LC87$	Operation Mode Change	Operation mode changed to RS-232C ON-LINE
			MJ01LD88$	Invalid Request	When in RS-485 mode
			MJ01LL90$	Invalid Request	When in LOCAL mode
	MJ01LF8A$	→		OFF-LINE Request	OFF-LINE request from RS-232C communication port
		←	MJ01LR96$	Operation Mode Change	Operation mode changed to REMOTE
			MJ01LD88$	Invalid Request	When in RS-485 mode
			MJ01LL90$	Invalid Request	When in LOCAL mode
TMP Operation	MJ01RT9E$	→		START Operation	START operation from RS-232C communication port
		←	MJ01RA8B$	Acceleration Start	
			MJ01RVA0$	Ineffective Operation	When START operation is ineffective or operation mode is not RS-232C
	MJ01RP9A$	→		STOP Operation	STOP operation from RS-232C communication port
		←	MJ01RB8C$	Deceleration Start	
			MJ01RVA0$	Ineffective Operation	When STOP operation is ineffective or operation mode is not RS-232C
	MJ01RR9C$	→		RESET Operation	RESET operation from RS-232C communication port
		←	MJ01RZA4$	Buzzer Off	When a buzzer sounded
			MJ01RF50F5$	Failure Occurrence	When the failure was not eliminated.
			MJ01RC8D$	Failure Eliminated	When the failure was eliminated.
			MJ01RVA0$	Ineffective Operation	When RESET operation is ineffective or operation mode is not RS-232C

Type	Computer **(Host)** *1	Send/ **Receive** *2	Power supply	Description	Remarks
Run Status	MJ01CS8E$	→		Run Status Check	
		←	MJ01NS00F9$	Stop	
			MJ01NA00E7$	Acceleration	
			MJ01NB00E8$	Deceleration	
			MJ01NN00F4$	Normal Rotation	
			MJ01FS1C05$	Failure Stop	LCD: “TMP:CAN NOT START”
			MJ01FF32E9$	Failure Idle	LCD: “EI:DC-DC OVERTEMP”
			MJ01FR15F6$	Failure Regeneration	LCD: “POWER FAILURE”
			MJ01FB60E6$	Failure Deceleration	LCD: “MB:VIBRATION 1 Z”
	MJ01CF01E2$	→		Read Alarm List	Confirm first alarm
		←	MJ01CA011543 $	Send Alarm List	Power failure occurred.
Parameter	MJ01PR03FD$	→		Read Parameter	Parameter 03 (rotational speed)
		←	MJ01PA032700 B5$	Send Parameter	Data = 2700 (27000 rpm)
	MJ01PR1500$	→		Read parameter	Parameter 10 (invalid number)
		←	MJ01PV1504$	Invalid parameter number	
Event		←	MJ01EF15E9$	Failure Occurrence	Power failure occurred.
	MJ01ECEF0B$	→		Event Confirmation	
		←	MJ01ER8F$	Rotation Start	
	MJ01ECER17$	→		Event Confirmation	
		←	MJ01ES90$	Rotation Stop	
	MJ01ECES18$	→		Event Confirmation	
		←	MJ01EN8B$	Normal Rotation	
	MJ01ECEN13$	→		Event Confirmation	
Timer	MJ01TR01FF$	→		Read Timer	Timer 01 (Run time)
		←	MJ01TA010013 503040515000 000000000B9$	Send Timer	Timer 01 = 135 (135 hours) Last update: 2003/4/5 15:00 Last reset: (invalid)
	MJ01TC03F2$	→		Clear Timer	Clear timer 03 (Number of power failure touch-downs).
		←	MJ01TA030000 003040515000 304051500C4$	Send Timer	Sets timer 03 = 0 to date/time cleared. Sets reset date/time to date/time when clear command was sent.
	MJ06TW06050 0003$	→		Write Timer	Timer 06 = 5000 Value set. (Maintenance call time)
		←	MJ01TA060500 003040515000 304051500CC$	Send Timer	Sets timer 06 = 5000 to date/ time updated. Sets reset date/ time to date/time when write command was sent.

Type	Computer **(Host)** *1	Send/ **Receive** *2	Power supply	Description	Remarks
History	MJ01GA01E1$	→		Read Alarm History	History 01
		←	MJ01GB01030401120015NNO1000010000275000400060003000300050005000200120098$	Send Alarm History	History: 01 Date & time: 2003/04/01 12:00 Alarm: power failure Status: normal rotation Rotational speed: 100 % Motor current: 1.0 A Pump temperature: 00 degrees C. (*3) Temperature control function: Power supply has no temperature control function Temperature control set temperature: 75 degrees C. (*3) Unbalance monitor Axis1: 4 %, Axis2: 6 % MB sensor output X1: 3 %, Y1: 3 %, X2: 5 %, Y2: 5 %, Z: 2 % Runtime: 1200 hours
	MJ01GA10E1$	→		Read Alarm History	History 10
		←	MJ01GV10F6$	No History Data	Less than 10 alarm data
Setting	MJ01SR02FF$	→		Read Settings	Settings number 02
		←	MJ01SA020000AE$	Send Settings Value	Settings number 02=0 →Speed display: %
	MJ01SW020001C5$	→		Write Settings	Overwrite settings number 02 = 1
		←	MJ01SA020001AF$	Send Settings Value	Settings number 02=1 →Speed display: rpm
Others	MJ01AA7A$	→		Undefined Command	When undefined command is received
		←	MJ01AN87$	Invalid Command	
	MJ01LS20$	→		Operation Mode Check	When command is correct, but checksum is not.
		←	MJ01AN87$	Invalid Command	

*1: "$" represents a carriage return code (0dH).

*2: → From computer to power supply unit.
← From power supply unit to computer.

*3: Invalid for a power supply without a pump temperature control function, but data is set.

A8 RELATION OF LOCAL MODE TO REMOTE MODE OPERATIONS

(1) Input of front panel switch is only effective when REMOTE lamp is in "LOCAL" mode (REMOTE lamp is OFF).

(2) When the REMOTE lamp is in "REMOTE" mode (REMOTE lamp is ON), "REMOTE" input signal only is effective under initial status.

(3) When the REMOTE lamp is in "REMOTE" mode (REMOTE lamp is ON),

  a. The operation mode is shifted to RS-232C ON-LINE in response to ON-LINE request of operation mode command from RS-232C communication port, only operation by the operation request command from computer is effective.

  b. The operation mode is shifted to RS-485 ON-LINE in response to ON-LINE request of operation mode command from RS-485 communication port, only operation by the operation request command from computer is effective.

(4) RESET switch input and "RESET" signals are all-time effective.

(5) When the selection switch is shifted to "LOCAL" under ON-LINE operation mode, the operation mode is force-shifted to LOCAL.

(6) Commands other than operation commands are all-time effective, and the power supply unit sends back an answer message to computer. In addition, event commands are all-time sent against event occurrence.

A9 TROUBLESHOOTING

A9.1 No Message can Transmit and Receive

(1) Start the pump in LOCAL mode and check if the event command of Rotation start can be received in the timing at which ROTATION lamp lights.

a. Could be received >> check if command from connected computer can be received or not, using another computer, etc.

b. A nonsensical character string was received >> Go to Section A9.2.

c. Not receivable >> (2)

(2) Check the connection of RS-232C cable in reference to Section A2 "INTERFACE SPECIFICATION".
Check the polarity of RS-485 interface, because there is the case that polarity is reverse.

(3) Check the transmission specification of RS-232C at computer side.

A9.2 Sending and Receiving are Done, But Receivable Messages are Invalid

(1) Check the transmission rate of the power supply unit and computer.

A9.3 Characters Get Disordered from Time to Time, Then Resulting in CHECKSUM Error

(1) Remove the cable from equipment as noise source if it runs near it.

(2) When the cable in use is not a shield cable, replace it with the latter cable.
When shield cable is used, be sure to check that it is connected to the frame gland of the connected computer.
Use twisted pair cable when RS-485 is used.

(3) When 10 m or longer cable is used, replace it with another cable as short as possible.

(4) Make the transmission rate smaller unless there is problem in application program.

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Declaration of Conformity

SHIMADZU CORPORATION INDUSTRIAL MACHINERY DIVISION
Address : 1, NISHINOKYO-KUWABARACHO, NAKAGYO-KU, KYOTO, 604-8511, JAPAN

as the Manufacturer
declares in sole responsibility that the following product

Product Name **Turbo Molecular Pump**
Part Number **Depend on configuration.**

Power Supply	
Model name	P/N
EI-R04M	262-78982-02
EI-R04M (232)	262-78990-02
EI-R04M (DN)	263-18047-02
EI-R04M (T17)	263-18676

referred to in this declaration conforms with following directives and standards

Machinery directive 2006/42/EC
EN ISO 12100:2010
EN1012-2:1996+A1:2009
EN61010-1:2010
EMC directive 2014/30/EU
EN61326-1:2013 Class A
RoHS Directive 2011/65/EU

Note 1) This declaration becomes invalid if technical or operational modifications are introduced without manufacturer's consent.

Note 2) This declaration is valid if this product is used alone or in combination with the accessories of this product or other instruments which fulfill with the requirement of mentioned directive.

Note 3) Authorised representative in EU is as follows:
Infraserv Vakuumservice GmbH
Address: Gleiwitzerstraße 8, 85386, Eching, Germany

Note 4) The translation version in the language of the denstination will be prepared by Shimadzu Corporation.

Kyoto, JAPAN 27. Jan. 2020
place and date of issue

H. Yamaguchi
signature

Hitoshi Yamaguchi
name

Manager of Quality Assurance Dept.
Industrial Machinery Division
Shimadzu Corporation
Position

FE9R-0021-02B

TMP overhaul / repair request form

Please fill out this request form and attach to the product before you send back to Shimadzu service center for overhaul or repair service. We ask that you fill out this form completely to expedite the service and return shipment.

Please mark the item box, and fill out the blank.

(1) Product	☐Pump (TMP-) s/n ( ) ☐Controller (EI-) s/n ( ) ☐Attachiment ( )
(2) Request	☐Overhaul ☐Repair ☐Other
(3) Details	
(4) Alarm code	(If status lamp indicates, No is)
(5) Date of failure occurrence ( )	(6) Date for return shipment request ( )
(7) Date of TMP operation start ( )	(8) Date of previous delivery ( )
(9) Parts exchange recommendation	If any parts exchange is recommended due to the excess of usage recommendation period, ☐(a) Please exchange the parts. ☐(b) Don't exchange unless any deterioration is found in the inspection. In case of (b), even if any failure occurs with the cause of parts for which customers choose continuous usage, it's out of warranty. Especially at rotor, it causes the material deterioration which even the fluorescent penetrant test can't detect after long term running. For this reason, periodical exchange is recommended in order to use the rotor safely.
(10) Declaration of contamination	Please fill out the following items to make sure of our safety. Components which have been contaminated by hazardius substrates will not be accepted or served without written evidence of decontamination. - Equipment process ☐Etch ☐PVD ☐CVD ☐Others ( ) - Materials the equipment has been exposed to ( ) (Eched material, CVD/PVD target, etc) - Gases the equipment has been exposed to ☐Air, Nitrogen, etc ( ) ☐Reactive/active gas ( ) ☐Hazardous gas ( ) - Is it hazardous to human ? ☐Yes ☐No ☐Inert gas such as helium, etc ( ) ☐Corrosive gas ( ) ☐Others ( ) - Is there color change or adhesion at inlet or outlet flange ? ☐Yes ☐No (If yes, please let us perform TMP cleaning at additional charge to keep appropriate performance. - Precaution and procedure for decontamination in case it's necessary at Shimadzu. ( )
(11) Gas purge	☐Used (sccm) ☐Not used

Date ( )

Company ( ) Job title ( )

Name ( ) Signature ( )

Phone ( ) Fax ( )

263-13450-03A

「電器電子製品有害物質使用制限管理弁法」（中華人民共和国 工業情報化部発行）に基づく，
「環境保護使用期限」と「製品中の有害物質の名称および含有量」表示

Markings regarding the "Environmental Protection Use Period" and the "Names and Contents of Hazardous Substances" for "Management Methods for the Restriction of the Use of Hazardous Substances in Electrical and Electronic Products" (Issued by: Ministry of Industry and Information Technology of the People's Republic of China)

Model Name : **Pump Unit : TMP-xx04 series and TMP-xx03 series**
Power Supply : EI-x04M/MT series

環境保護使用期限 [环保使用期限标识]
Environmental Protection Use Period

この環境保護使用期限標識（マーク）は【電器電子製品有害物質使用制限管理弁法】及び【電子電気製品有害物質使用制限の標識に関する要求】に基づき策定され，中国国内で販売される電気電子製品に適用されます。マーク中央の数字は，環境保護使用期限の年数です。この製品に関する安全や使用上の注意をお守りいただく限り，製造日から起算するこの年限内では，有害物質が外部へ漏れ出したり変化をおこしたりせず，重大な環境汚染や人体や財産に影響を及ぼすことはありません。なお，この標識の期限は，製品の品質や機能に関する保証期間と異なります。

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The Marking of Environmental Protection Use Period is formulated according to the stipulations in the "Management Methods for the Restriction of the Use of Hazardous Substances in Electrical and Electronic Products" and "Marking for the Restricted Use of Hazardous Substances in Electronic and Electrical Products", and applies to the Electrical and Electronic Products sold in the People's Republic of China.
The number in the middle of the logo means the unit of years for the actual environmental protection use period for this product. Within the indicated period from the date of manufacture, hazardous substances contained in the electronic information products will not leak or mutate under normal operating conditions so that the use of this electronic information products will not result in any severe environmental pollution, any bodily injury or any damage to assets.
In addition, note that a time limit in this marking is different from a guarantee period for quality and functions of this product.

製品中の有害物質の名称および含有量 [产品中有害物质的名称及含量]
Names and Contents of Hazardous Substances

部品名称 [部件名称] Part Name	有害物質 [有害物质] Hazardous Substances					
	鉛 [铅] Lead (Pb)	水銀 [汞] Mercury (Hg)	カドミウム [镉] Cadmium (Cd)	六価クロム [六价铬] Hexavalent Chromium (Cr (VI))	ポリ臭化ビフェニル [多溴联苯] Polybrominated biphenyls (PBB)	ポリ臭化ジフェニルエーテル [多溴二苯醚] Polybrominated diphenyl ethers (PBDE)
Pump Unit (04 series) [泵主体]	X	O	X	O	O	O
Pump Unit (03 series) [泵主体]	X	O	X	X	O	O
Power Supply [电源组件]	X	O	X	O	O	O
Cables [线束・联接电缆]	X	O	X	O	O	O

表は，SJ/T11364の規定に基づいて作成する。[本表格依据 SJ/T11364 的规定编制。]
This table is prepared in accordance with the provisions of SJ/T 11364.

O: 該当部品の中の全ての均質材料に含まれる有害物質の含有量がいずれもGB/T26572が定めた制限値を下回ったことを示す。
[表示该有害物质在该部件所有均质材料中的含量均在 GB/T26572 规定的限量要求以下。]
Indicates that said hazardous substance contained in all of the homogeneous materials for this part is below the limit requirement of GB/T 26572.

X: 該当部品の中の最低一つの均質材料に含まれる有害物質の含有量がGB/T26572が定めた制限値を上回ったことを示す。
[表示该有害物质至少在该部件的某一均质材料中的含量超出 GB/T26572 规定的限量要求。]
Indicates that said hazardous substance contained in at least one of the homogeneous materials used for this part is above the limit requirement of GB/T 26572.

To all user of Shimadzu equipment in the European Union :

Thank you for using Shimadzu equipment.

WEEE Mark

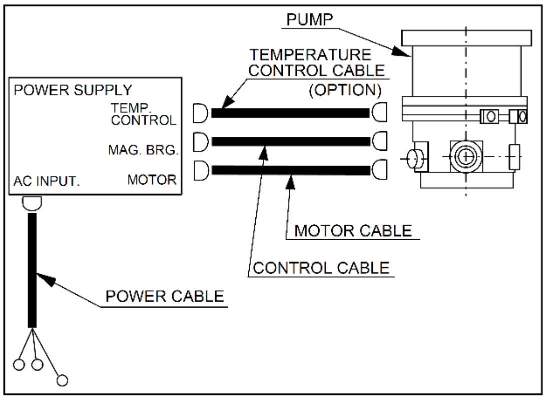
Equipment marked with this symbol indicates that it was sold on or after 13th August 2005, which means it should not be disposed of with general household waste. Please note that our equipment is for industrial/professional use only.

With your co-operation we are aiming to reduce contamination from waste electronic and electrical equipment and preserve natural resource through re-use and recycling. Please do not hesitate to ask your Shimadzu office or Shimadzu distributor, if you require further information.

Information for treatment facilities (TMP-03 series, EI-R04 series)

Cables (maximum configuration)

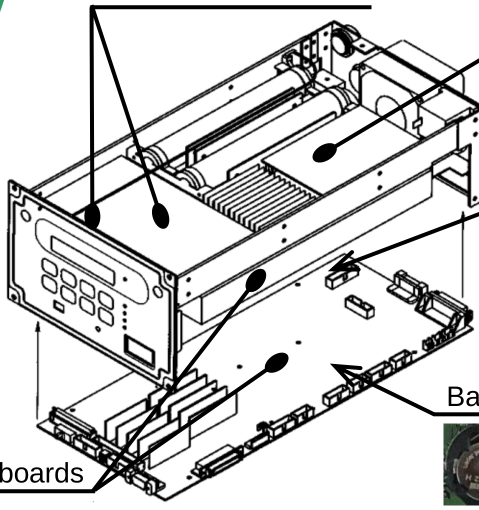
Power Supply (EI-R04M, EI-R04MT)



Pump (TMP-xxx3 series)

Printed circuit boards

Printed circuit board (Option)



Capacitors (4 pieces)

Battery

Printed circuit boards

For California, USA Only :

This product contains a battery that contains perchlorate material.
Perchlorate Material-special handling may apply.
See www.dtsc.ca.gov/hazardouswaste/perchlorate

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